Chapter I
Algebraic Integers

§ 1. The Gaussian Integers

The equations

$$2 = 1+1,\ 5 = 1+4,\ 13 = 4+9,\ 17 = 1+16,\ 29 = 4+25,\ 37 = 1+36$$

show the first prime numbers that can be represented as a sum of two squares. Except for 2, they are all $\equiv 1 \bmod 4$, and it is true in general that any odd prime number of the form $p = a^2 + b^2$ satisfies $p \equiv 1 \bmod 4$, because perfect squares are $\equiv 0$ or $\equiv 1 \bmod 4$. This is obvious. What is not obvious is the remarkable fact that the converse also holds:

(1.1) Theorem. *For all prime numbers $p \neq 2$, one has:*

$$p = a^2 + b^2 \quad (a, b \in \mathbb{Z}) \quad \Longleftrightarrow \quad p \equiv 1 \bmod 4 .$$

The natural explanation of this arithmetic law concerning the ring $\mathbb{Z}$ of rational integers is found in the larger domain of the **gaussian integers**

$$\mathbb{Z}[i] = \{ a + bi \mid a, b \in \mathbb{Z} \}, \quad i = \sqrt{-1}.$$

In this ring, the equation $p = x^2 + y^2$ turns into the product decomposition

$$p = (x + iy)(x - iy),$$

so that the problem is now when and how a prime number $p \in \mathbb{Z}$ factors in $\mathbb{Z}[i]$. The answer to this question is based on the following result about unique factorization in $\mathbb{Z}[i]$.

(1.2) Proposition. *The ring $\mathbb{Z}[i]$ is euclidean, therefore in particular factorial.*

Proof: We show that $\mathbb{Z}[i]$ is euclidean with respect to the function $\mathbb{Z}[i] \to \mathbb{N} \cup \{0\}$, $\alpha \mapsto |\alpha|^2$. So, for $\alpha, \beta \in \mathbb{Z}[i]$, $\beta \neq 0$, one has to verify the existence of gaussian integers γ, ρ such that

$$\alpha = \gamma\beta + \rho \quad \text{and} \quad |\rho|^2 < |\beta|^2 .$$

It clearly suffices to find $\gamma \in \mathbb{Z}[i]$ such that $\left| \frac{\alpha}{\beta} - \gamma \right| < 1$. Now, the gaussian integers form a **lattice** in the complex plane $\mathbb{C}$ (the points with integer coordinates with respect to the basis $1, i$). The complex number $\frac{\alpha}{\beta}$ lies in some mesh of the lattice and its distance from the nearest lattice point is not greater than half the length of the diagonal of the mesh, i.e. $\frac{1}{2}\sqrt{2}$. Therefore there exists an element $\gamma \in \mathbb{Z}[i]$ with $\left|\frac{\alpha}{\beta} - \gamma\right| \leq \frac{1}{2}\sqrt{2} < 1$. $\square$

Based on this result about the ring $\mathbb{Z}[i]$, theorem (1.1) now follows like this: it is sufficient to show that a prime number $p \equiv 1 \bmod 4$ of $\mathbb{Z}$ does not remain a prime element in the ring $\mathbb{Z}[i]$. Indeed, if this is proved, then there exists a decomposition

$$p = \alpha \cdot \beta$$

into two non-units α, β of $\mathbb{Z}[i]$. The **norm** of $z = x + iy$ is defined by

$$N(x + iy) = (x + iy)(x - iy) = x^2 + y^2,$$

i.e., by $N(z) = |z|^2$. It is multiplicative, so that one has

$$p^2 = N(\alpha) \cdot N(\beta).$$

Since α and β are not units, it follows that $N(\alpha), N(\beta) \neq 1$ (see exercise 1), and therefore $p = N(\alpha) = a^2 + b^2$, where we put $\alpha = a + bi$.

Finally, in order to prove that a rational prime of the form $p = 1 + 4n$ cannot be a prime element in $\mathbb{Z}[i]$, we note that the congruence

$$-1 \equiv x^2 \bmod p$$

admits a solution, namely $x = (2n)!$. Indeed, since $-1 \equiv (p-1)! \bmod p$ by Wilson's theorem, one has

$$\begin{aligned} -1 \equiv (p-1)! &= \big[1 \cdot 2 \cdots (2n)\big]\big[(p-1)(p-2)\cdots(p-2n)\big] \\ &\equiv \big[(2n)!\big]\big[(-1)^{2n}(2n)!\big] = \big[(2n)!\big]^2 \bmod p. \end{aligned}$$

Thus we have $p \,|\, x^2 + 1 = (x+i)(x-i)$. But since $\frac{x}{p} \pm \frac{i}{p} \notin \mathbb{Z}[i]$, p does not divide any of the factors $x+i, x-i$, and is therefore not a prime element in the factorial ring $\mathbb{Z}[i]$.

The example of the equation $p = x^2 + y^2$ shows that even quite elementary questions about rational integers may lead to the consideration of higher domains of integers. But it was not so much for this equation that we have introduced the ring $\mathbb{Z}[i]$, but rather in order to preface the general theory of algebraic integers with a concrete example. For the same reason we will now look at this ring a bit more closely.

When developing the theory of divisibility for a ring, two basic problems are most prominent: on the one hand, to determine the **units** of the ring in question, on the other, its **prime elements**. The answer to the first question in the present case is particularly easy. A number $\alpha = a + bi \in \mathbb{Z}[i]$ is a unit if and only if its norm is 1:

$$N(\alpha) := (a + ib)(a - ib) = a^2 + b^2 = 1$$

(exercise 1), i.e., if either $a^2 = 1$, $b^2 = 0$, or $a^2 = 0$, $b^2 = 1$. We thus obtain the

(1.3) Proposition. *The group of units of the ring $\mathbb{Z}[i]$ consists of the fourth roots of unity,*

$$\mathbb{Z}[i]^* = \{1, \ -1, i, \ -i\}.$$

In order to answer the question for primes, i.e., irreducible elements of the ring $\mathbb{Z}[i]$, we first recall that two elements α, β in a ring are called **associated**, symbolically $\alpha \sim \beta$, if they differ only by a unit factor, and that every element associated to an irreducible element π is also irreducible. Using theorem (1.1) we obtain the following precise list of all prime numbers of $\mathbb{Z}[i]$.

(1.4) Theorem. *The prime elements π of $\mathbb{Z}[i]$, up to associated elements, are given as follows.*

(1) $\pi = 1 + i$,

(2) $\pi = a + bi$ *with* $a^2 + b^2 = p$, $p \equiv 1 \bmod 4$, $a > |b| > 0$,

(3) $\pi = p$, $\quad p \equiv 3 \bmod 4$.

Here, p denotes a prime number of $\mathbb{Z}$.

Proof: Numbers as in (1) or (2) are prime because a decomposition $\pi = \alpha \cdot \beta$ in $\mathbb{Z}[i]$ implies an equation

$$p = N(\pi) = N(\alpha) \cdot N(\beta),$$

with some prime number p. Hence either $N(\alpha) = 1$ or $N(\beta) = 1$, so that either α or β is a unit.

Numbers $\pi = p$, where $p \equiv 3 \bmod 4$, are prime in $\mathbb{Z}[i]$, because a decomposition $p = \alpha \cdot \beta$ into non-units α, β would imply that $p^2 = N(\alpha) \cdot N(\beta)$, so that $p = N(\alpha) = N(a + bi) = a^2 + b^2$, which according to (1.1) would yield $p \equiv 1 \bmod 4$.

This being said, we have to check that an arbitrary prime element π of $\mathbb{Z}[i]$ is associated to one of those listed. First of all, the decomposition

$$N(\pi) = \pi \cdot \bar{\pi} = p_1 \cdots p_r\,,$$

with rational primes p_i, shows that $\pi \mid p$ for some $p = p_i$. This gives $N(\pi) \mid N(p) = p^2$, so that either $N(\pi) = p$ or $N(\pi) = p^2$. In the case $N(\pi) = p$ we get $\pi = a + bi$ with $a^2 + b^2 = p$, so π is of type (2) or, if $p = 2$, it is associated to $1 + i$. On the other hand, if $N(\pi) = p^2$, then π is associated to p since p/π is an integer with norm one and thus a unit. Moreover, $p \equiv 3 \bmod 4$ has to hold in this case because otherwise we would have $p = 2$ or $p \equiv 1 \bmod 4$ and because of (1.1) $p = a^2 + b^2 = (a + bi)(a - bi)$ could not be prime. This completes the proof. □

The proposition also settles completely the question of how prime numbers $p \in \mathbb{Z}$ decompose in $\mathbb{Z}[i]$. The prime $2 = (1+i)(1-i)$ is associated to the square of the prime element $1+i$. Indeed, the identity $1-i = -i(1+i)$ shows that $2 \sim (1+i)^2$. The prime numbers $p \equiv 1 \bmod 4$ split into two conjugate prime factors

$$p = (a + bi)(a - bi)\,,$$

and the prime numbers $p \equiv 3 \bmod 4$ remain prime in $\mathbb{Z}[i]$.

The gaussian integers play the same rôle in the field

$$\mathbb{Q}(i) = \big\{a + bi \mid a, b \in \mathbb{Q}\big\}$$

as the rational integers do in the field $\mathbb{Q}$. So they should be viewed as the "integers" in $\mathbb{Q}(i)$. This notion of integrality is relative to the coordinates of the basis $1, i$. However, we also have the following characterization of the gaussian integers, which is independent of a choice of basis.

(1.5) Proposition. *$\mathbb{Z}[i]$ consists precisely of those elements of the extension field $\mathbb{Q}(i)$ of $\mathbb{Q}$ which satisfy a monic polynomial equation*

$$x^2 + ax + b = 0$$

with coefficients $a, b \in \mathbb{Z}$.

Proof: An element $\alpha = c + id \in \mathbb{Q}(i)$ is a zero of the polynomial

$$x^2 + ax + b \in \mathbb{Q}[x] \quad \text{with} \quad a = -2c,\ b = c^2 + d^2.$$

If c and d are rational integers, then so are a and b. Conversely, if a and b are integers, then so are $2c$ and $2d$. From $(2c)^2 + (2d)^2 = 4b \equiv 0 \bmod 4$ it follows that $(2c)^2 \equiv (2d)^2 \equiv 0 \bmod 4$, since squares are always $\equiv 0$ or $\equiv 1$. Hence c and d are integers. □

The last proposition leads us to the general notion of an algebraic integer as being an element satisfying a monic polynomial equation with rational integer coefficients. For the domain of the gaussian integers we have obtained in this section a complete answer to the question of the units, the question of prime elements, and to the question of unique factorization.

These questions indicate already the fundamental problems in the general theory of algebraic integers. But the answers we found in the special case $\mathbb{Z}[i]$ are not typical. Novel features will present themselves instead.

Exercise 1. $\alpha \in \mathbb{Z}[i]$ is a unit if and only if $N(\alpha) = 1$.

Exercise 2. Show that, in the ring $\mathbb{Z}[i]$, the relation $\alpha\beta = \varepsilon\gamma^n$, for α, β relatively prime numbers and ε a unit, implies $\alpha = \varepsilon'\xi^n$ and $\beta = \varepsilon''\eta^n$, with ε', ε'' units.

Exercise 3. Show that the integer solutions of the equation

$$x^2 + y^2 = z^2$$

such that $x, y, z > 0$ and $(x, y, z) = 1$ ("pythagorean triples") are all given, up to possible permutation of x and y, by the formulæ

$$x = u^2 - v^2, \quad y = 2uv, \quad z = u^2 + v^2,$$

where $u, v \in \mathbb{Z}$, $u > v > 0$, $(u, v) = 1$, u, v not both odd.

Hint: Use exercise 2 to show that necessarily $x + iy = \varepsilon\alpha^2$ with a unit ε and with $\alpha = u + iv \in \mathbb{Z}[i]$.

Exercise 4. Show that the ring $\mathbb{Z}[i]$ cannot be ordered.

Exercise 5. Show that the only units of the ring $\mathbb{Z}[\sqrt{-d}] = \mathbb{Z} + \mathbb{Z}\sqrt{-d}$, for any rational integer $d > 1$, are ± 1.

Exercise 6. Show that the ring $\mathbb{Z}[\sqrt{d}] = \mathbb{Z} + \mathbb{Z}\sqrt{d}$, for any squarefree rational integer $d > 1$, has infinitely many units.

Exercise 7. Show that the ring $\mathbb{Z}[\sqrt{2}] = \mathbb{Z} + \mathbb{Z}\sqrt{2}$ is euclidean. Show furthermore that its units are given by $\pm(1 + \sqrt{2})^n$, $n \in \mathbb{Z}$, and determine its prime elements.

§ 2. Integrality

An **algebraic number field** is a finite field extension K of $\mathbb{Q}$. The elements of K are called **algebraic numbers**. An algebraic number is called **integral**, or an **algebraic integer**, if it is a zero of a monic polynomial $f(x) \in \mathbb{Z}[x]$. This notion of integrality applies not only to algebraic numbers, but occurs in many different contexts and therefore has to be treated in full generality.

In what follows, rings are always understood to be commutative rings with 1.

(2.1) Definition. *Let $A \subseteq B$ be an extension of rings. An element $b \in B$ is called* **integral** *over A, if it satisfies a monic equation*

$$x^n + a_1x^{n-1} + \cdots + a_n = 0, \quad n \geq 1,$$

with coefficients $a_i \in A$. The ring B is called **integral** *over A if all elements $b \in B$ are integral over A.*

It is desirable, but strangely enough not immediately obvious, that the sum and the product of two elements which are integral over A are again integral. This will be a consequence of the following abstract reinterpretation of the notion of integrality.

(2.2) Proposition. *Finitely many elements $b_1, \ldots, b_n \in B$ are all integral over A if and only if the ring $A[b_1, \ldots, b_n]$ viewed as an A-module is finitely generated.*

To prove this we make use of the following result of linear algebra.

(2.3) Proposition (Row-Column Expansion). *Let $A = (a_{ij})$ be an $(r \times r)$-matrix with entries in an arbitrary ring, and let $A^* = (a^*_{ij})$ be the adjoint matrix, i.e., $a^*_{ij} = (-1)^{i+j}\det(A_{ij})$, where the matrix A_{ij} is obtained from A by deleting the i-th column and the j-th row. Then one has*

$$AA^* = A^*A = \det(A)E\,,$$

where E denotes the unit matrix of rank r. For any vector $x = (x_1, \ldots, x_r)$, this yields the implication

$$Ax = 0 \implies (\det A)x = 0\,.$$

Proof of proposition (2.2): Let $b \in B$ be integral over A and $f(x) \in A[x]$ a monic polynomial of degree $n \geq 1$ such that $f(b) = 0$. For an arbitrary polynomial $g(x) \in A[x]$ we may then write

$$g(x) = q(x)f(x) + r(x),$$

with $q(x), r(x) \in A[x]$ and $\deg(r(x)) < n$, so that one has

$$g(b) = r(b) = a_0 + a_1b + \cdots + a_{n-1}b^{n-1}.$$

Thus $A[b]$ is generated as A-module by $1, b, \ldots, b^{n-1}$.

More generally, if $b_1, \ldots, b_n \in B$ are integral over A, then the fact that $A[b_1, \ldots, b_n]$ is of finite type over A follows by induction on n. Indeed, since b_n is integral over $R = A[b_1, \ldots, b_{n-1}]$, what we have just shown implies that $R[b_n] = A[b_1, \ldots, b_n]$ is finitely generated over R, hence also over A, if we assume, by induction, that R is an A-module of finite type.

Conversely, assume that the A-module $A[b_1, \ldots, b_n]$ is finitely generated and that $\omega_1, \ldots, \omega_r$ is a system of generators. Then, for any element $b \in A[b_1, \ldots, b_n]$, one finds that

$$b\,\omega_i = \sum_{j=1}^{r} a_{ij}\omega_j\,, \quad i = 1, \ldots, r\,, \quad a_{ij} \in A\,.$$

From (2.3) we see that $\det(bE - (a_{ij}))\,\omega_i = 0$, $i = 1, \ldots, r$ (here E is the unit matrix of rank r), and since 1 can be written $1 = c_1\omega_1 + \cdots + c_r\omega_r$, the identity $\det(bE - (a_{ij})) = 0$ gives us a monic equation for b with coefficients in A. This shows that b is indeed integral over A. □

According to this proposition, if $b_1, \ldots, b_n \in B$ are integral over A, then so is *any* element b of $A[b_1, \ldots, b_n]$, because $A[b_1, \ldots, b_n, b] = A[b_1, \ldots, b_n]$ is a finitely generated A-module. In particular, given two integral elements $b_1, b_2 \in B$, then $b_1 + b_2$ and $b_1 b_2$ are also integral over A. At the same time we obtain the

(2.4) Proposition. *Let $A \subseteq B \subseteq C$ be two ring extensions. If C is integral over B and B is integral over A, then C is integral over A.*

Proof: Take $c \in C$, and let $c^n + b_1 c^{n-1} + \cdots + b_n = 0$ be an equation with coefficients in B. Write $R = A[b_1, \ldots, b_n]$. Then $R[c]$ is a finitely generated R-module. If B is integral over A, then $R[c]$ is even finitely generated over A, since R is finitely generated over A. Thus c is integral over A. □

From what we have proven, the set of all elements

$$\bar{A} = \left\{ b \in B \;\middle|\; b \text{ integral over } A \right\}$$

in a ring extension $A \subseteq B$ forms a ring. It is called the **integral closure** of A in B. A is said to be **integrally closed** in B if $A = \bar{A}$. It is immediate from (2.4) that the integral closure $\bar{A}$ is itself integrally closed in B. If A is an integral domain with field of fractions K, then the integral closure $\bar{A}$ of A in K is called the **normalization** of A, and A is simply called integrally closed if $A = \bar{A}$. For instance, every **factorial** ring is integrally closed.

In fact, if $a/b \in K$ $(a, b \in A)$ is integral over A, i.e.,

$$(a/b)^n + a_1(a/b)^{n-1} + \cdots + a_n = 0,$$

with $a_i \in A$, then

$$a^n + a_1 b a^{n-1} + \cdots + a_n b^n = 0.$$

Therefore each prime element π which divides b also divides a. Assuming a/b to be reduced, this implies $a/b \in A$.

We now turn to a more specialized situation. Let A be an integral domain which is integrally closed, K its field of fractions, $L|K$ a finite field extension, and B the integral closure of A in L. According to (2.4), B is automatically integrally closed. Each element $\beta \in L$ is of the form

$$\beta = \frac{b}{a}, \quad b \in B, \ a \in A,$$

because if

$$a_n\beta^n + \cdots + a_1\beta + a_0 = 0, \quad a_i \in A, \ a_n \neq 0,$$

then $b = a_n\beta$ is integral over A, an integral equation

$$(a_n\beta)^n + \cdots + a_1'(a_n\beta) + a_0' = 0, \quad a_i' \in A,$$

being obtained from the equation for β by multiplication by a_n^{n-1}. Furthermore, the fact that A is integrally closed has the effect that an element $\beta \in L$ is integral over A if and only if its **minimal polynomial** $p(x)$ takes its coefficients in A. In fact, let β be a zero of the monic polynomial $g(x) \in A[x]$. Then $p(x)$ divides $g(x)$ in $K[x]$, so that all zeroes $\beta_1, \ldots, \beta_n$ of $p(x)$ are integral over A, hence the same holds for all the coefficients, in other words $p(x) \in A[x]$.

The trace and the norm in the field extension $L|K$ furnish important tools for the study of the integral elements in L. We recall the

(2.5) Definition. *The* **trace** *and* **norm** *of an element $x \in L$ are defined to be the trace and determinant, respectively, of the endomorphism*

$$T_x : L \to L, \quad T_x(\alpha) = x\alpha,$$

of the K-vector space L:

$$Tr_{L|K}(x) = Tr(T_x), \quad N_{L|K}(x) = \det(T_x).$$

In the characteristic polynomial

$$f_x(t) = \det(t\,\mathrm{id} - T_x) = t^n - a_1 t^{n-1} + \ldots + (-1)^n a_n \in K[t]$$

of T_x, $n = [L : K]$, we recognize the trace and the norm as

$$a_1 = Tr_{L|K}(x) \quad \text{and} \quad a_n = N_{L|K}(x).$$

Since $T_{x+y} = T_x + T_y$ and $T_{xy} = T_x \circ T_y$, we obtain homomorphisms

$$Tr_{L|K} : L \longrightarrow K \quad \text{and} \quad N_{L|K} : L^* \longrightarrow K^* .$$

In the case where the extension $L|K$ is separable, the trace and norm admit the following Galois-theoretic interpretation.

(2.6) Proposition. *If $L|K$ is a separable extension and $\sigma : L \to \overline{K}$ varies over the different K-embeddings of L into an algebraic closure $\overline{K}$ of K, then we have*

(i) $\quad f_x(t) = \prod\limits_{\sigma}(t - \sigma x)$,

(ii) $Tr_{L|K}(x) = \sum\limits_{\sigma} \sigma x$,

(iii) $N_{L|K}(x) = \prod\limits_{\sigma} \sigma x$.

Proof: The characteristic polynomial $f_x(t)$ is a power

$$f_x(t) = p_x(t)^d, \quad d = \big[L : K(x)\big],$$

of the minimal polynomial

$$p_x(t) = t^m + c_1 t^{m-1} + \cdots + c_m, \quad m = \big[K(x) : K\big],$$

of x. In fact, $1, x, \ldots, x^{m-1}$ is a basis of $K(x)|K$, and if $\alpha_1, \ldots, \alpha_d$ is a basis of $L|K(x)$, then

$$\alpha_1, \alpha_1 x, \ldots, \alpha_1 x^{m-1}; \ \ldots ; \ \alpha_d, \alpha_d x, \ldots, \alpha_d x^{m-1}$$

is a basis of $L|K$. The matrix of the linear transformation $T_x : y \mapsto xy$ with respect to this basis has obviously only blocks along the diagonal, each of them equal to

$$\begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & 0 & \cdots & 1 \\ -c_m & -c_{m-1} & -c_{m-2} & \cdots & -c_1 \end{pmatrix}.$$

The corresponding characteristic polynomial is easily checked to be

$$t^m + c_1 t^{m-1} + \cdots + c_m = p_x(t),$$

so that finally $f_x(t) = p_x(t)^d$.

The set $\mathrm{Hom}_K(L, \overline{K})$ of all K-embeddings of L is partitioned by the equivalence relation

$$\sigma \sim \tau \iff \sigma x = \tau x$$

into m equivalence classes of d elements each. If $\sigma_1, \ldots, \sigma_m$ is a system of representatives, then we find

$$p_x(t) = \prod_{i=1}^{m} (t - \sigma_i x),$$

and $f_x(t) = \prod_{i=1}^m (t - \sigma_i x)^d = \prod_{i=1}^m \prod_{\sigma \sim \sigma_i} (t - \sigma x) = \prod_\sigma (t - \sigma x)$. This proves (i), and therefore also (ii) and (iii), after Vietà. □

(2.7) Corollary. *In a tower of finite field extensions $K \subseteq L \subseteq M$, one has*

$$Tr_{L|K} \circ Tr_{M|L} = Tr_{M|K}\,, \quad N_{L|K} \circ N_{M|L} = N_{M|K}\,.$$

Proof: We assume that $M|K$ is separable. The set $\mathrm{Hom}_K(M, \overline{K})$ of K-embeddings of M is partitioned by the relation

$$\sigma \sim \tau \quad \iff \quad \sigma|_L = \tau|_L$$

into $m = [L : K]$ equivalence classes. If $\sigma_1, \ldots, \sigma_m$ is a system of representatives, then $\mathrm{Hom}_K(L, \overline{K}) = \{\sigma_i|_L \mid i = 1, \ldots, m\}$, and we find

$$\begin{aligned} Tr_{M|K}(x) &= \sum_{i=1}^m \sum_{\sigma \sim \sigma_i} \sigma x = \sum_{i=1}^m Tr_{\sigma_i M|\sigma_i L}(\sigma_i x) = \sum_{i=1}^m \sigma_i \, Tr_{M|L}(x) \\ &= Tr_{L|K}\big(Tr_{M|L}(x)\big)\,. \end{aligned}$$

Likewise for the norm.

We will not need the inseparable case for the sequel. However it follows easily from what we have shown above, by passing to the maximal separable subextension $M^s|K$. Indeed, for the inseparable degree $[M : K]_i = [M : M^s]$ one has $[M : K]_i = [M : L]_i \, [L : K]_i$ and

$$Tr_{M|K}(x) = [M : K]_i \, Tr_{M^s|K}(x), \quad N_{M|K}(x) = N_{M^s|K}(x)^{[M:K]_i}$$

(see [143], vol. I, chap. II, § 10). □

The **discriminant** of a basis $\alpha_1, \ldots, \alpha_n$ of a separable extension $L|K$ is defined by

$$d(\alpha_1, \ldots, \alpha_n) = \det((\sigma_i \alpha_j))^2,$$

where σ_i, $i = 1, \ldots, n$, varies over the K-embeddings $L \to \overline{K}$. Because of the relation

$$Tr_{L|K}(\alpha_i \alpha_j) = \sum_k (\sigma_k \alpha_i)(\sigma_k \alpha_j),$$

the matrix $(Tr_{L|K}(\alpha_i \alpha_j))$ is the product of the matrices $(\sigma_k \alpha_i)^t$ and $(\sigma_k \alpha_j)$. Thus one may also write

$$d(\alpha_1, \ldots, \alpha_n) = \det\big(Tr_{L|K}(\alpha_i \alpha_j)\big).$$

In the special case of a basis of type $1, \theta, \ldots, \theta^{n-1}$ one gets

$$d(1, \theta, \ldots, \theta^{n-1}) = \prod_{i<j} (\theta_i - \theta_j)^2,$$

where $\theta_i = \sigma_i \theta$. This is seen by successively multiplying each of the first $(n-1)$ columns in the **Vandermonde matrix**

$$\begin{pmatrix} 1 & \theta_1 & \theta_1^2 & \cdots & \theta_1^{n-1} \\ 1 & \theta_2 & \theta_2^2 & \cdots & \theta_2^{n-1} \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ 1 & \theta_n & \theta_n^2 & \cdots & \theta_n^{n-1} \end{pmatrix}$$

by θ_1 and subtracting it from the following.

(2.8) Proposition. *If $L|K$ is separable and $\alpha_1, \ldots, \alpha_n$ is a basis, then the discriminant*

$$d(\alpha_1, \ldots, \alpha_n) \neq 0,$$

and

$$(x, y) = Tr_{L|K}(xy)$$

is a nondegenerate bilinear form on the K-vector space L.

Proof: We first show that the bilinear form $(x, y) = Tr(xy)$ is nondegenerate. Let θ be a primitive element for $L|K$, i.e., $L = K(\theta)$. Then $1, \theta, \ldots, \theta^{n-1}$ is a basis with respect to which the form (x, y) is given by the matrix $M = (Tr_{L|K}(\theta^{i-1}\theta^{j-1}))_{i,j=1,\ldots,n}$. It is nondegenerate because, for $\theta_i = \sigma_i \theta$, we have

$$\det(M) = d(1, \theta, \ldots, \theta^{n-1}) = \prod_{i<j} (\theta_i - \theta_j)^2 \neq 0.$$

If $\alpha_1, \ldots, \alpha_n$ is an arbitrary basis of $L|K$, then the bilinear form (x, y) with respect to this basis is given by the matrix $M = (Tr_{L|K}(\alpha_i \alpha_j))$. From the above it follows that $d(\alpha_1, \ldots, \alpha_n) = \det(M) \neq 0$. □

After this review from the theory of fields, we return to the integrally closed integral domain A with field of fractions K, and to its integral closure B in the finite separable extension $L|K$. If $x \in B$ is an integral element of L, then all of its conjugates σx are also integral. Taking into account that A is integrally closed, i.e., $A = B \cap K$, (2.6) implies that

$$Tr_{L|K}(x), \quad N_{L|K}(x) \in A.$$

Furthermore, for the group of units of B over A, we obtain the relation

$$x \in B^* \iff N_{L|K}(x) \in A^*.$$

For if $aN_{L|K}(x) = 1$, $a \in A$, then $1 = a\prod_\sigma \sigma x = yx$ for some $y \in B$. The discriminant is often useful because of the following

(2.9) Lemma. *Let $\alpha_1, \dots, \alpha_n$ be a basis of $L|K$ which is contained in B, of discriminant $d = d(\alpha_1, \dots, \alpha_n)$. Then one has*

$$dB \subseteq A\alpha_1 + \cdots + A\alpha_n.$$

Proof: If $\alpha = a_1\alpha_1 + \cdots + a_n\alpha_n \in B$, $a_j \in K$, then the a_j are a solution of the system of linear equations

$$Tr_{L|K}(\alpha_i\alpha) = \sum_j Tr_{L|K}(\alpha_i\alpha_j)a_j,$$

and, as $Tr_{L|K}(\alpha_i\alpha) \in A$, they are given as the quotient of an element of A by the determinant $\det(Tr_{L|K}(\alpha_i\alpha_j)) = d$. Therefore $da_j \in A$, and thus

$$d\alpha \in A\alpha_1 + \cdots + A\alpha_n. \qquad \square$$

A system of elements $\omega_1, \dots, \omega_n \in B$ such that each $b \in B$ can be written uniquely as a linear combination

$$b = a_1\omega_1 + \cdots + a_n\omega_n$$

with coefficients $a_i \in A$, is called an **integral basis** of B over A (or: an A-basis of B). Since such an integral basis is automatically a basis of $L|K$, its length n always equals the degree $[L : K]$ of the field extension. The existence of an integral basis signifies that B is a **free A-module** of rank $n = [L : K]$. In general, such an integral basis does not exist. If, however, A is a principal ideal domain, then one has the following more general

(2.10) Proposition. *If $L|K$ is separable and A is a principal ideal domain, then every finitely generated B-submodule $M \neq 0$ of L is a free A-module of rank $[L : K]$. In particular, B admits an integral basis over A.*

Proof: Let $M \neq 0$ be a finitely generated B-submodule of L and $\alpha_1, \ldots, \alpha_n$ a basis of $L|K$. Multiplying by an element of A, we may arrange for the α_i to lie in B. By (2.9), we then have $dB \subseteq A\alpha_1 + \cdots + A\alpha_n$, in particular, $\operatorname{rank}(B) \leq [L : K]$, and since a system of generators of the A-module B is also a system of generators of the K-module L, we have $\operatorname{rank}(B) = [L : K]$. Let $\mu_1, \ldots, \mu_r \in M$ be a system of generators of the B-module M. There exists an $a \in A$, $a \neq 0$, such that $a\mu_i \in B$, $i = 1, \ldots, r$, so that $aM \subseteq B$. Then

$$adM \subseteq dB \subseteq A\alpha_1 + \cdots + A\alpha_n = M_0 .$$

According to the main theorem on finitely generated modules over principal ideal domains, since M_0 is a free A-module, so is adM, and hence also M. Finally,

$$[L : K] = \operatorname{rank}(B) \leq \operatorname{rank}(M) = \operatorname{rank}(adM) \leq \operatorname{rank}(M_0) = [L : K],$$

hence $\operatorname{rank}(M) = [L : K]$. □

It is in general a difficult problem to produce integral bases. In concrete situations it can also be an important one. This is why the following proposition is interesting. Instead of integral bases of the integral closure B of A in L, we will now simply speak of integral bases of the extension $L|K$.

(2.11) Proposition. *Let $L|K$ and $L'|K$ be two Galois extensions of degree n, resp. n', such that $L \cap L' = K$. Let $\omega_1, \ldots, \omega_n$, resp. $\omega'_1, \ldots, \omega'_{n'}$, be an integral basis of $L\,|\,K$, resp. $L'|K$, with discriminant d, resp. d'. Suppose that d and d' are relatively prime in the sense that $xd + x'd' = 1$, for suitable $x, x' \in A$. Then $\omega_i\,\omega'_j$ is an integral basis of LL', of discriminant $d^{n'}d'^{n}$.*

Proof: As $L \cap L' = K$, we have $[LL' : K] = nn'$, so the nn' products $\omega_i\omega'_j$ do form a basis of $LL'|K$. Now let α be an integral element of LL', and write

$$\alpha = \sum_{i,j} a_{ij}\,\omega_i\,\omega'_j , \quad a_{ij} \in K .$$

We have to show that $a_{ij} \in A$. Put $\beta_j = \sum_i a_{ij}\,\omega_i$. Let $G(LL'|L') = \{\sigma_1, \ldots, \sigma_n\}$ and $G(LL'|L) = \{\sigma'_1, \ldots, \sigma'_{n'}\}$. Thus

$$G(LL'|K) = \left\{\sigma_k\sigma'_\ell \;\middle|\; k = 1, \ldots, n,\ \ell = 1, \ldots, n'\right\} .$$

Putting

$$T = (\sigma'_\ell\omega'_j), \quad a = (\sigma'_1\alpha, \ldots, \sigma'_{n'}\alpha)^t , \quad b = (\beta_1, \ldots, \beta_{n'})^t ,$$

one finds $\det(T)^2 = d'$ and

$$a = T\,b\,.$$

Write T^* for the adjoint matrix of T. Then row-column expansion (2.3) gives

$$\det(T)\,b = T^*a\,.$$

Since T^* and a have integral entries in LL', the multiple $d'b$ has integral entries in L, namely $d'\beta_j = \sum_i d'a_{ij}\,\omega_i$. Thus $d'a_{ij} \in A$. Swapping the rôles of (ω_i) and (ω'_j), one checks in the same manner that $da_{ij} \in A$, so that

$$a_{ij} = xda_{ij} + x'd'a_{ij} \in A\,.$$

Therefore $\omega_i\,\omega'_j$ is indeed an integral basis of $LL'|K$. We compute the discriminant Δ of this integral basis. Since $G(LL'|K) = \{\sigma_k\sigma'_\ell \mid k = 1,\dots,n,\ \ell = 1,\dots,n'\}$, it is the square of the determinant of the $(nn' \times nn')$-matrix

$$M = (\sigma_k\sigma'_\ell\,\omega_i\omega'_j) = (\sigma_k\omega_i\,\sigma'_\ell\omega'_j)\,.$$

This matrix is itself an $(n'\times n')$-matrix with entries $(n\times n)$-matrices of which the (ℓ, j)-entry is the matrix $Q\sigma'_\ell\omega'_j$ where $Q = (\sigma_k\omega_i)$. In other words,

$$M = \begin{pmatrix} Q & & 0 \\ & \ddots & \\ 0 & & Q \end{pmatrix} \begin{pmatrix} E\sigma'_1\omega'_1 & \cdots & E\sigma'_{n'}\omega'_1 \\ \vdots & & \vdots \\ E\sigma'_1\omega'_{n'} & \cdots & E\sigma'_{n'}\omega'_{n'} \end{pmatrix}.$$

Here E denotes the $(n \times n)$-unit matrix. By changing indices the second matrix may be transformed to look like the first one. This yields

$$\Delta = \det(M)^2 = \det(Q)^{2n'}\det\big((\sigma'_\ell\omega'_j)\big)^{2n} = d^{n'}d'^{n}\,. \qquad \square$$

Remark: It follows from the proof that the proposition is valid for arbitrary separable extensions (not necessarily Galois), if one assumes instead of $L \cap L' = K$ that L and L' are linearly disjoint.

The chief application of our considerations on integrality will concern the integral closure $\mathcal{O}_K \subseteq K$ of $\mathbb{Z} \subseteq \mathbb{Q}$ in an algebraic number field K. By proposition (2.10), every finitely generated $\mathcal{O}_K$-submodule $\mathfrak{a}$ of K admits a $\mathbb{Z}$-basis $\alpha_1,\dots,\alpha_n$,

$$\mathfrak{a} = \mathbb{Z}\alpha_1 + \cdots + \mathbb{Z}\alpha_n\,.$$

The discriminant

$$d(\alpha_1,\dots,\alpha_n) = \det\big((\sigma_i\alpha_j)\big)^2$$

is independent of the choice of a $\mathbb{Z}$-basis; if $\alpha'_1, \ldots, \alpha'_n$ is another basis, then the base change matrix $T = (a_{ij})$, $\alpha'_i = \sum_j a_{ij}\alpha_j$, as well as its inverse, has integral entries. It therefore has determinant ± 1, so that indeed

$$d(\alpha'_1, \ldots, \alpha'_n) = \det(T)^2\, d(\alpha_1, \ldots, \alpha_n) = d(\alpha_1, \ldots, \alpha_n).$$

We may therefore write

$$d(\mathfrak{a}) = d(\alpha_1, \ldots, \alpha_n).$$

In the special case of an integral basis $\omega_1, \ldots, \omega_n$ of $\mathcal{O}_K$ we obtain the **discriminant of the algebraic number field** K,

$$d_K = d(\mathcal{O}_K) = d(\omega_1, \ldots, \omega_n).$$

In general, one has the

(2.12) Proposition. *If $\mathfrak{a} \subseteq \mathfrak{a}'$ are two nonzero finitely generated $\mathcal{O}_K$-submodules of K, then the index $(\mathfrak{a}' : \mathfrak{a})$ is finite and satisfies*

$$d(\mathfrak{a}) = (\mathfrak{a}' : \mathfrak{a})^2\, d(\mathfrak{a}').$$

All we have to show is that the index $(\mathfrak{a}' : \mathfrak{a})$ equals the absolute value of the determinant of the base change matrix passing from a $\mathbb{Z}$-basis of $\mathfrak{a}$ to a $\mathbb{Z}$-basis of $\mathfrak{a}'$. This proof is part of the well-known theory of finitely generated $\mathbb{Z}$-modules.

Exercise 1. Is $\frac{3+2\sqrt{6}}{1-\sqrt{6}}$ an algebraic integer?

Exercise 2. Show that, if the integral domain A is integrally closed, then so is the polynomial ring $A[t]$.

Exercise 3. In the polynomial ring $A = \mathbb{Q}[X,Y]$, consider the principal ideal $\mathfrak{p} = (X^2 - Y^3)$. Show that $\mathfrak{p}$ is a prime ideal, but $A/\mathfrak{p}$ is not integrally closed.

Exercise 4. Let D be a squarefree rational integer $\neq 0, 1$ and d the discriminant of the quadratic number field $K = \mathbb{Q}(\sqrt{D})$. Show that

$$d = D, \qquad \text{if } D \equiv 1 \bmod 4,$$

$$d = 4D, \qquad \text{if } D \equiv 2 \text{ or } 3 \bmod 4,$$

and that an integral basis of K is given by $\{1, \sqrt{D}\}$ in the second case, by $\{1, \frac{1}{2}(1+\sqrt{D})\}$ in the first case, and by $\{1, \frac{1}{2}(d+\sqrt{d})\}$ in both cases.

Exercise 5. Show that $\{1, \sqrt[3]{2}, \sqrt[3]{2^2}\}$ is an integral basis of $\mathbb{Q}(\sqrt[3]{2})$.

Exercise 6. Show that $\{1, \theta, \frac{1}{2}(\theta+\theta^2)\}$ is an integral basis of $\mathbb{Q}(\theta)$, $\theta^3 - \theta - 4 = 0$.

Exercise 7. The discriminant d_K of an algebraic number field K is always $\equiv 0 \bmod 4$ or $\equiv 1 \bmod 4$ (*Stickelberger's discriminant relation*).

Hint: The determinant $\det(\sigma_i\omega_j)$ of an integral basis ω_j is a sum of terms, each prefixed by a positive or a negative sign. Writing P, resp. N, for the sum of the positive, resp. negative terms, one finds $d_K = (P-N)^2 = (P+N)^2 - 4PN$.

§ 3. Ideals

Being a generalization of the ring $\mathbb{Z} \subseteq \mathbb{Q}$, the ring $\mathcal{O}_K$ of integers of an algebraic number field K is at the center of all our considerations. As in $\mathbb{Z}$, every non-unit $\alpha \neq 0$ can be factored in $\mathcal{O}_K$ into a product of irreducible elements. For if α is not itself irreducible, then it can be written as a product of two non-units $\alpha = \beta\gamma$. Then by § 2, one has

$$1 < \left|N_{K|\mathbb{Q}}(\beta)\right| < \left|N_{K|\mathbb{Q}}(\alpha)\right|, \quad 1 < \left|N_{K|\mathbb{Q}}(\gamma)\right| < \left|N_{K|\mathbb{Q}}(\alpha)\right|,$$

and the prime decomposition of α follows by induction from those of β and γ. However, contrary to what happens in the rings $\mathbb{Z}$ and $\mathbb{Z}[i]$, the uniqueness of prime factorization does not hold in general.

Example: The ring of integers of the field $K = \mathbb{Q}(\sqrt{-5})$ is given by § 2, exercise 4, as $\mathcal{O}_K = \mathbb{Z} + \mathbb{Z}\sqrt{-5}$. In this ring, the rational integer 21 can be decomposed in two ways,

$$21 = 3 \cdot 7 = (1 + 2\sqrt{-5}) \cdot (1 - 2\sqrt{-5}).$$

All factors occurring here are irreducible in $\mathcal{O}_K$. For if one had, for instance, $3 = \alpha\beta$, with α, β non-units, then $9 = N_{K|\mathbb{Q}}(\alpha)\, N_{K|\mathbb{Q}}(\beta)$ would imply $N_{K|\mathbb{Q}}(\alpha) = \pm 3$. But the equation

$$N_{K|\mathbb{Q}}(x + y\sqrt{-5}) = x^2 + 5y^2 = \pm 3$$

has no solutions in $\mathbb{Z}$. In the same way it is seen that 7, $1 + 2\sqrt{-5}$, and $1 - 2\sqrt{-5}$ are irreducible. As the fractions

$$\frac{1 \pm 2\sqrt{-5}}{3}, \quad \frac{1 \pm 2\sqrt{-5}}{7}$$

do not belong to $\mathcal{O}_K$, the numbers 3 and 7 are not associated to $1 + 2\sqrt{-5}$ or $1 - 2\sqrt{-5}$. The two prime factorizations of 21 are therefore essentially different.

Realizing the failure of unique factorization in general has led to one of the grand events in the history of number theory, the discovery of ideal theory by *Eduard Kummer*. Inspired by the discovery of complex numbers, Kummer's idea was that the integers of K would have to admit an embedding into a bigger domain of "ideal numbers" where **unique** factorization into "ideal prime numbers" would hold. For instance, in the example of

$$21 = 3 \cdot 7 = (1 + 2\sqrt{-5})(1 - 2\sqrt{-5}),$$

the factors on the right would be composed of ideal prime numbers $\mathfrak{p}_1, \mathfrak{p}_2, \mathfrak{p}_3, \mathfrak{p}_4$, subject to the rules

$$3 = \mathfrak{p}_1\mathfrak{p}_2, \quad 7 = \mathfrak{p}_3\mathfrak{p}_4, \quad 1 + 2\sqrt{-5} = \mathfrak{p}_1\mathfrak{p}_3, \quad 1 - 2\sqrt{-5} = \mathfrak{p}_2\mathfrak{p}_4.$$

This would resolve the above non-uniqueness into the wonderfully unique factorization

$$21 = (\mathfrak{p}_1\mathfrak{p}_2)(\mathfrak{p}_3\mathfrak{p}_4) = (\mathfrak{p}_1\mathfrak{p}_3)(\mathfrak{p}_2\mathfrak{p}_4).$$

Kummer's concept of "ideal numbers" was later replaced by that of **ideals** of the ring $\mathcal{O}_K$. The reason for this is easily seen: whatever an ideal number $\mathfrak{a}$ should be defined to be, it ought to be linked to certain numbers $a \in \mathcal{O}_K$ by a divisibility relation $\mathfrak{a} \,|\, a$ satisfying the following rules, for $a, b, \lambda \in \mathcal{O}_K$,

$$\mathfrak{a} \,|\, a \ \text{ and } \ \mathfrak{a} \,|\, b \ \Rightarrow\ \mathfrak{a} \,|\, a \pm b; \qquad \mathfrak{a} \,|\, a \ \Rightarrow\ \mathfrak{a} \,|\, \lambda a.$$

And an ideal number $\mathfrak{a}$ should be determined by the totality of its divisors in $\mathcal{O}_K$

$$\underline{\mathfrak{a}} = \big\{ a \in \mathcal{O}_K \;\big|\; \mathfrak{a} \,|\, a \big\}.$$

But in view of the rules for divisibility, this set is an ideal of $\mathcal{O}_K$.

This is the reason why *Richard Dedekind* re-introduced Kummer's "ideal numbers" as being the ideals of $\mathcal{O}_K$. Once this is done, the divisibility relation $\mathfrak{a} \,|\, a$ can simply be defined by the inclusion $a \in \mathfrak{a}$, and more generally the divisibility relation $\mathfrak{a} \,|\, \mathfrak{b}$ between two ideals by $\mathfrak{b} \subseteq \mathfrak{a}$. In what follows, we will study this notion of divisibility more closely. The basic theorem here is the following.

(3.1) Theorem. *The ring $\mathcal{O}_K$ is noetherian, integrally closed, and every prime ideal $\mathfrak{p} \neq 0$ is a maximal ideal.*

Proof: $\mathcal{O}_K$ is noetherian because every ideal $\mathfrak{a}$ is a finitely generated $\mathbb{Z}$-module by (2.10), and therefore *a fortiori* a finitely generated $\mathcal{O}_K$-module. By §2, $\mathcal{O}_K$ is also integrally closed, being the integral closure of $\mathbb{Z}$ in K. It thus remains to show that each prime ideal $\mathfrak{p} \neq 0$ is maximal. Now, $\mathfrak{p} \cap \mathbb{Z}$ is a nonzero prime ideal (p) in $\mathbb{Z}$: the primality is clear, and if $y \in \mathfrak{p}$, $y \neq 0$, and

$$y^n + a_1 y^{n-1} + \cdots + a_n = 0$$

is an equation for y with $a_i \in \mathbb{Z}$, $a_n \neq 0$, then $a_n \in \mathfrak{p} \cap \mathbb{Z}$. The integral domain $\overline{\mathcal{O}} = \mathcal{O}_K/\mathfrak{p}$ arises from $\kappa = \mathbb{Z}/p\mathbb{Z}$ by adjoining algebraic elements and is therefore again a field (recall the fact that $\kappa[\alpha] = \kappa(\alpha)$, if α is algebraic). Therefore $\mathfrak{p}$ is a maximal ideal. $\square$

The three properties of the ring $\mathcal{O}_K$ which we have just proven lay the foundation of the whole theory of divisibility of its ideals. This theory was developed by Dedekind, which suggested the following

(3.2) Definition. *A noetherian, integrally closed integral domain in which every nonzero prime ideal is maximal is called a* **Dedekind domain**.

Just as the rings of the form $\mathcal{O}_K$ may be viewed as generalizations of the ring $\mathbb{Z}$, the Dedekind domains may be viewed as generalized principal ideal domains. Indeed, if A is a principal ideal domain with field of fractions K, and $L|K$ is a finite field extension, then the integral closure B of A in L is, in general, not a principal ideal domain, but always a Dedekind domain, as we shall show further on.

Instead of the ring $\mathcal{O}_K$ we will now consider an arbitrary Dedekind domain $\mathcal{O}$, and we denote by K the field of fractions of $\mathcal{O}$. Given two ideals $\mathfrak{a}$ and $\mathfrak{b}$ of $\mathcal{O}$ (or more generally of an arbitrary ring), the divisibility relation $\mathfrak{a}|\mathfrak{b}$ is defined by $\mathfrak{b} \subseteq \mathfrak{a}$, and the sum of the ideals by

$$\mathfrak{a}+\mathfrak{b} = \{a+b \mid a \in \mathfrak{a},\ b \in \mathfrak{b}\}\,.$$

This is the smallest ideal containing $\mathfrak{a}$ as well as $\mathfrak{b}$, in other words, it is the greatest common divisor $\gcd(\mathfrak{a},\mathfrak{b})$ of $\mathfrak{a}$ and $\mathfrak{b}$. By the same token the intersection $\mathfrak{a}\cap\mathfrak{b}$ is the lcm (least common multiple) of $\mathfrak{a}$ and $\mathfrak{b}$. We define the **product** of $\mathfrak{a}$ and $\mathfrak{b}$ by

$$\mathfrak{a}\mathfrak{b} = \Big\{ \sum_i a_i b_i \;\Big|\; a_i \in \mathfrak{a},\ b_i \in \mathfrak{b}\Big\}\,.$$

With respect to this multiplication the ideals of $\mathcal{O}$ will grant us what the elements alone may refuse to provide: the **unique prime factorization.**

(3.3) Theorem. *Every ideal $\mathfrak{a}$ of $\mathcal{O}$ different from* (0) *and* (1) *admits a factorization*

$$\mathfrak{a} = \mathfrak{p}_1\cdots\mathfrak{p}_r$$

into nonzero prime ideals $\mathfrak{p}_i$ of $\mathcal{O}$ which is unique up to the order of the factors.

This theorem is of course perfectly in line with the invention of "ideal numbers". Still, the fact that it holds is remarkable because its proof is far from straightforward, and unveils a deeper principle governing the arithmetic in $\mathcal{O}$. We prepare the proof proper by two lemmas.

(3.4) Lemma. *For every ideal* $\mathfrak{a} \neq 0$ *of* $\mathcal{O}$ *there exist nonzero prime ideals* $\mathfrak{p}_1, \mathfrak{p}_2, \ldots, \mathfrak{p}_r$ *such that*

$$\mathfrak{a} \supseteq \mathfrak{p}_1\mathfrak{p}_2 \cdots \mathfrak{p}_r .$$

Proof: Suppose the set $\mathfrak{M}$ of those ideals which do not fulfill this condition is nonempty. As $\mathcal{O}$ is noetherian, every ascending chain of ideals becomes stationary. Therefore $\mathfrak{M}$ is inductively ordered with respect to inclusion and thus admits a maximal element $\mathfrak{a}$. This cannot be a prime ideal, so there exist elements $b_1, b_2 \in \mathcal{O}$ such that $b_1b_2 \in \mathfrak{a}$, but $b_1, b_2 \notin \mathfrak{a}$. Put $\mathfrak{a}_1 = (b_1) + \mathfrak{a}$, $\mathfrak{a}_2 = (b_2) + \mathfrak{a}$. Then $\mathfrak{a} \subsetneqq \mathfrak{a}_1$, $\mathfrak{a} \subsetneqq \mathfrak{a}_2$ and $\mathfrak{a}_1\mathfrak{a}_2 \subseteq \mathfrak{a}$. By the maximality of $\mathfrak{a}$, both $\mathfrak{a}_1$ and $\mathfrak{a}_2$ contain a product of prime ideals, and the product of these products is contained in $\mathfrak{a}$, a contradiction. □

(3.5) Lemma. *Let* $\mathfrak{p}$ *be a prime ideal of* $\mathcal{O}$ *and define*

$$\mathfrak{p}^{-1} = \{x \in K \mid x\mathfrak{p} \subseteq \mathcal{O}\} .$$

Then one has $\mathfrak{a}\mathfrak{p}^{-1} := \{\sum_i a_i x_i \mid a_i \in \mathfrak{a},\ x_i \in \mathfrak{p}^{-1}\} \neq \mathfrak{a}$, *for every ideal* $\mathfrak{a} \neq 0$.

Proof: Let $a \in \mathfrak{p}$, $a \neq 0$, and $\mathfrak{p}_1\mathfrak{p}_2 \cdots \mathfrak{p}_r \subseteq (a) \subseteq \mathfrak{p}$, with r as small as possible. Then one of the $\mathfrak{p}_i$, say $\mathfrak{p}_1$, is contained in $\mathfrak{p}$, and so $\mathfrak{p}_1 = \mathfrak{p}$ because $\mathfrak{p}_1$ is a maximal ideal. (Indeed, if none of the $\mathfrak{p}_i$ were contained in $\mathfrak{p}$, then for every i there would exist $a_i \in \mathfrak{p}_i \smallsetminus \mathfrak{p}$ such that $a_1 \cdots a_r \in \mathfrak{p}$. But $\mathfrak{p}$ is prime.) Since $\mathfrak{p}_2 \cdots \mathfrak{p}_r \not\subseteq (a)$, there exists $b \in \mathfrak{p}_2 \cdots \mathfrak{p}_r$ such that $b \notin a\mathcal{O}$, i.e., $a^{-1}b \notin \mathcal{O}$. On the other hand we have $b\mathfrak{p} \subseteq (a)$, i.e., $a^{-1}b\mathfrak{p} \subseteq \mathcal{O}$, and thus $a^{-1}b \in \mathfrak{p}^{-1}$. It follows that $\mathfrak{p}^{-1} \neq \mathcal{O}$.

Now let $\mathfrak{a} \neq 0$ be an ideal of $\mathcal{O}$ and $\alpha_1, \ldots, \alpha_n$ a system of generators. Let us assume that $\mathfrak{a}\mathfrak{p}^{-1} = \mathfrak{a}$. Then for every $x \in \mathfrak{p}^{-1}$,

$$x\alpha_i = \sum_j a_{ij}\,\alpha_j , \quad a_{ij} \in \mathcal{O} .$$

Writing A for the matrix $(x\delta_{ij} - a_{ij})$ we obtain $A(\alpha_1, \ldots, \alpha_n)^t = 0$. By (2.3), the determinant $d = \det(A)$ satisfies $d\alpha_1 = \cdots = d\alpha_n = 0$ and thus $d = 0$. It follows that x is integral over $\mathcal{O}$, being a zero of the monic polynomial $f(X) = \det(X\delta_{ij} - a_{ij}) \in \mathcal{O}[X]$. Therefore $x \in \mathcal{O}$. This means that $\mathfrak{p}^{-1} = \mathcal{O}$, a contradiction. □

Proof of (3.3): I. Existence of the prime ideal factorization. Let $\mathfrak{M}$ be the set of all ideals different from (0) and (1) which do not admit a prime ideal decomposition. If $\mathfrak{M}$ is nonempty, then we argue as for (3.4) that there exists a maximal element $\mathfrak{a}$ in $\mathfrak{M}$. It is contained in a maximal ideal $\mathfrak{p}$, and the inclusion $\mathcal{O} \subseteq \mathfrak{p}^{-1}$ gives us

$$\mathfrak{a} \subseteq \mathfrak{a}\mathfrak{p}^{-1} \subseteq \mathfrak{p}\mathfrak{p}^{-1} \subseteq \mathcal{O}.$$

By (3.5), one has $\mathfrak{a} \subsetneqq \mathfrak{a}\mathfrak{p}^{-1}$ and $\mathfrak{p} \subsetneqq \mathfrak{p}\mathfrak{p}^{-1} \subseteq \mathcal{O}$. Since $\mathfrak{p}$ is a maximal ideal, it follows that $\mathfrak{p}\mathfrak{p}^{-1} = \mathcal{O}$. In view of the maximality of $\mathfrak{a}$ in $\mathfrak{M}$ and since $\mathfrak{a} \neq \mathfrak{p}$, i.e., $\mathfrak{a}\mathfrak{p}^{-1} \neq \mathcal{O}$, the ideal $\mathfrak{a}\mathfrak{p}^{-1}$ admits a prime ideal decomposition $\mathfrak{a}\mathfrak{p}^{-1} = \mathfrak{p}_1 \cdots \mathfrak{p}_r$, and so does $\mathfrak{a} = \mathfrak{a}\mathfrak{p}^{-1}\mathfrak{p} = \mathfrak{p}_1 \cdots \mathfrak{p}_r\mathfrak{p}$, a contradiction.

II. Uniqueness of the prime ideal factorization. For a prime ideal $\mathfrak{p}$ one has: $\mathfrak{a}\mathfrak{b} \subseteq \mathfrak{p} \Rightarrow \mathfrak{a} \subseteq \mathfrak{p}$ or $\mathfrak{b} \subseteq \mathfrak{p}$, i.e., $\mathfrak{p} \mid \mathfrak{a}\mathfrak{b} \Rightarrow \mathfrak{p} \mid \mathfrak{a}$ or $\mathfrak{p} \mid \mathfrak{b}$. Let

$$\mathfrak{a} = \mathfrak{p}_1\mathfrak{p}_2 \cdots \mathfrak{p}_r = \mathfrak{q}_1\mathfrak{q}_2 \cdots \mathfrak{q}_s$$

be two prime ideal factorizations of $\mathfrak{a}$. Then $\mathfrak{p}_1$ divides a factor $\mathfrak{q}_i$, say $\mathfrak{q}_1$, and being maximal equals $\mathfrak{q}_1$. We multiply by $\mathfrak{p}_1^{-1}$ and obtain, in view of $\mathfrak{p}_1 \neq \mathfrak{p}_1\mathfrak{p}_1^{-1} = \mathcal{O}$, that

$$\mathfrak{p}_2 \cdots \mathfrak{p}_r = \mathfrak{q}_2 \cdots \mathfrak{q}_s .$$

Continuing like this we see that $r = s$ and, possibly after renumbering, $\mathfrak{p}_i = \mathfrak{q}_i$, for all $i = 1, \ldots, r$. □

Grouping together the occurrences of the same prime ideals in the prime ideal factorization of an ideal $\mathfrak{a} \neq 0$ of $\mathcal{O}$, gives a product representation

$$\mathfrak{a} = \mathfrak{p}_1^{\nu_1} \cdots \mathfrak{p}_r^{\nu_r}, \quad \nu_i > 0 .$$

In the sequel such an identity will be automatically understood to signify that the $\mathfrak{p}_i$ are pairwise distinct. If in particular $\mathfrak{a}$ is a principal ideal (a), then – following the tradition which tends to attribute to the ideals the rôle of "ideal numbers" – we will write with a slight abuse of notation

$$a = \mathfrak{p}_1^{\nu_1} \cdots \mathfrak{p}_r^{\nu_r} .$$

Similarly, the notation $\mathfrak{a} \mid a$ is often used instead of $\mathfrak{a} \mid (a)$ and $(\mathfrak{a}, \mathfrak{b}) = 1$ is written for two relatively prime ideals, instead of the correct formula $(\mathfrak{a}, \mathfrak{b}) = \mathfrak{a} + \mathfrak{b} = \mathcal{O}$. For a product $\mathfrak{a} = \mathfrak{a}_1 \cdots \mathfrak{a}_n$ of relatively prime ideals $\mathfrak{a}_1, \ldots, \mathfrak{a}_n$, one has an analogue of the well-known "Chinese Remainder Theorem" from elementary number theory. We may formulate this result for an arbitrary ring taking into account that

$$\mathfrak{a} = \bigcap_{i=1}^{n} \mathfrak{a}_i .$$

Indeed, since $\mathfrak{a}_i \mid \mathfrak{a}$, $i = 1, \ldots, n$, we find on the one hand that $\mathfrak{a} \subseteq \bigcap_{i=1}^{n} \mathfrak{a}_i$, and for $a \in \bigcap_i \mathfrak{a}_i$ we find that $\mathfrak{a}_i \mid a$, and therefore, the factors being relatively prime, we get $\mathfrak{a} = \mathfrak{a}_1 \cdots \mathfrak{a}_n \mid a$, i.e., $a \in \mathfrak{a}$.

(3.6) Chinese Remainder Theorem. *Let $\mathfrak{a}_1, \dots, \mathfrak{a}_n$ be ideals in a ring $\mathcal{O}$ such that $\mathfrak{a}_i + \mathfrak{a}_j = \mathcal{O}$ for $i \neq j$. Then, if $\mathfrak{a} = \bigcap_{i=1}^n \mathfrak{a}_i$, one has*

$$\mathcal{O}/\mathfrak{a} \cong \bigoplus_{i=1}^n \mathcal{O}/\mathfrak{a}_i .$$

Proof: The canonical homomorphism

$$\mathcal{O} \longrightarrow \bigoplus_{i=1}^n \mathcal{O}/\mathfrak{a}_i , \quad a \longmapsto \bigoplus_{i=1}^n a \bmod \mathfrak{a}_i ,$$

has kernel $\mathfrak{a} = \bigcap_i \mathfrak{a}_i$. It therefore suffices to show that it is surjective. For this, let $x_i \bmod \mathfrak{a}_i \in \mathcal{O}/\mathfrak{a}_i$, $i = 1, \dots, n$, be given. If $n = 2$, we may write $1 = a_1 + a_2$, $a_i \in \mathfrak{a}_i$, and putting $x = x_2a_1 + x_1a_2$ we get $x \equiv x_i \bmod \mathfrak{a}_i$, $i = 1, 2$.

If $n > 2$, we may find as before an element $y_1 \in \mathcal{O}$ such that

$$y_1 \equiv 1 \bmod \mathfrak{a}_1 , \quad y_1 \equiv 0 \bmod \bigcap_{i=2}^n \mathfrak{a}_i ,$$

and, by the same token, elements $y_2, \dots, y_n$ such that

$$y_i \equiv 1 \bmod \mathfrak{a}_i , \quad y_i \equiv 0 \bmod \mathfrak{a}_j \quad \text{for } i \neq j .$$

Putting $x = x_1y_1 + \cdots + x_ny_n$ we find $x \equiv x_i \bmod \mathfrak{a}_i$, $i = 1, \dots, n$. This proves the surjectivity. □

Now let $\mathcal{O}$ be again a Dedekind domain. Just as for nonzero numbers, we may obtain **inverses** for the nonzero ideals of $\mathcal{O}$ by introducing the notion of fractional ideal in the field of fractions K.

(3.7) Definition. *A* **fractional ideal** *of K is a finitely generated $\mathcal{O}$-submodule $\mathfrak{a} \neq 0$ of K.*

For instance, an element $a \in K^*$ defines the fractional "principal ideal" $(a) = a\mathcal{O}$. Obviously, since $\mathcal{O}$ is noetherian, an $\mathcal{O}$-submodule $\mathfrak{a} \neq 0$ of K is a fractional ideal if and only if there exists $c \in \mathcal{O}$, $c \neq 0$, such that $c\mathfrak{a} \subseteq \mathcal{O}$ is an ideal of the ring $\mathcal{O}$. Fractional ideals are multiplied in the same way as ideals in $\mathcal{O}$. For distinction the latter may henceforth be called **integral ideals** of K.

(3.8) Proposition. *The fractional ideals form an abelian group, the* **ideal group** *J_K of K. The identity element is $(1) = \mathcal{O}$, and the inverse of $\mathfrak{a}$ is*

$$\mathfrak{a}^{-1} = \{x \in K \mid x\mathfrak{a} \subseteq \mathcal{O}\} .$$

Proof: One obviously has associativity, commutativity and $\mathfrak{a}(1) = \mathfrak{a}$. For a prime ideal $\mathfrak{p}$, (3.5) says that $\mathfrak{p} \subsetneqq \mathfrak{p}\mathfrak{p}^{-1}$ and therefore $\mathfrak{p}\mathfrak{p}^{-1} = \mathcal{O}$ because $\mathfrak{p}$ is maximal. Consequently, if $\mathfrak{a} = \mathfrak{p}_1 \cdots \mathfrak{p}_r$ is an integral ideal, then $\mathfrak{b} = \mathfrak{p}_1^{-1} \cdots \mathfrak{p}_r^{-1}$ is an inverse. $\mathfrak{b}\mathfrak{a} = \mathcal{O}$ implies that $\mathfrak{b} \subseteq \mathfrak{a}^{-1}$. Conversely, if $x\mathfrak{a} \subseteq \mathcal{O}$, then $x\mathfrak{a}\mathfrak{b} \subseteq \mathfrak{b}$, so $x \in \mathfrak{b}$ because $\mathfrak{a}\mathfrak{b} = \mathcal{O}$. Thus we have $\mathfrak{b} = \mathfrak{a}^{-1}$. Finally, if $\mathfrak{a}$ is an arbitrary fractional ideal and $c \in \mathcal{O}$, $c \neq 0$, is such that $c\mathfrak{a} \subseteq \mathcal{O}$, then $(c\mathfrak{a})^{-1} = c^{-1}\mathfrak{a}^{-1}$ is the inverse of $c\mathfrak{a}$, so $\mathfrak{a}\mathfrak{a}^{-1} = \mathcal{O}$. □

(3.9) Corollary. *Every fractional ideal $\mathfrak{a}$ admits a unique representation as a product*

$$\mathfrak{a} = \prod_{\mathfrak{p}} \mathfrak{p}^{\nu_{\mathfrak{p}}}$$

with $\nu_{\mathfrak{p}} \in \mathbb{Z}$ and $\nu_{\mathfrak{p}} = 0$ for almost all $\mathfrak{p}$. In other words, J_K is the free abelian group on the set of nonzero prime ideals $\mathfrak{p}$ of $\mathcal{O}$.

Proof: Every fractional ideal $\mathfrak{a}$ is a quotient $\mathfrak{a} = \mathfrak{b}/\mathfrak{c}$ of two integral ideals $\mathfrak{b}$ and $\mathfrak{c}$, which by (3.3) have a prime decomposition. Therefore $\mathfrak{a}$ has a prime decomposition of the type stated in the corollary. By (3.3), it is unique if $\mathfrak{a}$ is integral, and therefore clearly also in general. □

The fractional principal ideals $(a) = a\mathcal{O}$, $a \in K^*$, form a subgroup of the group of ideals J_K, which will be denoted P_K. The quotient group

$$Cl_K = J_K/P_K$$

is called the **ideal class group**, or **class group** for short, of K. Along with the group of units $\mathcal{O}^*$ of $\mathcal{O}$, it fits into the exact sequence

$$1 \longrightarrow \mathcal{O}^* \longrightarrow K^* \longrightarrow J_K \longrightarrow Cl_K \longrightarrow 1,$$

where the arrow in the middle is given by $a \mapsto (a)$. So the class group Cl_K measures the expansion that takes place when we pass from numbers to ideals, whereas the unit group $\mathcal{O}^*$ measures the contraction in the same process. This immediately raises the problem of understanding these groups $\mathcal{O}^*$ and Cl_K more thoroughly. For general Dedekind domains they may turn out to be completely arbitrary groups. For the ring $\mathcal{O}_K$ of integers in a number field K, however, one obtains important finiteness theorems, which are fundamental for the further development of number theory. But these results cannot be had for nothing. They will be obtained by viewing the numbers geometrically as lattice points in space. For this we will now prepare the necessary concepts, which all come from linear algebra.

Exercise 1. Decompose $33+11\sqrt{-7}$ into irreducible integral elements of $\mathbb{Q}(\sqrt{-7})$.

Exercise 2. Show that

$$54 = 2\cdot 3^3 = \frac{13+\sqrt{-47}}{2}\cdot\frac{13-\sqrt{-47}}{2}$$

are two essentially different decompositions into irreducible integral elements of $\mathbb{Q}(\sqrt{-47})$.

Exercise 3. Let d be squarefree and p a prime number not dividing $2d$. Let $\mathcal{O}$ be the ring of integers of $\mathbb{Q}(\sqrt{d})$. Show that $(p) = p\,\mathcal{O}$ is a prime ideal of $\mathcal{O}$ if and only if the congruence $x^2 \equiv d \bmod p$ has no solution.

Exercise 4. A Dedekind domain with a finite number of prime ideals is a principal ideal domain.

Hint: If $\mathfrak{a} = \mathfrak{p}_1^{\nu_1}\cdots\mathfrak{p}_r^{\nu_r} \neq 0$ is an ideal, then choose elements $\pi_i \in \mathfrak{p}_i \smallsetminus \mathfrak{p}_i^2$ and apply the Chinese remainder theorem for the cosets $\pi_i^{\nu_i} \bmod \mathfrak{p}_i^{\nu_i+1}$.

Exercise 5. The quotient ring $\mathcal{O}/\mathfrak{a}$ of a Dedekind domain by an ideal $\mathfrak{a} \neq 0$ is a principal ideal domain.

Hint: For $\mathfrak{a} = \mathfrak{p}^n$ the only proper ideals of $\mathcal{O}/\mathfrak{a}$ are given by $\mathfrak{p}/\mathfrak{p}^n, \dots, \mathfrak{p}^{n-1}/\mathfrak{p}^n$. Choose $\pi \in \mathfrak{p} \smallsetminus \mathfrak{p}^2$ and show that $\mathfrak{p}^\nu = \mathcal{O}\pi^\nu + \mathfrak{p}^n$.

Exercise 6. Every ideal of a Dedekind domain can be generated by two elements.

Hint: Use exercise 5.

Exercise 7. In a noetherian ring R in which every prime ideal is maximal, each descending chain of ideals $\mathfrak{a}_1 \supseteq \mathfrak{a}_2 \supseteq \cdots$ becomes stationary.

Hint: Show as in (3.4) that (0) is a product $\mathfrak{p}_1\cdots\mathfrak{p}_r$ of prime ideals and that the chain $R \supseteq \mathfrak{p}_1 \supseteq \mathfrak{p}_1\mathfrak{p}_2 \supseteq \cdots \supseteq \mathfrak{p}_1\cdots\mathfrak{p}_r = (0)$ can be refined into a composition series.

Exercise 8. Let $\mathfrak{m}$ be a nonzero integral ideal of the Dedekind domain $\mathcal{O}$. Show that in every ideal class of Cl_K, there exists an integral ideal prime to $\mathfrak{m}$.

Exercise 9. Let $\mathcal{O}$ be an integral domain in which all nonzero ideals admit a unique factorization into prime ideals. Show that $\mathcal{O}$ is a Dedekind domain.

Exercise 10. The fractional ideals $\mathfrak{a}$ of a Dedekind domain $\mathcal{O}$ are projective $\mathcal{O}$-modules, i.e., given any surjective homomorphism $M \xrightarrow{f} N$ of $\mathcal{O}$-modules, each homomorphism $\mathfrak{a} \xrightarrow{g} N$ can be lifted to a homomorphism $h : \mathfrak{a} \to M$ such that $f \circ h = g$.

§ 4. Lattices

In § 1, when solving the basic problems concerning the gaussian integers, we used at a crucial place the inclusion

$$\mathbb{Z}[i] \subseteq \mathbb{C}$$

and considered the integers of $\mathbb{Q}(i)$ as lattice points in the complex plane. This point of view has been generalized to arbitrary number fields by HERMANN MINKOWSKI (1864–1909) and has led to results which make up an essential part of the foundations of algebraic number theory. In order to develop Minkowski's theory we first have to introduce the general notion of lattice and study some of its basic properties.

(4.1) Definition. *Let V be an n-dimensional $\mathbb{R}$-vector space. A* **lattice** *in V is a subgroup of the form*

$$\Gamma = \mathbb{Z}v_1 + \cdots + \mathbb{Z}v_m$$

with linearly independent vectors $v_1, \dots, v_m$ of V. The m-tuple $(v_1, \dots, v_m)$ is called a **basis** *and the set*

$$\Phi = \{x_1v_1 + \cdots + x_mv_m \mid x_i \in \mathbb{R},\ 0 \le x_i < 1\}$$

a **fundamental mesh** *of the lattice. The lattice is called* **complete** *or a $\mathbb{Z}$-***structure** *of V, if $m = n$.*

The completeness of the lattice is obviously tantamount to the fact that the set of all translates $\Phi + \gamma$, $\gamma \in \Gamma$, of the fundamental mesh covers the entire space V.

The above definition makes use of a choice of linearly independent vectors. But we will need a characterization of lattices which is independent of such a choice. Note that, first of all, a lattice is a finitely generated subgroup of V. But not every finitely generated subgroup is a lattice – for instance $\mathbb{Z} + \mathbb{Z}\sqrt{2} \subseteq \mathbb{R}$ is not. But each lattice $\Gamma = \mathbb{Z}v_1 + \cdots + \mathbb{Z}v_m$ has the special property of being a **discrete** subgroup of V. This is to say that every point $\gamma \in \Gamma$ is an isolated point in the sense that there exists a neighbourhood which contains no other points of Γ. In fact, if

$$\gamma = a_1v_1 + \cdots + a_mv_m \in \Gamma,$$

then, extending $v_1, \dots, v_m$ to a basis $v_1, \dots, v_n$ of V, the set

$$\{x_1v_1 + \cdots + x_nv_n \mid x_i \in \mathbb{R},\ |a_i - x_i| < 1 \text{ for } i = 1, \dots, m\}$$

clearly is such a neighbourhood. This property is indeed characteristic.

(4.2) Proposition. *A subgroup $\Gamma \subseteq V$ is a lattice if and only if it is discrete.*

Proof: Let Γ be a discrete subgroup of V. Then Γ is closed. For let U be an arbitrary neighbourhood of 0. Then there exists a neighbourhood $U' \subseteq U$ of 0 such that every difference of elements of U' lies in U. If there were an $x \notin \Gamma$ belonging to the closure of Γ, then we could find in the neighbourhood $x+U'$ of x two distinct elements $\gamma_1, \gamma_2 \in \Gamma$, so that $0 \neq \gamma_1 - \gamma_2 \in U' - U' \subseteq U$. Thus 0 would not be an isolated point, a contradiction.

Let V_0 be the linear subspace of V which is spanned by the set Γ, and let m be its dimension. Then we may choose a basis $u_1, \dots, u_m$ of V_0 which is contained in Γ, and form the complete lattice

$$\Gamma_0 = \mathbb{Z}u_1 + \cdots + \mathbb{Z}u_m \subseteq \Gamma$$

of V_0. We claim that the index $(\Gamma : \Gamma_0)$ is finite. To see this, let $\gamma_i \in \Gamma$ vary over a system of representatives of the cosets in Γ/Γ_0. Since Γ_0 is complete in V_0, the translates $\Phi_0 + \gamma$, $\gamma \in \Gamma_0$, of the fundamental mesh

$$\Phi_0 = \left\{x_1u_1 + \cdots + x_mu_m \mid x_i \in \mathbb{R},\ 0 \le x_i < 1\right\}$$

cover the entire space V_0. We may therefore write

$$\gamma_i = \mu_i + \gamma_{0i}, \quad \mu_i \in \Phi_0, \quad \gamma_{0i} \in \Gamma_0 \subseteq V_0.$$

As the $\mu_i = \gamma_i - \gamma_{0i} \in \Gamma$ lie discretely in the bounded set Φ_0, they have to be finite in number. In fact, the intersection of Γ with the closure of Φ_0 is compact and discrete, hence finite.

Putting now $q = (\Gamma : \Gamma_0)$, we have $q\Gamma \subseteq \Gamma_0$, whence

$$\Gamma \subseteq \frac{1}{q}\Gamma_0 = \mathbb{Z}\Big(\frac{1}{q}u_1\Big) + \cdots + \mathbb{Z}\Big(\frac{1}{q}u_m\Big).$$

By the main theorem on finitely generated abelian groups, Γ therefore admits a $\mathbb{Z}$-basis $v_1, \dots, v_r$, $r \le m$, i.e., $\Gamma = \mathbb{Z}v_1 + \cdots + \mathbb{Z}v_r$. The vectors $v_1, \dots, v_r$ are also $\mathbb{R}$-linearly independent because they span the m-dimensional space V_0. This shows that Γ is a lattice. □

Next we prove a criterion which will tell us when a lattice in the space V – given, say, as a discrete subgroup $\Gamma \subseteq V$ – is complete.

(4.3) Lemma. *A lattice Γ in V is complete if and only if there exists a bounded subset $M \subseteq V$ such that the collection of all translates $M + \gamma$, $\gamma \in \Gamma$, covers the entire space V.*

Proof: If $\Gamma = \mathbb{Z}v_1 + \cdots + \mathbb{Z}v_n$ is complete, then one may take M to be the fundamental mesh $\Phi = \{x_1v_1 + \cdots + x_nv_n \mid 0 \le x_i < 1\}$.

Conversely, let M be a bounded subset of V whose translates $M + \gamma$, for $\gamma \in \Gamma$, cover V. Let V_0 be the subspace spanned by Γ. We have to show that $V = V_0$. So let $v \in V$. Since $V = \bigcup_{\gamma \in \Gamma}(M + \gamma)$ we may write, for each $\nu \in \mathbb{N}$,

$$\nu v = a_\nu + \gamma_\nu, \quad a_\nu \in M, \quad \gamma_\nu \in \Gamma \subseteq V_0.$$

Since M is bounded, $\frac{1}{\nu}a_\nu$ converges to zero, and since V_0 is closed,

$$v = \lim_{\nu\to\infty} \frac{1}{\nu}a_\nu + \lim_{\nu\to\infty} \frac{1}{\nu}\gamma_\nu = \lim_{\nu\to\infty} \frac{1}{\nu}\gamma_\nu \in V_0. \qquad \square$$

Now let V be a *euclidean* vector space, i.e., an $\mathbb{R}$-vector space of finite dimension n equipped with a symmetric, positive definite bilinear form

$$\langle\ ,\ \rangle : V \times V \longrightarrow \mathbb{R}.$$

Then we have on V a notion of volume – more precisely a Haar measure. The cube spanned by an orthonormal basis $e_1, \ldots, e_n$ has volume 1, and more generally, the parallelepiped spanned by n linearly independent vectors $v_1, \ldots, v_n$,

$$\Phi = \left\{x_1v_1 + \cdots + x_nv_n \;\middle|\; x_i \in \mathbb{R},\ 0 \le x_i < 1\right\}$$

has volume

$$\mathrm{vol}(\Phi) = |\det A|,$$

where $A = (a_{ik})$ is the matrix of the base change from $e_1, \ldots, e_n$ to $v_1, \ldots, v_n$, so that $v_i = \sum_k a_{ik}e_k$. Since

$$\big(\langle v_i, v_j\rangle\big) = \Big(\sum_{k,\ell} a_{ik}\, a_{j\ell}\langle e_k, e_\ell\rangle\Big) = \Big(\sum_k a_{ik}\, a_{jk}\Big) = AA^t,$$

we also have the invariant notation

$$\mathrm{vol}(\Phi) = \left|\det(\langle v_i, v_j\rangle)\right|^{1/2}.$$

Let Γ be the lattice spanned by $v_1, \ldots, v_n$. Then Φ is a fundamental mesh of Γ, and we write for short

$$\mathrm{vol}(\Gamma) = \mathrm{vol}(\Phi).$$

This does not depend on the choice of a basis $v_1, \ldots, v_n$ of the lattice because the transition matrix passing to a different basis, as well as its inverse, has integer coefficients, and therefore has determinant ± 1 so that the set Φ is transformed into a set of the same volume.

We now come to the most important theorem about lattices. A subset X of V is called *centrally symmetric*, if, given any point $x \in X$, the point $-x$ also belongs to X. It is called *convex* if, given any two points $x, y \in X$, the whole line segment $\{ty + (1-t)x \mid 0 \le t \le 1\}$ joining x with y is contained in X. With these definitions we have

(4.4) Minkowski's Lattice Point Theorem. *Let Γ be a complete lattice in the euclidean vector space V and X a centrally symmetric, convex subset of V. Suppose that*

$$\operatorname{vol}(X) > 2^n \operatorname{vol}(\Gamma).$$

Then X contains at least one nonzero lattice point $\gamma \in \Gamma$.

Proof: It is enough to show that there exist two distinct lattice points $\gamma_1, \gamma_2 \in \Gamma$ such that

$$\big(\tfrac{1}{2}X + \gamma_1\big) \cap \big(\tfrac{1}{2}X + \gamma_2\big) \neq \emptyset.$$

In fact, choosing a point in this intersection,

$$\frac{1}{2}x_1 + \gamma_1 = \frac{1}{2}x_2 + \gamma_2, \quad x_1, x_2 \in X,$$

we obtain an element

$$\gamma = \gamma_1 - \gamma_2 = \frac{1}{2}x_2 - \frac{1}{2}x_1,$$

which is the center of the line segment joining x_2 and $-x_1$, and therefore belongs to $X \cap \Gamma$.

Now, if the sets $\frac{1}{2}X + \gamma$, $\gamma \in \Gamma$, were pairwise disjoint, then the same would be true of their intersections $\Phi \cap (\frac{1}{2}X + \gamma)$ with a fundamental mesh Φ of Γ, i.e., we would have

$$\operatorname{vol}(\Phi) \geq \sum_{\gamma \in \Gamma} \operatorname{vol}\big(\Phi \cap \big(\tfrac{1}{2}X + \gamma\big)\big).$$

But translation of $\Phi \cap (\frac{1}{2}X + \gamma)$ by $-\gamma$ creates the set $(\Phi - \gamma) \cap \frac{1}{2}X$ of equal volume, and the $\Phi - \gamma$, $\gamma \in \Gamma$, cover the entire space V, therefore also the set $\frac{1}{2}X$. Consequently we would obtain

$$\operatorname{vol}(\Phi) \geq \sum_{\gamma \in \Gamma} \operatorname{vol}\big((\Phi - \gamma) \cap \tfrac{1}{2}X\big) = \operatorname{vol}\big(\tfrac{1}{2}X\big) = \frac{1}{2^n} \operatorname{vol}(X),$$

which contradicts the hypothesis. □

Exercise 1. Show that a lattice Γ in $\mathbb{R}^n$ is complete if and only if the quotient $\mathbb{R}^n/\Gamma$ is compact.

Exercise 2. Show that Minkowski's lattice point theorem cannot be improved, by giving an example of a centrally symmetric convex set $X \subseteq V$ such that $\operatorname{vol}(X) = 2^n \operatorname{vol}(\Gamma)$ which does not contain any nonzero point of the lattice Γ. If X is compact, however, then the statement (4.4) does remain true in the case of equality.

Exercise 3 (Minkowski's Theorem on Linear Forms). Let

$$L_i(x_1, \ldots, x_n) = \sum_{j=1}^{n} a_{ij} x_j, \quad i = 1, \ldots, n,$$

be real linear forms such that $\det(a_{ij}) \neq 0$, and let $c_1, \ldots, c_n$ be positive real numbers such that $c_1 \cdots c_n > |\det(a_{ij})|$. Show that there exist integers $m_1, \ldots, m_n \in \mathbb{Z}$ such that

$$|L_i(m_1, \ldots, m_n)| < c_i, \quad i = 1, \ldots, n.$$

Hint: Use Minkowski's lattice point theorem.

§ 5. Minkowski Theory

The basic idea in Minkowski's treatment of an algebraic number field $K\,|\,\mathbb{Q}$ of degree n is to interpret its numbers as points in n-dimensional space. This explains why his theory has been called "Geometry of Numbers." It seems appropriate, however, to follow the current trend and call it "Minkowski Theory" instead, because in the meantime a geometric approach to number theory has been developed which is quite different in nature and much more comprehensive. We will explain this in § 13. In the present section, we consider the canonical mapping

$$j : K \longrightarrow K_{\mathbb{C}} := \prod_{\tau} \mathbb{C}, \quad a \longmapsto ja = (\tau a),$$

which results from the n complex embeddings $\tau : K \to \mathbb{C}$. The $\mathbb{C}$-vector space $K_{\mathbb{C}}$ is equipped with the *hermitian scalar product*

$$(*) \qquad \langle x, y \rangle = \sum_{\tau} x_\tau \bar{y}_\tau .$$

Let us recall that a hermitian scalar product is given by a form $H(x, y)$ which is linear in the first variable and satisfies $\overline{H(x, y)} = H(y, x)$ as well as $H(x, x) > 0$ for $x \neq 0$. In the sequel we always view $K_{\mathbb{C}}$ as a hermitian space, with respect to the "standard metric" $(*)$.

The Galois group $G(\mathbb{C}\,|\,\mathbb{R})$ is generated by complex conjugation

$$F : z \longmapsto \bar{z} .$$

The notation F will be justified only later (see chap. III, § 4). F acts on the one hand on the factors of the product $\prod_\tau \mathbb{C}$, but on the other hand it also acts on the indexing set of τ's; to each embedding $\tau : K \to \mathbb{C}$ corresponds its complex conjugate $\bar{\tau} : K \to \mathbb{C}$. Altogether, this defines an involution

$$F : K_{\mathbb{C}} \longrightarrow K_{\mathbb{C}}$$

which, on the points $z = (z_\tau) \in K_{\mathbb{C}}$, is given by

$$(Fz)_\tau = \bar{z}_{\bar{\tau}}.$$

The scalar product $\langle\ ,\ \rangle$ is equivariant under F, that is

$$\langle Fx, Fy\rangle = F\langle x, y\rangle .$$

Finally, we have on the $\mathbb{C}$-vector space $K_{\mathbb{C}} = \prod_\tau \mathbb{C}$ the linear map

$$Tr : K_{\mathbb{C}} \longrightarrow \mathbb{C}\,,$$

given as the sum of the coordinates. It is also F-invariant. The composite

$$K \xrightarrow{\ j\ } K_{\mathbb{C}} \xrightarrow{\ Tr\ } \mathbb{C}$$

gives the usual trace of $K|\mathbb{Q}$ (see (2.6), (ii)),

$$Tr_{K|\mathbb{Q}}(a) = Tr(ja)\,.$$

We now concentrate on the $\mathbb{R}$-vector space

$$K_{\mathbb{R}} = K_{\mathbb{C}}^{+} = \big[\prod_\tau \mathbb{C}\big]^{+}$$

consisting of the $G(\mathbb{C}|\mathbb{R})$-invariant, i.e., F-invariant, points of $K_{\mathbb{C}}$. These are the points (z_τ) such that $z_{\bar{\tau}} = \bar{z}_\tau$. An explicit description of $K_{\mathbb{R}}$ will be given anon. Since $\bar{\tau}a = \overline{\tau a}$ for $a \in K$, one has $F(ja) = ja$. This yields a mapping

$$j : K \longrightarrow K_{\mathbb{R}}.$$

The restriction of the hermitian scalar product $\langle\, ,\rangle$ from $K_{\mathbb{C}}$ to $K_{\mathbb{R}}$ gives a scalar product

$$\langle\ ,\ \rangle : K_{\mathbb{R}} \times K_{\mathbb{R}} \longrightarrow \mathbb{R}$$

on the $\mathbb{R}$-vector space $K_{\mathbb{R}}$. Indeed, for $x, y \in K_{\mathbb{R}}$, one has $\langle x, y\rangle \in \mathbb{R}$ in view of the relations $F\langle x, y\rangle = \langle Fx, Fy\rangle = \langle x, y\rangle$, $\overline{\langle x, y\rangle} = \overline{\langle x, y\rangle} = \langle y, x\rangle$, and, in any case, $\langle x, x\rangle > 0$ for $x \neq 0$.

We call the *euclidean* vector space

$$K_{\mathbb{R}} = \big[\prod_\tau \mathbb{C}\big]^{+}$$

the **Minkowski space**, its scalar product $\langle\ ,\ \rangle$ the **canonical metric**, and the associated Haar measure (see §4, p. 26) the **canonical measure**. Since $Tr \circ F = F \circ Tr$ we have on $K_{\mathbb{R}}$ the $\mathbb{R}$-linear map

$$Tr : K_{\mathbb{R}} \longrightarrow \mathbb{R}\,,$$

and its composite with $j : K \to K_{\mathbb{R}}$ is again the usual trace of $K\,|\,\mathbb{Q}$,

$$Tr_{K|\mathbb{Q}}(a) = Tr(ja).$$

Remark: We mention in passing – it will not be used in the sequel – that the mapping $j : K \to K_{\mathbb{R}}$ identifies the vector space $K_{\mathbb{R}}$ with the tensor product $K \otimes_{\mathbb{Q}} \mathbb{R}$,

$$K \otimes_{\mathbb{Q}} \mathbb{R} \xrightarrow{\sim} K_{\mathbb{R}}, \quad a \otimes x \longmapsto (ja)x.$$

Likewise, $K \otimes_{\mathbb{Q}} \mathbb{C} \xrightarrow{\sim} K_{\mathbb{C}}$. In this approach, the inclusion $K_{\mathbb{R}} \subseteq K_{\mathbb{C}}$ corresponds to the canonical mapping $K \otimes_{\mathbb{Q}} \mathbb{R} \to K \otimes_{\mathbb{Q}} \mathbb{C}$ which is induced by the inclusion $\mathbb{R} \hookrightarrow \mathbb{C}$. F corresponds to $F(a \otimes z) = a \otimes \bar{z}$.

An explicit description of the Minkowski space $K_{\mathbb{R}}$ can be given in the following manner. Some of the embeddings $\tau : K \to \mathbb{C}$ are real in that they land already in $\mathbb{R}$, and others are complex, i.e., not real. Let

$$\rho_1, \ldots, \rho_r : K \longrightarrow \mathbb{R}$$

be the real embeddings. The complex ones come in pairs

$$\sigma_1, \bar{\sigma}_1, \ldots, \sigma_s, \bar{\sigma}_s : K \longrightarrow \mathbb{C}$$

of complex conjugate embeddings. Thus $n = r + 2s$. We choose from each pair some fixed complex embedding, and let ρ vary over the family of real embeddings and σ over the family of chosen complex embeddings. Since F leaves the ρ invariant, but exchanges the $\sigma, \bar{\sigma}$, we have

$$K_{\mathbb{R}} = \Big\{ (z_\tau) \in \prod_\tau \mathbb{C} \;\Big|\; z_\rho \in \mathbb{R},\ z_{\bar{\sigma}} = \bar{z}_\sigma \Big\}.$$

This gives the

(5.1) Proposition. *There is an isomorphism*

$$f : K_{\mathbb{R}} \longrightarrow \prod_\tau \mathbb{R} = \mathbb{R}^{r+2s}$$

given by the rule $(z_\tau) \mapsto (x_\tau)$ *where*

$$x_\rho = z_\rho, \quad x_\sigma = \operatorname{Re}(z_\sigma), \quad x_{\bar{\sigma}} = \operatorname{Im}(z_\sigma).$$

This isomorphism transforms the canonical metric $\langle\ ,\ \rangle$ *into the scalar product*

$$(x, y) = \sum_\tau \alpha_\tau x_\tau y_\tau,$$

where $\alpha_\tau = 1$, *resp.* $\alpha_\tau = 2$, *if* τ *is real, resp. complex.*

Proof: The map is clearly an isomorphism. If $z = (z_\tau) = (x_\tau + iy_\tau)$, $z' = (z'_\tau) = (x'_\tau + iy'_\tau) \in K_{\mathbb{R}}$, then $z_\rho \bar{z}'_\rho = x_\rho x'_\rho$, and in view of $y_\sigma = x_{\bar{\sigma}}$ and $y'_\sigma = x'_{\bar{\sigma}}$, one gets

$$z_\sigma \bar{z}'_\sigma + z_{\bar{\sigma}} \bar{z}'_{\bar{\sigma}} = z_\sigma \bar{z}'_\sigma + \bar{z}_\sigma z'_\sigma = 2\,\mathrm{Re}(z_\sigma \bar{z}'_\sigma) = 2(x_\sigma x'_\sigma + x_{\bar{\sigma}} x'_{\bar{\sigma}}).$$

This proves the claim concerning the scalar products. □

The scalar product $(x, y) = \sum_\tau \alpha_\tau x_\tau y_\tau$ transfers the canonical measure from $K_{\mathbb{R}}$ to $\mathbb{R}^{r+2s}$. It obviously differs from the standard Lebesgue measure by

$$\mathrm{vol}_{\mathrm{canonical}}(X) = 2^s \,\mathrm{vol}_{\mathrm{Lebesgue}}\big(f(X)\big).$$

Minkowski himself worked with the Lebesgue measure on $\mathbb{R}^{r+2s}$, and most textbooks follow suit. The corresponding measure on $K_{\mathbb{R}}$ is the one determined by the scalar product

$$(x, y) = \sum_\tau \frac{1}{\alpha_\tau} x_\tau \bar{y}_\tau .$$

This scalar product may therefore be called the **Minkowski metric** on $K_{\mathbb{R}}$. But we will systematically work with the canonical metric, and denote by vol the corresponding canonical measure.

The mapping $j : K \to K_{\mathbb{R}}$ gives us the following lattices in Minkowski space $K_{\mathbb{R}}$.

(5.2) Proposition. *If $\mathfrak{a} \neq 0$ is an ideal of $\mathcal{O}_K$, then $\Gamma = j\mathfrak{a}$ is a complete lattice in $K_{\mathbb{R}}$. Its fundamental mesh has volume*

$$\mathrm{vol}(\Gamma) = \sqrt{|d_K|}\,(\mathcal{O}_K : \mathfrak{a}).$$

Proof: Let $\alpha_1, \ldots, \alpha_n$ be a $\mathbb{Z}$-basis of $\mathfrak{a}$, so that $\Gamma = \mathbb{Z}j\alpha_1 + \cdots + \mathbb{Z}j\alpha_n$. We choose a numbering of the embeddings $\tau : K \to \mathbb{C}$, $\tau_1, \ldots, \tau_n$, and form the matrix $A = (\tau_\ell \alpha_i)$. Then, according to (2.12), we have

$$d(\mathfrak{a}) = d(\alpha_1, \ldots, \alpha_n) = (\det A)^2 = (\mathcal{O}_K : \mathfrak{a})^2 d(\mathcal{O}_K) = (\mathcal{O}_K : \mathfrak{a})^2 d_K,$$

and on the other hand

$$\big(\langle j\alpha_i, j\alpha_k\rangle\big) = \Big(\sum_{\ell=1}^{n} \tau_\ell\, \alpha_i\, \bar{\tau}_\ell\, \alpha_k\Big) = A\bar{A}^t .$$

This indeed yields

$$\mathrm{vol}(\Gamma) = \big|\det(\langle j\alpha_i, j\alpha_k\rangle)\big|^{1/2} = |\det A| = \sqrt{|d_K|}\,(\mathcal{O}_K : \mathfrak{a}). \qquad \square$$

Using this proposition, Minkowski's lattice point theorem now gives the following result, which is what we chiefly intend to use in our applications to number theory.

(5.3) Theorem. *Let $\mathfrak{a} \neq 0$ be an integral ideal of K, and let $c_\tau > 0$, for $\tau \in \operatorname{Hom}(K,\mathbb{C})$, be real numbers such that $c_\tau = c_{\bar\tau}$ and*

$$\prod_\tau c_\tau > A(\mathcal{O}_K : \mathfrak{a}),$$

where $A = \left(\frac{2}{\pi}\right)^s \sqrt{|d_K|}$. Then there exists $a \in \mathfrak{a}$, $a \neq 0$, such that

$$|\tau a| < c_\tau \quad \textit{for all} \quad \tau \in \operatorname{Hom}(K,\mathbb{C}).$$

Proof: The set $X = \{(z_\tau) \in K_\mathbb{R} \mid |z_\tau| < c_\tau\}$ is centrally symmetric and convex. Its volume $\operatorname{vol}(X)$ can be computed via the map (5.1)

$$f : K_\mathbb{R} \xrightarrow{\sim} \prod_\tau \mathbb{R}, \quad (z_\tau) \longmapsto (x_\tau),$$

given by $x_\rho = z_\rho$, $x_\sigma = \operatorname{Re}(z_\sigma)$, $x_{\bar\sigma} = \operatorname{Im}(z_\sigma)$. It comes out to be 2^s times the Lebesgue-volume of the image

$$f(X) = \left\{ (x_\tau) \in \prod_\tau \mathbb{R} \;\middle|\; |x_\rho| < c_\rho,\ x_\sigma^2 + x_{\bar\sigma}^2 < c_\sigma^2 \right\}.$$

This gives

$$\operatorname{vol}(X) = 2^s \operatorname{vol}_{\text{Lebesgue}}\big(f(X)\big) = 2^s \prod_\rho (2c_\rho) \prod_\sigma (\pi c_\sigma^2) = 2^{r+s}\pi^s \prod_\tau c_\tau.$$

Now using (5.2), we obtain

$$\operatorname{vol}(X) > 2^{r+s}\pi^s \left(\frac{2}{\pi}\right)^s \sqrt{|d_K|}\,(\mathcal{O}_K : \mathfrak{a}) = 2^n \operatorname{vol}(\Gamma).$$

Thus the hypothesis of Minkowski's lattice point theorem is satisfied. So there does indeed exist a lattice point $ja \in X$, $a \neq 0$, $a \in \mathfrak{a}$; in other words $|\tau a| < c_\tau$. □

There is also a **multiplicative version** of Minkowski theory. It is based on the homomorphism

$$j : K^* \longrightarrow K_\mathbb{C}^* = \prod_\tau \mathbb{C}^*.$$

The multiplicative group $K_\mathbb{C}^*$ admits the homomorphism

$$N : K_\mathbb{C}^* \longrightarrow \mathbb{C}^*$$

given by the product of the coordinates. The composite

$$K^* \xrightarrow{j} K_\mathbb{C}^* \xrightarrow{N} \mathbb{C}^*$$

is the usual norm of $K|\mathbb{Q}$,

$$N_{K|\mathbb{Q}}(a) = N(ja).$$

In order to produce a lattice from the multiplicative theory, we use the logarithm to pass from multiplicative to additive groups

$$\ell : \mathbb{C}^* \longrightarrow \mathbb{R}, \quad z \longmapsto \log|z|.$$

It induces a surjective homomorphism

$$\ell : K_{\mathbb{C}}^* \longrightarrow \prod_\tau \mathbb{R},$$

and we obtain the commutative diagram

$$\begin{array}{ccccc} K^* & \xrightarrow{\ j\ } & K_{\mathbb{C}}^* & \xrightarrow{\ \ell\ } & \prod_\tau \mathbb{R} \\ {\scriptstyle N_{K|\mathbb{Q}}}\downarrow & & \downarrow{\scriptstyle N} & & \downarrow{\scriptstyle Tr} \\ \mathbb{Q}^* & \longrightarrow & \mathbb{C}^* & \xrightarrow{\ \ell\ } & \mathbb{R}. \end{array}$$

The involution $F \in G(\mathbb{C}|\mathbb{R})$ acts on all groups in this diagram, trivially on K^*, on $K_{\mathbb{C}}^*$ as before, and on the points $x = (x_\tau) \in \prod_\tau \mathbb{R}$ by $(Fx)_\tau = x_{\bar\tau}$. One clearly has

$$F \circ j = j, \quad F \circ \ell = \ell \circ F, \quad N \circ F = F \circ N, \quad Tr \circ F = Tr,$$

i.e., the homomorphisms of the diagram are $G(\mathbb{C}|\mathbb{R})$-homomorphisms. We now pass everywhere to the fixed modules under $G(\mathbb{C}|\mathbb{R})$ and obtain the diagram

$$\begin{array}{ccccc} K^* & \xrightarrow{\ j\ } & K_{\mathbb{R}}^* & \xrightarrow{\ \ell\ } & [\prod_\tau \mathbb{R}]^+ \\ {\scriptstyle N_{K|\mathbb{Q}}}\downarrow & & \downarrow{\scriptstyle N} & & \downarrow{\scriptstyle Tr} \\ \mathbb{Q}^* & \longrightarrow & \mathbb{R}^* & \xrightarrow{\ \ell\ } & \mathbb{R}. \end{array}$$

The $\mathbb{R}$-vector space $\left[\prod_\tau \mathbb{R}\right]^+$ is explicitly given as follows. Separate as before the embeddings $\tau : K \to \mathbb{C}$ into real ones, $\rho_1, \dots, \rho_r$, and pairs of complex conjugate ones, $\sigma_1, \bar\sigma_1, \dots, \sigma_s, \bar\sigma_s$. We obtain a decomposition which is analogous to the one we saw above for $\left[\prod_\tau \mathbb{C}\right]^+$,

$$\Big[\prod_\tau \mathbb{R}\Big]^+ = \prod_\rho \mathbb{R} \times \prod_\sigma [\mathbb{R} \times \mathbb{R}]^+.$$

The factor $[\mathbb{R} \times \mathbb{R}]^+$ now consists of the points (x,x), and we identify it with $\mathbb{R}$ by the map $(x,x) \mapsto 2x$. In this way we obtain an isomorphism

$$\Big[\prod_\tau \mathbb{R}\Big]^+ \cong \mathbb{R}^{r+s},$$

which again transforms the map $Tr : \left[\prod_\tau \mathbb{R}\right]^+ \to \mathbb{R}$ into the usual map

$$Tr : \mathbb{R}^{r+s} \longrightarrow \mathbb{R}$$

given by the sum of the coordinates. Identifying $\left[\prod_\tau \mathbb{R}\right]^+$ with $\mathbb{R}^{r+s}$, the homomorphism

$$\ell : K_{\mathbb{R}}^* \longrightarrow \mathbb{R}^{r+s}$$

is given by

$$\ell(x) = \left(\log|x_{\rho_1}|, \ldots, \log|x_{\rho_r}|, \log|x_{\sigma_1}|^2, \ldots, \log|x_{\sigma_s}|^2\right),$$

where we write $x \in K_{\mathbb{R}}^* \subseteq \prod_\tau \mathbb{C}^*$ as $x = (x_\tau)$.

Exercise 1. Write down a constant A which depends only on K such that every integral ideal $\mathfrak{a} \neq 0$ of K contains an element $a \neq 0$ satisfying

$$|\tau a| < A(\mathcal{O}_K : \mathfrak{a})^{1/n} \quad \text{for all } \tau \in \operatorname{Hom}(K, \mathbb{C}),\ n = [K : \mathbb{Q}].$$

Exercise 2. Show that the convex, centrally symmetric set

$$X = \Big\{ (z_\tau) \in K_{\mathbb{R}} \;\Big|\; \sum_\tau |z_\tau| < t \Big\}$$

has volume $\operatorname{vol}(X) = 2^r \pi^s \frac{t^n}{n!}$ (see chap. III, (2.15)).

Exercise 3. Show that in every ideal $\mathfrak{a} \neq 0$ of $\mathcal{O}_K$ there exists an $a \neq 0$ such that

$$|N_{K|\mathbb{Q}}(a)| \leq M(\mathcal{O}_K : \mathfrak{a}),$$

where $M = \frac{n!}{n^n}\left(\frac{4}{\pi}\right)^s \sqrt{|d_K|}$ (the so-called **Minkowski bound**).

Hint: Use exercise 2 to proceed as in (5.3), and make use of the inequality between arithmetic and geometric means,

$$\frac{1}{n} \sum_\tau |z_\tau| \geq \Big(\prod_\tau |z_\tau|\Big)^{1/n}.$$

§ 6. The Class Number

As a first application of Minkowski theory, we are going to show that the ideal class group $Cl_K = J_K/P_K$ of an algebraic number field K is finite. In order to count the ideals $\mathfrak{a} \neq 0$ of the ring $\mathcal{O}_K$ we consider their **absolute norm**

$$\mathfrak{N}(\mathfrak{a}) = (\mathcal{O}_K : \mathfrak{a}).$$

(Throughout this book the case of the zero ideal $\mathfrak{a} = 0$ is often tacitly excluded, when its consideration would visibly make no sense.) This index is finite by (2.12), and the name is justified by the special case of a principal ideal (α) of $\mathcal{O}_K$, where we have the identity

$$\mathfrak{N}\big((\alpha)\big) = \big|N_{K|\mathbb{Q}}(\alpha)\big|.$$

Indeed, if $\omega_1, \dots, \omega_n$ is a $\mathbb{Z}$-basis of $\mathcal{O}_K$, then $\alpha\,\omega_1, \dots, \alpha\,\omega_n$ is a $\mathbb{Z}$-basis of $(\alpha) = \alpha\mathcal{O}_K$, and if $A = (a_{ij})$ denotes the transition matrix, $\alpha\,\omega_i = \sum a_{ij}\,\omega_j$, then, as was pointed out already in §2, one has $|\det(A)| = (\mathcal{O}_K : (\alpha))$ as well as $\det(A) = N_{K|\mathbb{Q}}(\alpha)$ by definition.

(6.1) Proposition. *If* $\mathfrak{a} = \mathfrak{p}_1^{\nu_1} \cdots \mathfrak{p}_r^{\nu_r}$ *is the prime factorization of an ideal* $\mathfrak{a} \neq 0$, *then one has*

$$\mathfrak{N}(\mathfrak{a}) = \mathfrak{N}(\mathfrak{p}_1)^{\nu_1} \cdots \mathfrak{N}(\mathfrak{p}_r)^{\nu_r}.$$

Proof: By the Chinese remainder theorem (3.6), one has

$$\mathcal{O}_K/\mathfrak{a} = \mathcal{O}_K/\mathfrak{p}_1^{\nu_1} \oplus \cdots \oplus \mathcal{O}_K/\mathfrak{p}_r^{\nu_r}.$$

We are thus reduced to considering the case where $\mathfrak{a}$ is a prime power $\mathfrak{p}^\nu$. In the chain

$$\mathfrak{p} \supseteq \mathfrak{p}^2 \supseteq \cdots \supseteq \mathfrak{p}^\nu$$

one has $\mathfrak{p}^i \neq \mathfrak{p}^{i+1}$ because of the unique prime factorization, and each quotient $\mathfrak{p}^i/\mathfrak{p}^{i+1}$ is an $\mathcal{O}_K/\mathfrak{p}$-vector space of dimension 1. In fact, if $a \in \mathfrak{p}^i \smallsetminus \mathfrak{p}^{i+1}$ and $\mathfrak{b} = (a) + \mathfrak{p}^{i+1}$, then $\mathfrak{p}^i \supseteq \mathfrak{b} \supsetneqq \mathfrak{p}^{i+1}$ and consequently $\mathfrak{p}^i = \mathfrak{b}$, because otherwise $\mathfrak{b}' = \mathfrak{b}\mathfrak{p}^{-i}$ would be a proper divisor of $\mathfrak{p} = \mathfrak{p}^{i+1}\mathfrak{p}^{-i}$. Thus $\bar{a} = a \bmod \mathfrak{p}^{i+1}$ is a basis of the $\mathcal{O}_K/\mathfrak{p}$-vector space $\mathfrak{p}^i/\mathfrak{p}^{i+1}$. So we have $\mathfrak{p}^i/\mathfrak{p}^{i+1} \cong \mathcal{O}_K/\mathfrak{p}$ and therefore

$$\mathfrak{N}(\mathfrak{p}^\nu) = (\mathcal{O}_K : \mathfrak{p}^\nu) = (\mathcal{O}_K : \mathfrak{p})(\mathfrak{p} : \mathfrak{p}^2) \cdots (\mathfrak{p}^{\nu-1} : \mathfrak{p}^\nu) = \mathfrak{N}(\mathfrak{p})^\nu. \qquad \square$$

The proposition immediately implies the multiplicativity

$$\mathfrak{N}(\mathfrak{a}\mathfrak{b}) = \mathfrak{N}(\mathfrak{a})\mathfrak{N}(\mathfrak{b})$$

of the absolute norm. It may therefore be extended to a homomorphism

$$\mathfrak{N} : J_K \longrightarrow \mathbb{R}_+^*$$

defined on all fractional ideals $\mathfrak{a} = \prod_{\mathfrak{p}} \mathfrak{p}^{\nu_{\mathfrak{p}}}$, $\nu_{\mathfrak{p}} \in \mathbb{Z}$. The following lemma, a consequence of (5.3), is crucial for the finiteness of the ideal class group.

(6.2) Lemma. *In every ideal* $\mathfrak{a} \neq 0$ *of* $\mathcal{O}_K$ *there exists an* $a \in \mathfrak{a}$, $a \neq 0$, *such that*

$$\big|N_{K|\mathbb{Q}}(a)\big| \le \Big(\frac{2}{\pi}\Big)^s \sqrt{|d_K|}\,\mathfrak{N}(\mathfrak{a}).$$

Proof: Given $\varepsilon > 0$, we choose positive real numbers c_τ, for $\tau \in \mathrm{Hom}(K, \mathbb{C})$, such that $c_\tau = c_{\bar\tau}$ and

$$\prod_\tau c_\tau = \left(\frac{2}{\pi}\right)^s \sqrt{|d_K|}\,\mathfrak{N}(\mathfrak{a}) + \varepsilon.$$

Then by (5.3) we find an element $a \in \mathfrak{a}$, $a \neq 0$, satisfying $|\tau a| < c_\tau$. Thus

$$\left|N_{K|\mathbb{Q}}(a)\right| = \prod_\tau |\tau a| < \left(\frac{2}{\pi}\right)^s \sqrt{|d_K|}\,\mathfrak{N}(\mathfrak{a}) + \varepsilon\,.$$

This being true for all $\varepsilon > 0$ and since $|N_{K|\mathbb{Q}}(a)|$ is always a positive integer, there has to exist an $a \in \mathfrak{a}$, $a \neq 0$, such that

$$\left|N_{K|\mathbb{Q}}(a)\right| \leq \left(\frac{2}{\pi}\right)^s \sqrt{|d_K|}\,\mathfrak{N}(\mathfrak{a})\,. \qquad \square$$

(6.3) Theorem. *The ideal class group $Cl_K = J_K/P_K$ is finite. Its order*

$$h_K = (J_K : P_K)$$

is called the **class number** *of K.*

Proof: If $\mathfrak{p} \neq 0$ is a prime ideal of $\mathcal{O}_K$ and $\mathfrak{p} \cap \mathbb{Z} = p\mathbb{Z}$, then $\mathcal{O}_K/\mathfrak{p}$ is a finite field extension of $\mathbb{Z}/p\mathbb{Z}$ of degree, say, $f \geq 1$, and we have

$$\mathfrak{N}(\mathfrak{p}) = p^f\,.$$

Given p, there are only finitely many prime ideals $\mathfrak{p}$ such that $\mathfrak{p} \cap \mathbb{Z} = p\mathbb{Z}$, because this means that $\mathfrak{p} \,|\, (p)$. It follows that there are only finitely many prime ideals $\mathfrak{p}$ of bounded absolute norm. Since every integral ideal admits a representation $\mathfrak{a} = \mathfrak{p}_1^{\nu_1} \cdots \mathfrak{p}_r^{\nu_r}$ where $\nu_i > 0$ and

$$\mathfrak{N}(\mathfrak{a}) = \mathfrak{N}(\mathfrak{p}_1)^{\nu_1} \cdots \mathfrak{N}(\mathfrak{p}_r)^{\nu_r},$$

there are altogether only a finite number of ideals $\mathfrak{a}$ of $\mathcal{O}_K$ with bounded absolute norm $\mathfrak{N}(\mathfrak{a}) \leq M$.

It therefore suffices to show that each class $[\mathfrak{a}] \in Cl_K$ contains an integral ideal $\mathfrak{a}_1$ satisfying

$$\mathfrak{N}(\mathfrak{a}_1) \leq M = \left(\frac{2}{\pi}\right)^s \sqrt{|d_K|}\,.$$

For this, choose an arbitrary representative $\mathfrak{a}$ of the class, and a $\gamma \in \mathcal{O}_K$, $\gamma \neq 0$, such that $\mathfrak{b} = \gamma\,\mathfrak{a}^{-1} \subseteq \mathcal{O}_K$. By (6.2), there exists $\alpha \in \mathfrak{b}$, $\alpha \neq 0$, such that

$$\left|N_{K|\mathbb{Q}}(\alpha)\right| \cdot \mathfrak{N}(\mathfrak{b})^{-1} = \mathfrak{N}\big((\alpha)\mathfrak{b}^{-1}\big) = \mathfrak{N}(\alpha\mathfrak{b}^{-1}) \leq M\,.$$

The ideal $\mathfrak{a}_1 = \alpha\mathfrak{b}^{-1} = \alpha\gamma^{-1}\mathfrak{a} \in [\mathfrak{a}]$ therefore has the required property. $\square$

The theorem of the finiteness of the class number h_K means that passing from numbers to ideals has not thrust us into unlimited new territory. The most favourable case occurs of course when $h_K = 1$. This means that $\mathcal{O}_K$ is a principal ideal domain, i.e., that prime factorization of elements in the classical sense holds. In general, however, one has $h_K > 1$. For instance, we know now that the only imaginary quadratic fields $\mathbb{Q}(\sqrt{d})$, d squarefree and < 0, which have class number 1 are those with

$$d = -1, \; -2, \; -3, \; -7, \; -11, \; -19, \; -43, \; -67, \; -163.$$

Among real quadratic fields, class number 1 is more common. In the range $2 \le d < 100$ for instance, it occurs for

$$\begin{aligned} d = {} & 2,3,5,6,7,11,13,14,17,19,21,22,23,29,\\ & 31,33,37,38,41,43,46,47,53,57,59,61,\\ & 62,67,69,71,73,77,83,86,89,93,94,97. \end{aligned}$$

It is conjectured that there are infinitely many real quadratic fields of class number 1. But we do not even yet know whether there are infinitely many algebraic number fields (of arbitrary degree) with class number 1. It was found time and again in innumerable investigations that the ideal class groups Cl_K behave completely unpredictably, both in their size and their structure. An exception to this lack of rule is *Kenkichi Iwasawa*'s discovery that the p-part of the class number of the field of p^n-th roots of unity obeys a very strict law when n varies (see [136], th. 13.13).

In the case of the field of p-th roots of unity, the question whether the class number is divisible by p has played a very important rôle because it is intimately linked to the celebrated **Fermat's Last Theorem** according to which the equation

$$x^p + y^p = z^p$$

for $p \ge 3$ has no solutions in integers $\neq 0$. In a similar way as the sums of two squares $x^2+y^2 = (x+iy)(x-iy)$ lead to studying the gaussian integers, the decomposition of x^p+y^p by means of a p-th root of unity $\zeta \neq 1$ leads to a problem in the ring $\mathbb{Z}[\zeta]$ of integers of $\mathbb{Q}(\zeta)$. The equation $y^p = z^p - x^p$ there turns into the identity

$$y \cdot y \cdots y = (z-x)(z-\zeta x)\cdots(z-\zeta^{p-1}x).$$

Thus, assuming the existence of a solution, one obtains two multiplicative decompositions of the same number in $\mathbb{Z}[\zeta]$. One can show that this contradicts the unique factorization – provided that this holds in the ring $\mathbb{Z}[\zeta]$. Supposing erroneously that this was the case in general – in other words that the class number h_p of the field $\mathbb{Q}(\zeta)$ were always equal to 1 – some actually thought they had proved "Fermat's Last Theorem" in this way. *Kummer*, however, did not fall into this trap. Instead, he proved that the arguments we have indicated can be salvaged if one only assumes $p \nmid h_p$ instead of $h_p = 1$. In this case he called a prime number p **regular**, otherwise **irregular**. He even showed that p is regular if and only if the numerators of the **Bernoulli numbers** $B_2, B_4, \ldots, B_{p-3}$ are not divisible by p. Among the first 25 prime numbers < 100 only three are irregular: 37, 59, and 67. We still do not know today whether there are infinitely many regular prime numbers.

The connection with Fermat's last theorem has at last become obsolete. Following a surprising discovery by the mathematician *Gerhard Frey*, who established a link with the theory of *elliptic curves*, it was *Kenneth Ribet*, who managed to reduce Fermat's statement to another, much more important conjecture, the **Taniyama-Shimura-Weil Conjecture**. This was proved in sufficient generality in 1994 by *Andrew Wiles*, after many years of work, and with a helping hand from *Richard Taylor*. See [144].

The regular and irregular prime numbers do however continue to be important.

Exercise 1. How many integral ideals $\mathfrak{a}$ are there with the given norm $\mathfrak{N}(\mathfrak{a}) = n$?

Exercise 2. Show that the quadratic fields with discriminant 5, 8, 11, -3, -4, -7, -8, -11 have class number 1.

Exercise 3. Show that in every ideal class of an algebraic number field K of degree n, there exists an integral ideal $\mathfrak{a}$ such that

$$\mathfrak{N}(\mathfrak{a}) \leq \frac{n!}{n^n}\Big(\frac{4}{\pi}\Big)^s \sqrt{|d_K|}\,.$$

Hint: Using exercise 3, §5, proceed as in the proof of (6.3).

Exercise 4. Show that the absolute value of the discriminant $|d_K|$ is > 1 for every algebraic number field $K \neq \mathbb{Q}$ (Minkowski's theorem on the discriminant, see chap. III, (2.17)).

Exercise 5. Show that the absolute value of the discriminant $|d_K|$ tends to ∞ with the degree n of the field.

Exercise 6. Let $\mathfrak{a}$ be an integral ideal of K and $\mathfrak{a}^m = (a)$. Show that $\mathfrak{a}$ becomes a principal ideal in the field $L = K(\sqrt[m]{a})$, in the sense that $\mathfrak{a}\mathcal{O}_L = (\alpha)$.

Exercise 7. Show that, for every number field K, there exists a finite extension L such that every ideal of K becomes a principal ideal.

§ 7. Dirichlet's Unit Theorem

After considering the ideal class group Cl_K, we now turn to the second main problem posed by the ring $\mathcal{O}_K$ of integers of an algebraic number field K, the group of units $\mathcal{O}_K^*$. It contains the finite group $\mu(K)$ of the roots of unity that lie in K, but in general is not itself finite. Its size is in fact determined by the number r of real embeddings $\rho : K \to \mathbb{R}$ and the number s of pairs $\sigma, \bar{\sigma} : K \to \mathbb{C}$ of complex conjugate embeddings. In order to describe the group, we use the diagram which was set up in §5:

$$\begin{array}{ccccc} K^* & \xrightarrow{\ j\ } & K_{\mathbb{R}}^* & \xrightarrow{\ \ell\ } & [\prod_\tau \mathbb{R}]^+ \\ {\scriptstyle N_{K|\mathbb{Q}}}\downarrow & & \downarrow{\scriptstyle N} & & \downarrow{\scriptstyle Tr} \\ \mathbb{Q}^* & \longrightarrow & \mathbb{R}^* & \xrightarrow{\log|\ |} & \mathbb{R}\,. \end{array}$$

In the upper part of the diagram we consider the subgroups

$$\mathcal{O}_K^* = \big\{\varepsilon \in \mathcal{O}_K \mid N_{K|\mathbb{Q}}(\varepsilon) = \pm 1\big\}, \quad \text{the group of units,}$$

$$S = \big\{y \in K_{\mathbb{R}}^* \mid N(y) = \pm 1\big\}, \quad \text{the "norm-one surface",}$$

$$H = \big\{x \in [\prod_\tau \mathbb{R}]^+ \mid Tr(x) = 0\big\}, \quad \text{the "trace-zero hyperplane".}$$

We obtain the homomorphisms

$$\mathcal{O}_K^* \xrightarrow{\ j\ } S \xrightarrow{\ \ell\ } H$$

and the composite $\lambda := \ell \circ j : \mathcal{O}_K^* \to H$. The image will be denoted by

$$\Gamma = \lambda(\mathcal{O}_K^*) \subseteq H,$$

and we obtain the

(7.1) Proposition. *The sequence*

$$1 \longrightarrow \mu(K) \longrightarrow \mathcal{O}_K^* \xrightarrow{\ \lambda\ } \Gamma \longrightarrow 0$$

is exact.

Proof: We have to show that $\mu(K)$ is the kernel of λ. For $\zeta \in \mu(K)$ and $\tau : K \to \mathbb{C}$ any embedding, we find $\log|\tau\zeta| = \log 1 = 0$, so that certainly $\mu(K) \subseteq \ker(\lambda)$. Conversely, let $\varepsilon \in \mathcal{O}_K^*$ be an element in the kernel, so that $\lambda(\varepsilon) = \ell(j\varepsilon) = 0$. This means that $|\tau\varepsilon| = 1$ for each embedding

$\tau : K \to \mathbb{C}$, so that $j\varepsilon = (\tau\varepsilon)$ lies in a bounded domain of the $\mathbb{R}$-vector space $K_{\mathbb{R}}$. On the other hand, $j\varepsilon$ is a point of the lattice $j\mathcal{O}_K$ of $K_{\mathbb{R}}$ (see (5.2)). Therefore the kernel of λ can contain only a finite number of elements, and thus, being a finite group, contains only roots of unity in K^*. □

Given this proposition, it remains to determine the group Γ. For this, we need the following

(7.2) Lemma. *Up to multiplication by units there are only finitely many elements $\alpha \in \mathcal{O}_K$ of given norm $N_{K|\mathbb{Q}}(\alpha) = a$.*

Proof: Let $a \in \mathbb{Z}$, $a > 1$. In every one of the finitely many cosets of $\mathcal{O}_K/a\mathcal{O}_K$ there exists, up to multiplication by units, at most one element α such that $|N(\alpha)| = |N_{K|\mathbb{Q}}(\alpha)| = a$. For if $\beta = \alpha + a\gamma$, $\gamma \in \mathcal{O}_K$, is another one, then

$$\frac{\alpha}{\beta} = 1 \pm \frac{N(\beta)}{\beta}\gamma \in \mathcal{O}_K$$

because $N(\beta)/\beta \in \mathcal{O}_K$, and by the same token $\frac{\beta}{\alpha} = 1 \pm \frac{N(\alpha)}{\alpha}\gamma \in \mathcal{O}_K$, i.e., β is associated to α. Therefore, up to multiplication by units, there are at most $(\mathcal{O}_K : a\mathcal{O}_K)$ elements of norm $\pm a$. □

(7.3) Theorem. *The group Γ is a complete lattice in the $(r+s-1)$-dimensional vector space H, and is therefore isomorphic to $\mathbb{Z}^{r+s-1}$.*

Proof: We first show that $\Gamma = \lambda(\mathcal{O}_K^*)$ is a lattice in H, i.e., a discrete subgroup. The mapping $\lambda : \mathcal{O}_K^* \to H$ arises by restricting the mapping

$$K^* \xrightarrow{j} \prod_\tau \mathbb{C}^* \xrightarrow{\ell} \prod_\tau \mathbb{R},$$

and it suffices to show that, for any $c > 0$, the bounded domain $\{(x_\tau) \in \prod_\tau \mathbb{R} \mid |x_\tau| \le c\}$ contains only finitely many points of $\Gamma = \ell(j\mathcal{O}_K^*)$. Since $\ell((z_\tau)) = (\log|z_\tau|)$, the preimage of this domain with respect to ℓ is the bounded domain

$$\Big\{(z_\tau) \in \prod_\tau \mathbb{C}^* \mid e^{-c} \le |z_\tau| \le e^c\Big\}.$$

It contains only finitely many elements of the set $j\mathcal{O}_K^*$ because this is a subset of the lattice $j\mathcal{O}_K$ in $\big[\prod_\tau \mathbb{C}\big]^+$ (see (5.2)). Therefore Γ is a lattice.

We now show that Γ is a complete lattice in H. This is the principal claim of the theorem. We apply the criterion (4.3). So we have to find a bounded set $M \subseteq H$ such that

$$H = \bigcup_{\gamma \in \Gamma} (M + \gamma).$$

We construct this set through its preimage with respect to the surjective homomorphism

$$\ell : S \longrightarrow H.$$

More precisely, we will construct a bounded set T in the norm-one surface S, the *multiplicative* translations $Tj\varepsilon$, $\varepsilon \in \mathcal{O}_K^*$, of which cover all of S:

$$S = \bigcup_{\varepsilon \in \mathcal{O}_K^*} Tj\varepsilon.$$

For $x = (x_\tau) \in T$, it will follow that the absolute values $|x_\tau|$ are bounded from above and also away from zero, because $\prod_\tau |x_\tau| = 1$. Thus $M = \ell(T)$ will also be bounded. We choose real numbers $c_\tau > 0$, for $\tau \in \mathrm{Hom}(K, \mathbb{C})$, satisfying $c_\tau = c_{\bar\tau}$ and

$$C = \prod_\tau c_\tau > \left(\frac{2}{\pi}\right)^s \sqrt{|d_K|},$$

and we consider the set

$$X = \left\{ (z_\tau) \in K_{\mathbb{R}} \;\middle|\; |z_\tau| < c_\tau \right\}.$$

For an arbitrary point $y = (y_\tau) \in S$, it follows that

$$Xy = \left\{ (z_\tau) \in K_{\mathbb{R}} \;\middle|\; |z_\tau| < c'_\tau \right\}$$

where $c'_\tau = c_\tau |y_\tau|$, and one has $c'_\tau = c'_{\bar\tau}$ and $\prod_\tau c'_\tau = \prod_\tau c_\tau = C$ because $\prod_\tau |y_\tau| = |N(y)| = 1$. Then, by (5.3), there is a point

$$ja = (\tau a) \in Xy, \quad a \in \mathcal{O}_K, \quad a \neq 0.$$

Now, according to lemma (7.2), we may pick a system $\alpha_1, \dots, \alpha_N \in \mathcal{O}_K$, $\alpha_i \neq 0$, in such a way that every $a \in \mathcal{O}_K$ with $0 < |N_{K|\mathbb{Q}}(a)| \leq C$ is associated to one of these numbers. The set

$$T = S \cap \bigcup_{i=1}^{N} X(j\alpha_i)^{-1}$$

then has the required property: since X is bounded, so is $X(j\alpha_i)^{-1}$ and therefore also T, and we have

$$S = \bigcup_{\varepsilon \in \mathcal{O}_K^*} Tj\varepsilon.$$

In fact, if $y \in S$, we find by the above an $a \in \mathcal{O}_K$, $a \neq 0$, such that $ja \in Xy^{-1}$, so $ja = xy^{-1}$ for some $x \in X$. Since

$$\left|N_{K|\mathbb{Q}}(a)\right| = \left|N(xy^{-1})\right| = \left|N(x)\right| < \prod_\tau c_\tau = C,$$

a is associated to some α_i, $\alpha_i = \varepsilon a$, $\varepsilon \in \mathcal{O}_K^*$. Consequently

$$y = xja^{-1} = xj(\alpha_i^{-1}\varepsilon).$$

Since $y, j\varepsilon \in S$, one finds $xj\alpha_i^{-1} \in S \cap Xj\alpha_i^{-1} \subseteq T$, and thus $y \in Tj\varepsilon$. $\square$

From proposition (7.1) and theorem (7.3) we immediately deduce **Dirichlet's unit theorem** in its classical form.

(7.4) Theorem. *The group of units $\mathcal{O}_K^*$ of $\mathcal{O}_K$ is the direct product of the finite cyclic group $\mu(K)$ and a free abelian group of rank $r + s - 1$.*

In other words: there exist units $\varepsilon_1, \dots, \varepsilon_t$, $t = r + s - 1$, called **fundamental units**, such that any other unit ε can be written uniquely as a product

$$\varepsilon = \zeta\, \varepsilon_1^{\nu_1} \cdots \varepsilon_t^{\nu_t}$$

with a root of unity ζ and integers ν_i.

Proof: In the exact sequence

$$1 \longrightarrow \mu(K) \longrightarrow \mathcal{O}_K^* \xrightarrow{\lambda} \Gamma \longrightarrow 0$$

Γ is a free abelian group of rank $t = r + s - 1$ by (7.3). Let $v_1, \dots, v_t$ be a $\mathbb{Z}$-basis of Γ, let $\varepsilon_1, \dots, \varepsilon_t \in \mathcal{O}_K^*$ be preimages of the v_i, and let $A \subseteq \mathcal{O}_K^*$ be the subgroup generated by the ε_i. Then A is mapped isomorphically onto Γ by λ, i.e., one has $\mu(K) \cap A = \{1\}$ and therefore $\mathcal{O}_K^* = \mu(K) \times A$. $\square$

Identifying $\left[\prod_\tau \mathbb{R}\right]^+ = \mathbb{R}^{r+s}$ (see §5, p.33), H becomes a subspace of the euclidean space $\mathbb{R}^{r+s}$ and thus itself a euclidean space. We may therefore speak of the volume of the fundamental mesh $\mathrm{vol}(\lambda(\mathcal{O}_K^*))$ of the unit lattice $\Gamma = \lambda(\mathcal{O}_K^*) \subseteq H$, and will now compute it. Let $\varepsilon_1, \dots, \varepsilon_t$, $t = r + s - 1$, be a system of fundamental units and Φ the fundamental mesh of the unit lattice $\lambda(\mathcal{O}_K^*)$, spanned by the vectors $\lambda(\varepsilon_1), \dots, \lambda(\varepsilon_t) \in H$. The vector

$$\lambda_0 = \frac{1}{\sqrt{r+s}}(1, \dots, 1) \in \mathbb{R}^{r+s}$$

is obviously orthogonal to H and has length 1. The t-dimensional volume of Φ therefore equals the $(t+1)$-dimensional volume of the parallelepiped spanned by $\lambda_0, \lambda(\varepsilon_1), \ldots, \lambda(\varepsilon_t)$ in $\mathbb{R}^{t+1}$. But this has volume

$$\pm\det\begin{pmatrix} \lambda_{01} & \lambda_1(\varepsilon_1) & \cdots & \lambda_1(\varepsilon_t) \\ \vdots & \vdots & & \vdots \\ \lambda_{0t+1} & \lambda_{t+1}(\varepsilon_1) & \cdots & \lambda_{t+1}(\varepsilon_t) \end{pmatrix}.$$

Adding all rows to a fixed one, say the i-th row, this row has only zeroes, except for the first entry, which equals $\sqrt{r+s}$. We therefore get the

(7.5) Proposition. *The volume of the fundamental mesh of the unit lattice $\lambda(\mathcal{O}_K^*)$ in H is*

$$\operatorname{vol}\big(\lambda(\mathcal{O}_K^*)\big) = \sqrt{r+s}\,R,$$

where R is the absolute value of the determinant of an arbitrary minor of rank $t = r+s-1$ of the following matrix:

$$\begin{pmatrix} \lambda_1(\varepsilon_1) & \cdots & \lambda_1(\varepsilon_t) \\ \vdots & & \vdots \\ \lambda_{t+1}(\varepsilon_1) & \cdots & \lambda_{t+1}(\varepsilon_t) \end{pmatrix}.$$

This absolute value R is called the **regulator** *of the field K.*

The importance of the regulator will only be demonstrated later (see chap. VII, §5).

Exercise 1. Let $D > 1$ be a squarefree integer and d the discriminant of the real quadratic number field $K = \mathbb{Q}(\sqrt{D})$ (see §2, exercise 4). Let x_1, y_1 be the uniquely determined rational integer solution of the equation

$$x^2 - dy^2 = -4,$$

or – in case this equation has no rational integer solutions – of the equation

$$x^2 - dy^2 = 4,$$

for which $x_1, y_1 > 0$ are as small as possible. Then

$$\varepsilon_1 = \frac{x_1 + y_1\sqrt{d}}{2}$$

is a fundamental unit of K. (The pair of equations $x^2 - dy^2 = \pm 4$ is called **Pell's equation.**)

Exercise 2. Check the following table of fundamental units ε_1 for $\mathbb{Q}(\sqrt{D})$:

D	2	3	5	6	7	10
ε_1	$1+\sqrt{2}$	$2+\sqrt{3}$	$(1+\sqrt{5})/2$	$5+2\sqrt{6}$	$8+3\sqrt{7}$	$3+\sqrt{10}$

Hint: Check one by one for $y = 1, 2, 3, \ldots$, whether one of the numbers $dy^2 \mp 4$ is a square x^2. By the unit theorem this is bound to happen, with the plus sign. However, for fixed y, let preference be given to the minus sign. Then the first case, in this order, where $dy_1^2 \mp 4 = x_1^2$, gives the fundamental unit $\varepsilon_1 = (x_1 + y_1\sqrt{d})/2$.

Exercise 3. The Battle of Hastings (October 14, 1066).
"The men of Harold stood well together, as their wont was, and formed thirteen squares, with a like number of men in every square thereof, and woe to the hardy Norman who ventured to enter their redoubts; for a single blow of a Saxon war-hatched would break his lance and cut through his coat of mail... When Harold threw himself into the fray the Saxons were one mighty square of men, shouting the battle-cries, 'Ut!', 'Olicrosse!', 'Godemite!'." [Fictitious historical text, following essentially problem no. 129 in: H.E. Dundeney, *Amusements in Mathematics*, 1917 (Dover reprints 1958 and 1970).]
Question. How many troops does this suggest Harold II had at the battle of Hastings?

Exercise 4. Let ζ be a primitive p-th root of unity, p an odd prime number. Show that $\mathbb{Z}[\zeta]^* = (\zeta)\mathbb{Z}[\zeta + \zeta^{-1}]^*$. Show that $\mathbb{Z}[\zeta]^* = \{\pm\zeta^k(1+\zeta)^n \mid 0 \le k < 5,\ n \in \mathbb{Z}\}$, if $p = 5$.

Exercise 5. Let ζ be a primitive m-th root of unity, $m \ge 3$. Show that the numbers $\frac{1-\zeta^k}{1-\zeta}$ for $(k, m) = 1$ are units in the ring of integers of the field $\mathbb{Q}(\zeta)$. The subgroup of the group of units they generate is called the group of **cyclotomic units**.

Exercise 6. Let K be a totally real number field, i.e., $X = \mathrm{Hom}(K, \mathbb{C}) = \mathrm{Hom}(K, \mathbb{R})$, and let T be a proper nonempty subset of X. Then there exists a unit ε satisfying $0 < \tau\varepsilon < 1$ for $\tau \in T$, and $\tau\varepsilon > 1$ for $\tau \notin T$.

Hint: Apply Minkowski's lattice point theorem to the unit lattice in trace-zero space.

§ 8. Extensions of Dedekind Domains

Having studied the ideal class group and the group of units of the ring $\mathcal{O}_K$ of integers of a number field K, we now propose to make a first survey of the set of prime ideals of $\mathcal{O}_K$. They are often referred to as the prime ideals of K – an imprecise manner of speaking which is, however, not likely to cause any misunderstanding.

Every prime ideal $\mathfrak{p} \neq 0$ of $\mathcal{O}_K$ contains a rational prime number p (see §3, p. 17) and is therefore a divisor of the ideal $p\mathcal{O}_K$. Hence the question arises as to how a prime number p factors into prime ideals of the ring $\mathcal{O}_K$. We treat this problem in a more general context, starting from an arbitrary Dedekind domain $\mathcal{O}$ at the base instead of $\mathbb{Z}$, and taking instead of $\mathcal{O}_K$ the integral closure $\mathcal{O}$ of $\mathcal{O}$ in a finite extension of its field of fractions.

(8.1) Proposition. *Let $\mathcal{o}$ be a Dedekind domain with field of fractions K, let $L|K$ be a finite extension of K and $\mathcal{O}$ the integral closure of $\mathcal{o}$ in L. Then $\mathcal{O}$ is again a Dedekind domain.*

Proof: Being the integral closure of $\mathcal{o}$, $\mathcal{O}$ is integrally closed. The fact that the nonzero prime ideals $\mathfrak{P}$ of $\mathcal{O}$ are maximal is proved similarly as in the case $\mathcal{o} = \mathbb{Z}$ (see (3.1)): $\mathfrak{p} = \mathfrak{P} \cap \mathcal{o}$ is a nonzero prime ideal of $\mathcal{o}$. Thus the integral domain $\mathcal{O}/\mathfrak{P}$ is an extension of the field $\mathcal{o}/\mathfrak{p}$, and therefore has itself to be a field, because if it were not, then it would admit a nonzero prime ideal whose intersection with $\mathcal{o}/\mathfrak{p}$ would again be a nonzero prime ideal in $\mathcal{o}/\mathfrak{p}$. It remains to show that $\mathcal{O}$ is noetherian. In the case that is of chief interest to us, namely, if $L|K$ is a separable extension, the proof is very easy. Let $\alpha_1, \dots, \alpha_n$ be a basis of $L|K$ contained in $\mathcal{O}$, of discriminant $d = d(\alpha_1, \dots, \alpha_n)$. Then $d \neq 0$ by (2.8), and (2.9) tells us that $\mathcal{O}$ is contained in the finitely generated $\mathcal{o}$-module $\mathcal{o}\alpha_1/d + \cdots + \mathcal{o}\alpha_n/d$. Every ideal of $\mathcal{O}$ is also contained in this finitely generated $\mathcal{o}$-module, and therefore is itself an $\mathcal{o}$-module of finite type, hence *a fortiori* a finitely generated $\mathcal{O}$-module. This shows that $\mathcal{O}$ is noetherian, provided $L|K$ is separable. We ask the reader's permission to content ourselves for the time being with this case. We shall come back to the general case on a more convenient occasion. In fact, we shall give the proof in a more general framework in § 12 (see (12.8)). □

For a prime ideal $\mathfrak{p}$ of $\mathcal{o}$ one always has

$$\mathfrak{p}\mathcal{O} \neq \mathcal{O}.$$

In fact, let $\pi \in \mathfrak{p} \setminus \mathfrak{p}^2$ ($\mathfrak{p} \neq 0$), so that $\pi\mathcal{o} = \mathfrak{p}\mathfrak{a}$ with $\mathfrak{p} \nmid \mathfrak{a}$, hence $\mathfrak{p} + \mathfrak{a} = \mathcal{o}$. Writing $1 = b + s$, with $b \in \mathfrak{p}$ and $s \in \mathfrak{a}$, we find $s \notin \mathfrak{p}$ and $s\mathfrak{p} \subseteq \mathfrak{p}\mathfrak{a} = \pi\mathcal{o}$. If one had $\mathfrak{p}\mathcal{O} = \mathcal{O}$, then it would follow that $s\mathcal{O} = s\mathfrak{p}\mathcal{O} \subseteq \pi\mathcal{O}$, so that $s = \pi x$ for some $x \in \mathcal{O} \cap K = \mathcal{o}$, i.e., $s \in \mathfrak{p}$, a contradiction.

A prime ideal $\mathfrak{p} \neq 0$ of the ring $\mathcal{o}$ decomposes in $\mathcal{O}$ in a unique way into a product of prime ideals,

$$\mathfrak{p}\mathcal{O} = \mathfrak{P}_1^{e_1} \cdots \mathfrak{P}_r^{e_r}.$$

Instead of $\mathfrak{p}\mathcal{O}$ we will often write simply $\mathfrak{p}$. The prime ideals $\mathfrak{P}_i$ occurring in the decomposition are precisely those prime ideals $\mathfrak{P}$ of $\mathcal{O}$ which lie over $\mathfrak{p}$ in the sense that one has the relation

$$\mathfrak{p} = \mathfrak{P} \cap \mathcal{o}.$$

This we also denote for short by $\mathfrak{P} \,|\, \mathfrak{p}$, and we call $\mathfrak{P}$ a prime divisor of $\mathfrak{p}$. The exponent e_i is called the **ramification index**, and the degree of the field extension

$$f_i = \left[\mathcal{O}/\mathfrak{P}_i : \mathcal{o}/\mathfrak{p}\right]$$

is called the **inertia degree** of $\mathfrak{P}_i$ over $\mathfrak{p}$. If the extension $L|K$ is separable, the numbers e_i, f_i and the degree $n = [L : K]$ are connected by the following law.

(8.2) Proposition. *Let $L|K$ be separable. Then we have the* **fundamental identity**

$$\sum_{i=1}^{r} e_i f_i = n .$$

Proof: The proof is based on the Chinese remainder theorem

$$\mathcal{O}/\mathfrak{p}\mathcal{O} \;\cong\; \bigoplus_{i=1}^{r} \mathcal{O}/\mathfrak{P}_i^{e_i} .$$

$\mathcal{O}/\mathfrak{p}\mathcal{O}$ and $\mathcal{O}/\mathfrak{P}_i^{e_i}$ are vector spaces over the field $\kappa = \mathcal{o}/\mathfrak{p}$, and it suffices to show that

$$\dim_\kappa(\mathcal{O}/\mathfrak{p}\mathcal{O}) = n \quad \text{and} \quad \dim_\kappa(\mathcal{O}/\mathfrak{P}_i^{e_i}) = e_i f_i .$$

In order to prove the first identity, let $\omega_1, \dots, \omega_m \in \mathcal{O}$ be representatives of a basis $\overline{\omega}_1, \dots, \overline{\omega}_m$ of $\mathcal{O}/\mathfrak{p}\mathcal{O}$ over κ (we have seen in the proof of (8.1) that $\mathcal{O}$ is a finitely generated $\mathcal{o}$-module, so certainly $\dim_\kappa(\mathcal{O}/\mathfrak{p}\mathcal{O}) < \infty$). It is sufficient to show that $\omega_1, \dots, \omega_m$ is a basis of $L|K$. Assume the $\omega_1, \dots, \omega_m$ are linearly dependent over K, and hence also over $\mathcal{o}$. Then there are elements $a_1, \dots, a_m \in \mathcal{o}$ not all zero such that

$$a_1\omega_1 + \dots + a_m\omega_m = 0 .$$

Consider the ideal $\mathfrak{a} = (a_1, \dots, a_m)$ of $\mathcal{o}$ and find $a \in \mathfrak{a}^{-1}$ such that $a \notin \mathfrak{a}^{-1}\mathfrak{p}$, hence $a\mathfrak{a} \not\subseteq \mathfrak{p}$. Then the elements $aa_1, \dots, aa_m$ lie in $\mathcal{o}$, but not all belong to $\mathfrak{p}$. The congruence

$$aa_1\omega_1 + \dots + aa_m\omega_m \equiv 0 \bmod \mathfrak{p}$$

thus gives us a linear dependence among the $\overline{\omega}_1, \dots, \overline{\omega}_m$ over κ, a contradiction. The $\omega_1, \dots, \omega_m$ are therefore linearly independent over K.

In order to show that the ω_i are a basis of $L|K$, we consider the $\mathcal{o}$-modules $M = \mathcal{o}\omega_1 + \dots + \mathcal{o}\omega_m$ and $N = \mathcal{O}/M$. Since $\mathcal{O} = M + \mathfrak{p}\mathcal{O}$, we have $\mathfrak{p}N = N$. As $L|K$ is separable, $\mathcal{O}$, and hence also N, are finitely generated $\mathcal{o}$-modules (see p. 45). If $\alpha_1, \dots, \alpha_s$ is a system of generators of N, then

$$\alpha_i = \sum_j a_{ij}\alpha_j \quad \text{for } a_{ij} \in \mathfrak{p} .$$

Let A be the matrix $(a_{ij}) - I$, where I is the unit matrix of rank s, and let B be the **adjoint** matrix of A, whose entries are the minors of rank $(s-1)$

of A. Then one has $A(\alpha_1, \dots, \alpha_s)^t = 0$ and $BA = dI$, with $d = \det(A)$, (see (2.3)). Hence

$$0 = BA(\alpha_1, \dots, \alpha_s)^t = (d\alpha_1, \dots, d\alpha_s)^t,$$

and therefore $dN = 0$, i.e., $d\mathcal{O} \subseteq M = \mathcal{o}\omega_1 + \cdots + \mathcal{o}\omega_m$. We have $d \neq 0$, because expanding the determinant $d = \det((a_{ij}) - I)$ we find $d \equiv (-1)^s \bmod \mathfrak{p}$ because $a_{ij} \in \mathfrak{p}$. It follows that $L = dL = K\omega_1 + \cdots + K\omega_m$. $\omega_1, \dots, \omega_m$ is therefore indeed a basis of $L|K$.

In order to prove the second identity, let us consider the descending chain

$$\mathcal{O}/\mathfrak{P}_i^{e_i} \supseteq \mathfrak{P}_i/\mathfrak{P}_i^{e_i} \supseteq \cdots \supseteq \mathfrak{P}_i^{e_i - 1}/\mathfrak{P}_i^{e_i} \supseteq (0)$$

of κ-vector spaces. The successive quotients $\mathfrak{P}_i^\nu/\mathfrak{P}_i^{\nu+1}$ in this chain are isomorphic to $\mathcal{O}/\mathfrak{P}_i$, for if $\alpha \in \mathfrak{P}_i^\nu \setminus \mathfrak{P}_i^{\nu+1}$, then the homomorphism

$$\mathcal{O} \longrightarrow \mathfrak{P}_i^\nu/\mathfrak{P}_i^{\nu+1}, \quad a \longmapsto a\alpha,$$

has kernel $\mathfrak{P}_i$ and is surjective because $\mathfrak{P}_i^\nu$ is the gcd of $\mathfrak{P}_i^{\nu+1}$ and $(\alpha) = \alpha\mathcal{O}$ so that $\mathfrak{P}_i^\nu = \alpha\mathcal{O} + \mathfrak{P}_i^{\nu+1}$. Since $f_i = [\mathcal{O}/\mathfrak{P}_i : \kappa]$, we obtain $\dim_\kappa(\mathfrak{P}_i^\nu/\mathfrak{P}_i^{\nu+1}) = f_i$ and therefore

$$\dim_\kappa(\mathcal{O}/\mathfrak{P}_i^{e_i}) = \sum_{\nu=0}^{e_i - 1} \dim_\kappa(\mathfrak{P}_i^\nu/\mathfrak{P}_i^{\nu+1}) = e_i f_i . \qquad \square$$

Suppose now that the separable extension $L|K$ is given by a primitive element $\theta \in \mathcal{O}$ with minimal polynomial

$$p(X) \in \mathcal{o}[X],$$

so that $L = K(\theta)$. We may then deduce a result about the nature of the decomposition of $\mathfrak{p}$ in $\mathcal{O}$ which, albeit not complete, does show characteristic phenomena and a striking simplicity. It is incomplete in that a finite number of prime ideals are excluded; only those relatively prime to the **conductor** of the ring $\mathcal{o}[\theta]$ can be considered. This conductor is defined to be the biggest ideal $\mathfrak{F}$ of $\mathcal{O}$ which is contained in $\mathcal{o}[\theta]$. In other words

$$\mathfrak{F} = \{\alpha \in \mathcal{O} \mid \alpha\mathcal{O} \subseteq \mathcal{o}[\theta]\}.$$

Since $\mathcal{O}$ is a finitely generated $\mathcal{o}$-module (see proof of (8.1)), one has $\mathfrak{F} \neq 0$.

(8.3) Proposition. *Let $\mathfrak{p}$ be a prime ideal of $\mathcal{o}$ which is relatively prime to the conductor $\mathfrak{F}$ of $\mathcal{o}[\theta]$, and let*

$$\bar{p}(X) = \bar{p}_1(X)^{e_1} \cdots \bar{p}_r(X)^{e_r}$$

be the factorization of the polynomial $\overline{p}(X) = p(X) \bmod \mathfrak{p}$ *into irreducibles* $\overline{p}_i(X) = p_i(X) \bmod \mathfrak{p}$ *over the residue class field* $\mathcal{o}/\mathfrak{p}$*, with all* $p_i(X) \in \mathcal{o}[X]$ *monic. Then*

$$\mathfrak{P}_i = \mathfrak{p}\mathcal{O} + p_i(\theta)\mathcal{O}, \quad i = 1, \dots, r,$$

are the different prime ideals of $\mathcal{O}$ *above* $\mathfrak{p}$*. The inertia degree* f_i *of* $\mathfrak{P}_i$ *is the degree of* $\overline{p}_i(X)$*, and one has*

$$\mathfrak{p} = \mathfrak{P}_1^{e_1} \cdots \mathfrak{P}_r^{e_r}.$$

Proof: Writing $\mathcal{O}' = \mathcal{o}[\theta]$ and $\overline{\mathcal{o}} = \mathcal{o}/\mathfrak{p}$, we have a canonical isomorphism

$$\mathcal{O}/\mathfrak{p}\mathcal{O} \;\cong\; \mathcal{O}'/\mathfrak{p}\mathcal{O}' \;\cong\; \overline{\mathcal{o}}[X]/\big(\overline{p}(X)\big).$$

The first isomorphism follows from the relative primality $\mathfrak{p}\mathcal{O} + \mathfrak{F} = \mathcal{O}$. As $\mathfrak{F} \subseteq \mathcal{O}'$, it follows that $\mathcal{O} = \mathfrak{p}\mathcal{O} + \mathcal{O}'$, i.e., the homomorphism $\mathcal{O}' \to \mathcal{O}/\mathfrak{p}\mathcal{O}$ is surjective. It has kernel $\mathfrak{p}\mathcal{O} \cap \mathcal{O}'$, which equals $\mathfrak{p}\mathcal{O}'$. Since $(\mathfrak{p}, \mathfrak{F} \cap \mathcal{o}) = 1$, it follows that $\mathfrak{p}\mathcal{O} \cap \mathcal{O}' = (\mathfrak{p} + \mathfrak{F})(\mathfrak{p}\mathcal{O} \cap \mathcal{O}') \subseteq \mathfrak{p}\mathcal{O}'$.

The second isomorphism is deduced from the surjective homomorphism

$$\mathcal{o}[X] \longrightarrow \overline{\mathcal{o}}[X]/\big(\overline{p}(X)\big).$$

Its kernel is the ideal generated by $\mathfrak{p}$ and $p(X)$, and in view of $\mathcal{O}' = \mathcal{o}[\theta] = \mathcal{o}[X]/(p(X))$, we have $\mathcal{O}'/\mathfrak{p}\mathcal{O}' \cong \overline{\mathcal{o}}[X]/(\overline{p}(X))$.

Since $\overline{p}(X) = \prod_{i=1}^r \overline{p}_i(X)^{e_i}$, the Chinese remainder theorem finally gives the isomorphism

$$\overline{\mathcal{o}}[X]/\big(\overline{p}(X)\big) \;\cong\; \bigoplus_{i=1}^{r} \overline{\mathcal{o}}[X]\big/(\overline{p}_i(X))^{e_i}.$$

This shows that the prime ideals of the ring $R = \overline{\mathcal{o}}[X]/(\overline{p}(X))$ are the principal ideals $(\overline{p}_i)$ generated by the $\overline{p}_i(X) \bmod \overline{p}(X)$, for $i = 1, \dots, r$, that the degree $[R/(\overline{p}_i) : \overline{\mathcal{o}}]$ equals the degree of the polynomial $\overline{p}_i(X)$, and that

$$(0) = (\overline{p}) = \bigcap_{i=1}^{r} (\overline{p}_i)^{e_i}.$$

In view of the isomorphism $\overline{\mathcal{o}}[X]/(\overline{p}(X)) \cong \mathcal{O}/\mathfrak{p}\mathcal{O}$, $f(X) \mapsto f(\theta)$, the same situation holds in the ring $\overline{\mathcal{O}} = \mathcal{O}/\mathfrak{p}\mathcal{O}$. Thus the prime ideals $\overline{\mathfrak{P}}_i$ of $\overline{\mathcal{O}}$ correspond to the prime ideals $(\overline{p}_i)$, and they are the principal ideals generated by the $p_i(\theta) \bmod \mathfrak{p}\mathcal{O}$. The degree $[\overline{\mathcal{O}}/\overline{\mathfrak{P}}_i : \overline{\mathcal{o}}]$ is the degree of the polynomial $\overline{p}_i(X)$, and we have $(0) = \bigcap_{i=1}^r \overline{\mathfrak{P}}_i^{e_i}$. Now let $\mathfrak{P}_i = \mathfrak{p}\mathcal{O} + p_i(\theta)\mathcal{O}$ be the preimage of $\overline{\mathfrak{P}}_i$ with respect to the canonical homomorphism

$$\mathcal{O} \longrightarrow \mathcal{O}/\mathfrak{p}\mathcal{O}.$$

Then $\mathfrak{P}_i$, for $i = 1, \dots, r$, varies over the prime ideals of $\mathcal{O}$ above $\mathfrak{p}$. $f_i = [\mathcal{O}/\mathfrak{P}_i : \mathcal{o}/\mathfrak{p}]$ is the degree of the polynomial $\overline{p}_i(X)$. Furthermore $\mathfrak{P}_i^{e_i}$ is the preimage of $\overline{\mathfrak{P}}_i^{e_i}$ (because $e_i = \#\{\overline{\mathfrak{P}}^{\nu} \mid \nu \in \mathbb{N}\}$), and $\mathfrak{p}\mathcal{O} \supseteq \bigcap_{i=1}^r \mathfrak{P}_i^{e_i}$, so that $\mathfrak{p}\,\mathcal{O} \mid \prod_{i=1}^r \mathfrak{P}_i^{e_i}$ and therefore $\mathfrak{p}\mathcal{O} = \prod_{i=1}^r \mathfrak{P}_i^{e_i}$ because $\sum e_i f_i = n$. □

The prime ideal $\mathfrak{p}$ is said to **split completely** (or to be **totally split**) in L, if in the decomposition

$$\mathfrak{p} = \mathfrak{P}_1^{e_1} \cdots \mathfrak{P}_r^{e_r},$$

one has $r = n = [L : K]$, so that $e_i = f_i = 1$ for all $i = 1, \dots, r$. $\mathfrak{p}$ is called **nonsplit**, or **indecomposed**, if $r = 1$, i.e., if there is only a single prime ideal of L over $\mathfrak{p}$. From the fundamental identity

$$\sum_{i=1}^{r} e_i f_i = n$$

we now understand the name of inertia degree: the smaller this degree is, the more the ideal $\mathfrak{p}$ will be tend to factor into different prime ideals.

The prime ideal $\mathfrak{P}_i$ in the decomposition $\mathfrak{p} = \prod_{i=1}^r \mathfrak{P}_i^{e_i}$ is called **unramified** over $\mathcal{o}$ (or over K) if $e_i = 1$ and if the residue class field extension $\mathcal{O}/\mathfrak{P}_i | \mathcal{o}/\mathfrak{p}$ is separable. If not, it is called **ramified**, and **totally ramified** if furthermore $f_i = 1$. The prime ideal $\mathfrak{p}$ is called unramified if all $\mathfrak{P}_i$ are unramified, otherwise it is called ramified. The extension $L|K$ itself is called unramified if all prime ideals $\mathfrak{p}$ of K are unramified in L.

The case where a prime ideal $\mathfrak{p}$ of K is ramified in L is an exceptional phenomenon. In fact, we have the

(8.4) Proposition. *If $L|K$ is separable, then there are only finitely many prime ideals of K which are ramified in L.*

Proof: Let $\theta \in \mathcal{O}$ be a primitive element for L, i.e., $L = K(\theta)$, and let $p(X) \in \mathcal{o}[X]$ be its minimal polynomial. Let

$$d = d(1, \theta, \dots, \theta^{n-1}) = \prod_{i<j} (\theta_i - \theta_j)^2 \in \mathcal{o}$$

be the discriminant of $p(X)$ (see §2, p. 11). Then every prime ideal $\mathfrak{p}$ of K which is relatively prime to d and to the conductor $\mathfrak{F}$ of $\mathcal{o}[\theta]$ is unramified. In fact, by (8.3), the ramification indices e_i equal 1 as soon as they are equal to 1 in the factorization of $\bar{p}(X) = p(X) \bmod \mathfrak{p}$ in $\mathcal{o}/\mathfrak{p}$, so certainly if $\bar{p}(X)$ has no multiple roots. But this is the case since the discriminant $\bar{d} = d \bmod \mathfrak{p}$ of $\bar{p}(X)$ is nonzero. The residue class field extensions $\mathcal{O}/\mathfrak{P}_i | \mathcal{o}/\mathfrak{p}$ are generated by $\bar{\theta} = \theta \bmod \mathfrak{P}_i$ and therefore separable. Hence $\mathfrak{p}$ is unramified. □

The precise description of the ramified prime ideals is given by the **discriminant** of $\mathcal{O}|\mathcal{o}$. It is defined to be the ideal $\mathfrak{d}$ of $\mathcal{o}$ which is generated by the discriminants $d(\omega_1, \dots, \omega_n)$ of all bases $\omega_1, \dots, \omega_n$ of $L|K$ contained in $\mathcal{O}$. We will show in chapter III, § 2 that the prime divisors of $\mathfrak{d}$ are exactly the prime ideals which ramify in L.

Example: The law of decomposition of prime numbers p in a **quadratic number field** $\mathbb{Q}(\sqrt{a})$ is intimately related to Gauss's famous **quadratic reciprocity law**. The latter concerns the problem of integer solutions of the equation

$$x^2 + by = a, \quad (a, b \in \mathbb{Z}),$$

the simplest among the nontrivial diophantine equations. The theory of this equation reduces immediately to the case where b is an odd prime number p and $(a, p) = 1$ (exercise 6). Let us assume this for the sequel. We are then facing the question as to whether a is a **quadratic residue** mod p, i.e., whether the congruence

$$x^2 \equiv a \bmod p$$

does or does not have a solution. In other words, we want to know if the equation $\bar{x}^2 = \bar{a}$, for a given element $\bar{a} = a \bmod p \in \mathbb{F}_p^*$, admits a solution in the field $\mathbb{F}_p$ or not. For this one introduces the **Legendre symbol** $\left(\frac{a}{p}\right)$, which, for every rational number a relatively prime to p, is defined to be $\left(\frac{a}{p}\right) = 1$ or -1, according as $x^2 \equiv a \bmod p$ has or does not have a solution. This symbol is multiplicative,

$$\left(\frac{ab}{p}\right) = \left(\frac{a}{p}\right)\left(\frac{b}{p}\right).$$

This is because the group $\mathbb{F}_p^*$ is cyclic of order $p-1$ and the subgroup $\mathbb{F}_p^{*2}$ of squares has index 2, i.e., $\mathbb{F}_p^*/\mathbb{F}_p^{*2} \cong \mathbb{Z}/2\mathbb{Z}$. Since $\left(\frac{a}{p}\right) = 1 \iff \bar{a} \in \mathbb{F}_p^{*2}$, one also has

$$\left(\frac{a}{p}\right) \equiv a^{\frac{p-1}{2}} \bmod p.$$

In the case of squarefree a, the Legendre symbol $\left(\frac{a}{p}\right)$ bears the following relation with prime factorization. $\left(\frac{a}{p}\right) = 1$ signifies that

$$x^2 - a \equiv (x - \alpha)(x + \alpha) \bmod p$$

for some $\alpha \in \mathbb{Z}$. The conductor of $\mathbb{Z}[\sqrt{a}\,]$ in the ring of integers of $\mathbb{Q}(\sqrt{a})$ is a divisor of 2 (see § 2, exercise 4). We may therefore apply proposition (8.3) and obtain the

(8.5) Proposition. *For squarefree a and $(p, 2a) = 1$, we have the equivalence*

$$\left(\frac{a}{p}\right) = 1 \iff p \textit{ is totally split in } \mathbb{Q}(\sqrt{a}).$$

For the Legendre symbol, one has the following remarkable law, which like none other has left its mark on the development of algebraic number theory.

(8.6) Theorem (Gauss's Reciprocity Law). *For two distinct odd prime numbers ℓ and p, the following identity holds:*

$$\left(\frac{\ell}{p}\right)\left(\frac{p}{\ell}\right) = (-1)^{\frac{\ell-1}{2}\frac{p-1}{2}}.$$

One also has the two "supplementary theorems"

$$\left(\frac{-1}{p}\right) = (-1)^{\frac{p-1}{2}}, \quad \left(\frac{2}{p}\right) = (-1)^{\frac{p^2-1}{8}}.$$

Proof: $\left(\frac{-1}{p}\right) \equiv (-1)^{\frac{p-1}{2}} \bmod p$ implies $\left(\frac{-1}{p}\right) = (-1)^{\frac{p-1}{2}}$ since $p \neq 2$.

In order to determine $\left(\frac{2}{p}\right)$, we work in the ring $\mathbb{Z}[i]$ of gaussian integers. Since $(1+i)^2 = 2i$, we find

$$(1+i)^p = (1+i)\big((1+i)^2\big)^{\frac{p-1}{2}} = (1+i)i^{\frac{p-1}{2}}2^{\frac{p-1}{2}},$$

and since $(1+i)^p \equiv 1 + i^p \bmod p$ and $\left(\frac{2}{p}\right) \equiv 2^{\frac{p-1}{2}} \bmod p$, it follows that

$$\left(\frac{2}{p}\right)(1+i)\,i^{\frac{p-1}{2}} \equiv 1 + i(-1)^{\frac{p-1}{2}} \bmod p.$$

From this, an easy computation yields

$$\left(\frac{2}{p}\right) \equiv (-1)^{\frac{p-1}{4}} \bmod p, \quad \text{resp.} \quad \left(\frac{2}{p}\right) \equiv (-1)^{\frac{p+1}{4}} \bmod p,$$

if $\frac{p-1}{2}$ is even, resp. odd. Since $\frac{p^2-1}{8} = \frac{p-1}{4}\frac{p+1}{2} = \frac{p+1}{4}\frac{p-1}{2}$, we deduce $\left(\frac{2}{p}\right) = (-1)^{\frac{p^2-1}{8}}$.

In order to prove the first formula, we work in the ring $\mathbb{Z}[\zeta]$, where ζ is a primitive ℓ-th root of unity. We consider the **Gauss sum**

$$\tau = \sum_{a \in (\mathbb{Z}/\ell\mathbb{Z})^*} \left(\frac{a}{\ell}\right)\zeta^a$$

and show that

$$\tau^2 = \left(\frac{-1}{\ell}\right)\ell.$$

For this, let a and b vary over the group $(\mathbb{Z}/\ell\mathbb{Z})^*$, put $c = ab^{-1}$ and deduce from the identity $\left(\frac{b}{\ell}\right) = \left(\frac{b^{-1}}{\ell}\right)$ that

$$\left(\frac{-1}{\ell}\right)\tau^2 = \sum_{a,b}\left(\frac{-ab}{\ell}\right)\zeta^{a+b} = \sum_{a,b}\left(\frac{ab^{-1}}{\ell}\right)\zeta^{a-b} = \sum_{b,c}\left(\frac{c}{\ell}\right)\zeta^{bc-b}$$
$$= \sum_{c\neq 1}\left(\frac{c}{\ell}\right)\sum_{b}\zeta^{b(c-1)} + \sum_{b}\left(\frac{1}{\ell}\right).$$

Now $\sum_c\left(\frac{c}{\ell}\right) = 0$, as one sees by multiplying the sum with a symbol $\left(\frac{x}{\ell}\right) = -1$, and putting $\xi = \zeta^{c-1}$ gives $\sum_b \zeta^{b(c-1)} = \xi + \xi^2 + \cdots + \xi^{\ell-1} = -1$, from which we indeed find that

$$\left(\frac{-1}{\ell}\right)\tau^2 = (-1)(-1) + \ell - 1 = \ell.$$

This, together with the congruence $\left(\frac{\ell}{p}\right) \equiv \ell^{\frac{p-1}{2}} \bmod p$ and the identity $\left(\frac{-1}{\ell}\right) = (-1)^{\frac{\ell-1}{2}}$, implies

$$\tau^p = \tau(\tau^2)^{\frac{p-1}{2}} \equiv \tau(-1)^{\frac{\ell-1}{2}\frac{p-1}{2}}\left(\frac{\ell}{p}\right) \bmod p.$$

On the other hand one has

$$\tau^p \equiv \sum_a\left(\frac{a}{\ell}\right)\zeta^{ap} \equiv \left(\frac{p}{\ell}\right)\sum_a\left(\frac{ap}{\ell}\right)\zeta^{ap} \equiv \left(\frac{p}{\ell}\right)\tau \bmod p,$$

so that

$$\tau\left(\frac{p}{\ell}\right) \equiv \tau(-1)^{\frac{\ell-1}{2}\frac{p-1}{2}}\left(\frac{\ell}{p}\right) \bmod p.$$

Multiplying by τ and dividing by $\pm\ell$ yields the claim. □

We have proved Gauss's reciprocity law by a rather contrived calculation. In § 10, however, we will recognize the true reason why it holds in the law of decomposition of primes in the field $\mathbb{Q}(\zeta)$ of ℓ-th roots of unity. The Gauss sums do have a higher theoretical significance, though, as will become apparent later (see VII, § 2 and § 6).

Exercise 1. If $\mathfrak{a}$ and $\mathfrak{b}$ are ideals of $\mathcal{O}$, then one has $\mathfrak{a} = \mathfrak{a}\,\mathcal{O} \cap \mathcal{O}$ and $\mathfrak{a} \mid \mathfrak{b} \iff \mathfrak{a}\mathcal{O} \mid \mathfrak{b}\mathcal{O}$.

Exercise 2. For every integral ideal $\mathfrak{A}$ of $\mathcal{O}$, there exists a $\theta \in \mathcal{O}$ such that the conductor $\mathfrak{F} = \{\alpha \in \mathcal{O} \mid \alpha\mathcal{O} \subseteq o[\theta]\}$ is prime to $\mathfrak{A}$ and such that $L = K(\theta)$.

Exercise 3. If a prime ideal $\mathfrak{p}$ of K is totally split in two separable extensions $L|K$ and $L'|K$, then it is also totally split in the composite extension.

Exercise 4. A prime ideal $\mathfrak{p}$ of K is totally split in the separable extension $L|K$ if and only if it is totally split in the Galois closure $N|K$ of $L|K$.

Exercise 5. For a number field K the statement of proposition (8.3) concerning the prime decomposition in the extension $K(\theta)$ holds for all prime ideals $\mathfrak{p} \nmid (\mathcal{O} : \mathcal{o}[\theta])$.

Exercise 6. Given a positive integer $b > 1$, an integer a relatively prime to b is a quadratic residue $\bmod b$ if and only if it is a quadratic residue modulo each prime divisor p of b, and if $a \equiv 1 \bmod 4$ when $4|b$, $8 \nmid b$, resp. $a \equiv 1 \bmod 8$ when $8|b$.

Exercise 7. Let $(a, p) = 1$ and $a\nu \equiv r_\nu \bmod p$, $\nu = 1, \dots, p-1$, $0 < r_\nu < p$. Then the r_ν give a permutation π of the numbers $1, \dots, p-1$. Show that $\operatorname{sgn} \pi = \left(\frac{a}{p}\right)$.

Exercise 8. Let $a_n = \dfrac{\varepsilon^n - \varepsilon'^n}{\sqrt{5}}$, where $\varepsilon = \dfrac{1+\sqrt{5}}{2}$, $\varepsilon' = \dfrac{1-\sqrt{5}}{2}$ (a_n is the n-th Fibonacci number). If p is a prime number $\neq 2, 5$, then one has

$$a_p \equiv \left(\frac{p}{5}\right) \bmod p.$$

Exercise 9. Study the Legendre symbol $\left(\frac{3}{p}\right)$ as a function of $p > 3$. Show that the property of 3 being a quadratic residue or nonresidue $\bmod p$ depends only on the class of $p \bmod 12$.

Exercise 10. Show that the number of solutions of $x^2 \equiv a \bmod p$ equals $1 + \left(\frac{a}{p}\right)$.

Exercise 11. Show that the number of solutions of the congruence $ax^2 + bx + c \equiv 0 \bmod p$, where $(a, p) = 1$, equals $1 + \left(\frac{b^2 - 4ac}{p}\right)$.

§ 9. Hilbert's Ramification Theory

The question of prime decomposition in a finite extension $L|K$ takes a particularly interesting and important turn once we assume $L|K$ to be a Galois extension. The prime ideals are then subject to the action of the Galois group

$$G = G(L|K).$$

The "ramification theory" that arises from this assumption has been introduced into number theory by *David Hilbert* (1862–1943). Given a in the ring $\mathcal{O}$ of integral elements of L, the conjugate σa, for every $\sigma \in G$, also belongs to $\mathcal{O}$, i.e., G acts on $\mathcal{O}$. If $\mathfrak{P}$ is a prime ideal of $\mathcal{O}$ above $\mathfrak{p}$, then so is $\sigma\mathfrak{P}$, for each $\sigma \in G$, because

$$\sigma\mathfrak{P} \cap \mathcal{o} = \sigma(\mathfrak{P} \cap \mathcal{o}) = \sigma\mathfrak{p} = \mathfrak{p}.$$

The ideals $\sigma\mathfrak{P}$, for $\sigma \in G$, are called the prime ideals **conjugate** to $\mathfrak{P}$.

(9.1) Proposition. *The Galois group G acts transitively on the set of all prime ideals $\mathfrak{P}$ of $\mathcal{O}$ lying above $\mathfrak{p}$, i.e., these prime ideals are all conjugates of each other.*

Proof: Let $\mathfrak{P}$ and $\mathfrak{P}'$ be two prime ideals above $\mathfrak{p}$. Assume $\mathfrak{P}' \neq \sigma\mathfrak{P}$ for any $\sigma \in G$. By the Chinese remainder theorem there exists $x \in \mathcal{O}$ such that

$$x \equiv 0 \bmod \mathfrak{P}' \quad \text{and} \quad x \equiv 1 \bmod \sigma\mathfrak{P} \quad \text{for all} \quad \sigma \in G.$$

Then the norm $N_{L|K}(x) = \prod_{\sigma \in G} \sigma x$ belongs to $\mathfrak{P}' \cap \mathcal{o} = \mathfrak{p}$. On the other hand, $x \notin \sigma\mathfrak{P}$ for any $\sigma \in G$, hence $\sigma x \notin \mathfrak{P}$ for any $\sigma \in G$. Consequently $\prod_{\sigma \in G} \sigma x \notin \mathfrak{P} \cap \mathcal{o} = \mathfrak{p}$, a contradiction. □

(9.2) Definition. *If $\mathfrak{P}$ is a prime ideal of $\mathcal{O}$, then the subgroup*

$$G_{\mathfrak{P}} = \{\sigma \in G \mid \sigma\mathfrak{P} = \mathfrak{P}\}$$

is called the **decomposition group** *of $\mathfrak{P}$ over K. The fixed field*

$$Z_{\mathfrak{P}} = \{x \in L \mid \sigma x = x \quad \text{for all } \sigma \in G_{\mathfrak{P}}\}$$

is called the **decomposition field** *of $\mathfrak{P}$ over K.*

The decomposition group encodes in group-theoretic language the number of different prime ideals into which a prime ideal $\mathfrak{p}$ of $\mathcal{o}$ decomposes in $\mathcal{O}$. For if $\mathfrak{P}$ is one of them and σ varies over a system of representatives of the cosets in $G/G_{\mathfrak{P}}$, then $\sigma\mathfrak{P}$ varies over the different prime ideals above $\mathfrak{p}$, each one occurring precisely once, i.e., their number equals the index $(G : G_{\mathfrak{P}})$. In particular, one has

$$G_{\mathfrak{P}} = 1 \iff Z_{\mathfrak{P}} = L \iff \mathfrak{p} \text{ is totally split,}$$

$$G_{\mathfrak{P}} = G \iff Z_{\mathfrak{P}} = K \iff \mathfrak{p} \text{ is nonsplit.}$$

The decomposition group of a prime ideal $\sigma\mathfrak{P}$ conjugate to $\mathfrak{P}$ is the conjugate subgroup

$$G_{\sigma\mathfrak{P}} = \sigma G_{\mathfrak{P}} \sigma^{-1}.$$

In fact, for $\tau \in G$, one has the equivalences

$$\tau \in G_{\sigma\mathfrak{P}} \iff \tau\sigma\mathfrak{P} = \sigma\mathfrak{P} \iff \sigma^{-1}\tau\sigma\mathfrak{P} = \mathfrak{P}$$
$$\iff \sigma^{-1}\tau\sigma \in G_{\mathfrak{P}} \iff \tau \in \sigma G_{\mathfrak{P}} \sigma^{-1}.$$

Remark: The decomposition group regulates the prime decomposition also in the case of a non-Galois extension. For subgroups U and V of a group G, consider the equivalence relation in G defined by

$$\sigma \sim \sigma' \iff \sigma' = u\sigma v \quad \text{for } u \in U,\ v \in V.$$

The corresponding equivalence classes

$$U\sigma V = \{u\sigma v \mid u \in U,\ v \in V\}$$

are called the **double cosets** of G modd U, V. The set of these double cosets, which form a partition of G, is denoted $U\backslash G/V$.

Now let $L|K$ be an arbitrary separable extension, and embed it into a Galois extension $N|K$ with Galois group G. In G, consider the subgroup $H = G(N|L)$. Let $\mathfrak{p}$ be a prime ideal of K and $P_{\mathfrak{p}}$ the set of prime ideals of L above $\mathfrak{p}$. If $\mathfrak{P}$ is a prime ideal of N above $\mathfrak{p}$, then the rule

$$H\backslash G/G_{\mathfrak{P}} \longrightarrow P_{\mathfrak{p}}, \quad H\sigma G_{\mathfrak{P}} \longmapsto \sigma\mathfrak{P} \cap L,$$

gives a well-defined bijection. The proof is left to the reader.

In the Galois case, the inertia degrees $f_1, \ldots, f_r$ and the ramification indices $e_1, \ldots, e_r$ in the prime decomposition

$$\mathfrak{p} = \mathfrak{P}_1^{e_1} \cdots \mathfrak{P}_r^{e_r}$$

of a prime ideal $\mathfrak{p}$ of K are both independent of i,

$$f_1 = \cdots = f_r = f, \quad e_1 = \cdots = e_r = e.$$

In fact, writing $\mathfrak{P} = \mathfrak{P}_1$, we find $\mathfrak{P}_i = \sigma_i\mathfrak{P}$ for suitable $\sigma_i \in G$, and the isomorphism $\sigma_i : \mathcal{O} \to \mathcal{O}$ induces an isomorphism

$$\mathcal{O}/\mathfrak{P} \xrightarrow{\sim} \mathcal{O}/\sigma_i\mathfrak{P}, \quad a \bmod \mathfrak{P} \longmapsto \sigma_i a \bmod \sigma_i\mathfrak{P},$$

so that

$$f_i = [\mathcal{O}/\sigma_i\mathfrak{P} : \mathfrak{o}/\mathfrak{p}] = [\mathcal{O}/\mathfrak{P} : \mathfrak{o}/\mathfrak{p}], \quad i = 1, \ldots, r.$$

Furthermore, since $\sigma_i(\mathfrak{p}\mathcal{O}) = \mathfrak{p}\mathcal{O}$, we deduce from

$$\mathfrak{P}^\nu \mid \mathfrak{p}\mathcal{O} \iff \sigma_i(\mathfrak{P}^\nu) \mid \sigma_i(\mathfrak{p}\mathcal{O}) \iff (\sigma_i\mathfrak{P})^\nu \mid \mathfrak{p}\mathcal{O}$$

the equality of the e_i, $i = 1, \ldots, r$. Thus the prime decomposition of $\mathfrak{p}$ in $\mathcal{O}$ takes on the following simple form in the Galois case:

$$\mathfrak{p} = \Big(\prod_\sigma \sigma\mathfrak{P}\Big)^e,$$

where σ varies over a system of representatives of $G/G_{\mathfrak{P}}$. The decomposition field $Z_{\mathfrak{P}}$ of $\mathfrak{P}$ over K has the following significance for the decomposition of $\mathfrak{p}$ and the invariants e and f.

(9.3) Proposition. *Let $\mathfrak{P}_Z = \mathfrak{P} \cap Z_{\mathfrak{P}}$ be the prime ideal of $Z_{\mathfrak{P}}$ below $\mathfrak{P}$. Then we have:*

(i) *$\mathfrak{P}_Z$ is nonsplit in L, i.e., $\mathfrak{P}$ is the only prime ideal of L above $\mathfrak{P}_Z$.*

(ii) *$\mathfrak{P}$ over $Z_{\mathfrak{P}}$ has ramification index e and inertia degree f.*

(iii) *The ramification index and the inertia degree of $\mathfrak{P}_Z$ over K both equal 1.*

Proof: (i) Since $G(L|Z_{\mathfrak{P}}) = G_{\mathfrak{P}}$, the prime ideals above $\mathfrak{P}_Z$ are the $\sigma\mathfrak{P}$, for $\sigma \in G(L|Z_{\mathfrak{P}})$, and they are all equal to $\mathfrak{P}$.
(ii) Since in the Galois case, ramification indices and inertia degrees are independent of the prime divisor, the fundamental identity in this case reads

$$n = efr,$$

where $n := \#G$, $r = (G : G_{\mathfrak{P}})$. We see therefore that $\#G_{\mathfrak{P}} = [L : Z_{\mathfrak{P}}] = ef$. Let e', resp. e'', be the ramification index of $\mathfrak{P}$ over $Z_{\mathfrak{P}}$, resp. of $\mathfrak{P}_Z$ over K. Then $\mathfrak{p} = \mathfrak{P}_Z^{e''} \cdot \ldots$ in $Z_{\mathfrak{P}}$ and $\mathfrak{P}_Z = \mathfrak{P}^{e'}$ in L, so that $\mathfrak{p} = \mathfrak{P}^{e''e'} \cdot \ldots$, i.e., $e = e'e''$. One also obviously gets the analogous identity for the inertia degrees $f = f'f''$. The fundamental identity for the decomposition of $\mathfrak{P}_Z$ in L then reads $[L : Z_{\mathfrak{P}}] = e'f'$, i.e., we have $e'f' = ef$, and therefore $e' = e$, $f' = f$, $e'' = f'' = 1$. □

The ramification index e and the inertia degree f admit a further interesting group-theoretic interpretation. Since $\sigma\mathcal{O} = \mathcal{O}$ and $\sigma\mathfrak{P} = \mathfrak{P}$, every $\sigma \in G_{\mathfrak{P}}$ induces an automorphism

$$\bar{\sigma} : \mathcal{O}/\mathfrak{P} \longrightarrow \mathcal{O}/\mathfrak{P}, \quad a \bmod \mathfrak{P} \longmapsto \sigma a \bmod \mathfrak{P},$$

of the residue class field $\mathcal{O}/\mathfrak{P}$. Putting $\kappa(\mathfrak{P}) = \mathcal{O}/\mathfrak{P}$ and $\kappa(\mathfrak{p}) = o/\mathfrak{p}$, we obtain the

(9.4) Proposition. *The extension $\kappa(\mathfrak{P})|\kappa(\mathfrak{p})$ is normal and admits a surjective homomorphism*

$$G_{\mathfrak{P}} \longrightarrow G\big(\kappa(\mathfrak{P})|\kappa(\mathfrak{p})\big).$$

Proof: The inertia degree of $\mathfrak{P}_Z$ over K equals 1, i.e., $Z_{\mathfrak{P}}$ has the same residue class field $\kappa(\mathfrak{p})$ as K with respect to $\mathfrak{p}$. Therefore we may, and do, assume that $Z_{\mathfrak{P}} = K$, i.e., $G_{\mathfrak{P}} = G$. Let $\theta \in \mathcal{O}$ be a representative of an element $\bar{\theta} \in \kappa(\mathfrak{P})$ and $f(X)$, resp. $\bar{g}(X)$, the minimal polynomial of θ over K, resp. of $\bar{\theta}$ over $\kappa(\mathfrak{p})$. Then $\bar{\theta} = \theta \bmod \mathfrak{P}$ is a zero of the polynomial $\bar{f}(X) = f(X) \bmod \mathfrak{p}$, i.e., $\bar{g}(X)$ divides $\bar{f}(X)$. Since $L|K$ is normal, $f(X)$ splits over $\mathcal{O}$ into linear factors. Hence $\bar{f}(X)$ splits into linear factors over $\kappa(\mathfrak{P})$, and the same is true of $\bar{g}(X)$. In other words, $\kappa(\mathfrak{P})|\kappa(\mathfrak{p})$ is a normal extension.

Now let $\bar{\theta}$ be a primitive element for the maximal separable subextension of $\kappa(\mathfrak{P})|\kappa(\mathfrak{p})$ and

$$\bar{\sigma} \in G\big(\kappa(\mathfrak{P})|\kappa(\mathfrak{p})\big) = G\big(\kappa(\mathfrak{p})(\bar{\theta})|\kappa(\mathfrak{p})\big).$$

Then $\bar{\sigma}\,\bar{\theta}$ is a root of $\bar{g}(X)$, and hence of $\bar{f}(X)$, i.e., there exists a zero θ' of $f(X)$ such that $\theta' \equiv \bar{\sigma}\,\bar{\theta} \bmod \mathfrak{P}$. θ' is a conjugate of θ, i.e., $\theta' = \sigma\,\theta$ for some $\sigma \in G(L|K)$. Since $\sigma\,\theta \equiv \bar{\sigma}\,\bar{\theta} \bmod \mathfrak{P}$, the automorphism σ is mapped by the homomorphism in question to $\bar{\sigma}$. This proves the surjectivity. □

(9.5) Definition. *The kernel $I_{\mathfrak{P}} \subseteq G_{\mathfrak{P}}$ of the homomorphism*

$$G_{\mathfrak{P}} \longrightarrow G\big(\kappa(\mathfrak{P})|\kappa(\mathfrak{p})\big)$$

is called the **inertia group** *of $\mathfrak{P}$ over K. The fixed field*

$$T_{\mathfrak{P}} = \big\{x \in L \,\big|\, \sigma x = x \quad \text{for all } \sigma \in I_{\mathfrak{P}}\big\}$$

is called the **inertia field** *of $\mathfrak{P}$ over K.*

This inertia field $T_{\mathfrak{P}}$ appears in the tower of fields

$$K \subseteq Z_{\mathfrak{P}} \subseteq T_{\mathfrak{P}} \subseteq L,$$

and we have the exact sequence

$$1 \longrightarrow I_{\mathfrak{P}} \longrightarrow G_{\mathfrak{P}} \longrightarrow G\big(\kappa(\mathfrak{P})|\kappa(\mathfrak{p})\big) \longrightarrow 1.$$

Its properties are expressed in the

(9.6) Proposition. *The extension $T_{\mathfrak{P}}|Z_{\mathfrak{P}}$ is normal, and one has*

$$G(T_{\mathfrak{P}}|Z_{\mathfrak{P}}) \cong G\big(\kappa(\mathfrak{P})|\kappa(\mathfrak{p})\big), \quad G(L|T_{\mathfrak{P}}) = I_{\mathfrak{P}}.$$

If the residue field extension $\kappa(\mathfrak{P})|\kappa(\mathfrak{p})$ is separable, then one has

$$\# I_{\mathfrak{P}} = [L : T_{\mathfrak{P}}] = e, \quad (G_{\mathfrak{P}} : I_{\mathfrak{P}}) = [T_{\mathfrak{P}} : Z_{\mathfrak{P}}] = f.$$

In this case one finds for the prime ideal $\mathfrak{P}_T$ of $T_{\mathfrak{P}}$ below $\mathfrak{P}$:

(i) *The ramification index of $\mathfrak{P}$ over $\mathfrak{P}_T$ is e and the inertia degree is 1.*

(ii) *The ramification index of $\mathfrak{P}_T$ over $\mathfrak{P}_Z$ is 1, and the inertia degree is f.*

Proof: The first two claims follow from the identity $\# G_{\mathfrak{P}} = ef$. So we only have to show statements (i) and (ii). Using the fundamental identity, they all follow from $\kappa(\mathfrak{P}_T) = \kappa(\mathfrak{P})$. As the inertia group $I_{\mathfrak{P}}$ of $\mathfrak{P}$ over K is also the inertia group of $\mathfrak{P}$ over $T_{\mathfrak{P}}$, it follows from an application of proposition (9.4) to the extension $L|T_{\mathfrak{P}}$ that $G(\kappa(\mathfrak{P})|\kappa(\mathfrak{P}_T)) = 1$, hence $\kappa(\mathfrak{P}_T) = \kappa(\mathfrak{P})$. □

In the diagram

$$K \xrightarrow[1]{1} Z_{\mathfrak{P}} \xrightarrow[f]{1} T_{\mathfrak{P}} \xrightarrow[1]{e} L$$

we have indicated the ramification indices of the individual field extensions on top, and the inertia degrees on the bottom. In the special case where the residue field extension $\kappa(\mathfrak{P})|\kappa(\mathfrak{p})$ is separable we find

$$I_{\mathfrak{P}} = 1 \iff T_{\mathfrak{P}} = L \iff \mathfrak{p} \text{ is unramified in } L.$$

In this case the Galois group $G(\kappa(\mathfrak{P})|\kappa(\mathfrak{p})) \cong G_{\mathfrak{P}}$ of the residue class field extension may be viewed as a subgroup of $G = G(L|K)$.

Hilbert's ramification theory, with its various refinements and generalizations, belongs naturally to the theory of valuations, which we will develop in the next chapter (see chap. II, §9).

Exercise 1. If $L|K$ is a Galois extension of algebraic number fields with noncyclic Galois group, then there are at most finitely many nonsplit prime ideals of K.

Exercise 2. If $L|K$ is a Galois extension of algebraic number fields, and $\mathfrak{P}$ a prime ideal which is unramified over K (i.e., $\mathfrak{p} = \mathfrak{P} \cap K$ is unramified in L), then there is one and only one automorphism $\varphi_{\mathfrak{P}} \in G(L|K)$ such that

$$\varphi_{\mathfrak{P}} a \equiv a^q \bmod \mathfrak{P} \quad \text{for all } a \in \mathcal{O},$$

where $q = [\kappa(\mathfrak{P}) : \kappa(\mathfrak{p})]$. It is called the **Frobenius automorphism**. The decomposition group $G_{\mathfrak{P}}$ is cyclic and $\varphi_{\mathfrak{P}}$ is a generator of $G_{\mathfrak{P}}$.

Exercise 3. Let $L|K$ be a solvable extension of prime degree p (not necessarily Galois). If the unramified prime ideal $\mathfrak{p}$ in L has two prime factors $\mathfrak{P}$ and $\mathfrak{P}'$ of degree 1, then it is already totally split (theorem of *F.K. Schmidt*).

Hint: Use the following result of *Galois* (see [75], chap. II, §3): if G is a transitive solvable permutation group of prime degree p, then there is no nontrivial permutation $\sigma \in G$ which fixes two distinct letters.

Exercise 4. Let $L|K$ be a finite (not necessarily Galois) extension of algebraic number fields and $N|K$ the normal closure of $L|K$. Show that a prime ideal $\mathfrak{p}$ of K is totally split in L if and only if it is totally split in N.

Hint: Use the double coset decomposition $H\backslash G/G_{\mathfrak{P}}$, where $G = G(N|K)$, $H = G(N|L)$ and $G_{\mathfrak{P}}$ is the decomposition group of a prime ideal $\mathfrak{P}$ over $\mathfrak{p}$.

§10. Cyclotomic Fields

The concepts and results of the theory as far as it has now been developed have reached a degree of abstraction which we will now balance by something more concrete. We will put the insights of the general theory to the task and make them more explicit in the example of the **n-th cyclotomic field** $\mathbb{Q}(\zeta)$, where ζ is a **primitive n-th root of unity**. Among all number fields, this field occupies a special, central place. So studying it does not only furnish a worthwhile example but in fact an essential building block for the further theory.

It will be our first goal to determine explicitly the ring of integers of the field $\mathbb{Q}(\zeta)$. For this we need the

(10.1) Lemma. *Let n be a prime power ℓ^ν and put $\lambda = 1-\zeta$. Then the principal ideal (λ) in the ring $\mathcal{O}$ of integers of $\mathbb{Q}(\zeta)$ is a prime ideal of degree 1, and we have*

$$\ell\,\mathcal{O} = (\lambda)^d, \quad \textit{where} \quad d = \varphi(\ell^\nu) = [\mathbb{Q}(\zeta):\mathbb{Q}].$$

Furthermore, the basis $1, \zeta, \dots, \zeta^{d-1}$ of $\mathbb{Q}(\zeta)|\mathbb{Q}$ has the discriminant

$$d(1, \zeta, \dots, \zeta^{d-1}) = \pm\ell^s, \quad s = \ell^{\nu-1}(\nu\ell - \nu - 1).$$

Proof: The minimal polynomial of ζ over $\mathbb{Q}$ is the n-th cyclotomic polynomial

$$\phi_n(X) = (X^{\ell^\nu} - 1)/(X^{\ell^{\nu-1}} - 1) = X^{\ell^{\nu-1}(\ell-1)} + \cdots + X^{\ell^{\nu-1}} + 1.$$

Putting $X = 1$, we obtain the identity

$$\ell = \prod_{g \in (\mathbb{Z}/n\mathbb{Z})^*} (1 - \zeta^g).$$

But $1 - \zeta^g = \varepsilon_g(1-\zeta)$, for the algebraic integer $\varepsilon_g = \frac{1-\zeta^g}{1-\zeta} = 1 + \zeta + \cdots + \zeta^{g-1}$. If g' is an integer such that $gg' \equiv 1 \bmod \ell^\nu$, then

$$\frac{1-\zeta}{1-\zeta^g} = \frac{1-(\zeta^g)^{g'}}{1-\zeta^g} = 1 + \zeta^g + \cdots + (\zeta^g)^{g'-1}$$

is integral as well, i.e., ε_g is a unit. Consequently $\ell = \varepsilon(1-\zeta)^{\varphi(\ell^\nu)}$, with the unit $\varepsilon = \prod_g \varepsilon_g$, hence $\ell\,\mathcal{O} = (\lambda)^{\varphi(\ell^\nu)}$. Since $[\mathbb{Q}(\zeta):\mathbb{Q}] = \varphi(\ell^\nu)$, the fundamental identity (8.2) shows that (λ) is a prime ideal of degree 1.

Let $\zeta = \zeta_1, \dots, \zeta_d$ be the conjugates of ζ. Then the cyclotomic polynomial is $\phi_n(X) = \prod_{i=1}^d (X - \zeta_i)$ and (see § 2, p. 11)

$$\pm d(1, \zeta, \dots, \zeta^{d-1}) = \prod_{i \neq j} (\zeta_i - \zeta_j) = \prod_{i=1}^{d} \phi_n'(\zeta_i) = N_{\mathbb{Q}(\zeta)|\mathbb{Q}}\big(\phi_n'(\zeta)\big).$$

Differentiating the equation

$$(X^{\ell^{\nu-1}} - 1)\phi_n(X) = X^{\ell^\nu} - 1$$

and substituting ζ for X yields

$$(\xi - 1)\phi_n'(\zeta) = \ell^\nu \zeta^{-1},$$

with the primitive ℓ-th root of unity $\xi = \zeta^{\ell^{\nu-1}}$. But $N_{\mathbb{Q}(\xi)|\mathbb{Q}}(\xi - 1) = \pm\ell$, so that

$$N_{\mathbb{Q}(\zeta)|\mathbb{Q}}(\xi - 1) = N_{\mathbb{Q}(\xi)|\mathbb{Q}}(\xi - 1)^{\ell^{\nu-1}} = \pm\ell^{\ell^{\nu-1}}.$$

Observing that ζ^{-1} has norm ± 1 we obtain

$$d(1, \zeta, \ldots, \zeta^{d-1}) = \pm N_{\mathbb{Q}(\zeta)|\mathbb{Q}}\big(\phi_n'(\zeta)\big) = \pm\ell^{\nu\ell^{\nu-1}(\ell-1)-\ell^{\nu-1}} = \pm\ell^s$$

with $s = \ell^{\nu-1}(\nu\ell - \nu - 1)$. □

The ring of integers of $\mathbb{Q}(\zeta)$ is now determined, for arbitrary n, as follows.

(10.2) Proposition. *A $\mathbb{Z}$-basis of the ring $\mathcal{O}$ of integers of $\mathbb{Q}(\zeta)$ is given by $1, \zeta, \ldots, \zeta^{d-1}$, with $d = \varphi(n)$, in other words,*

$$\mathcal{O} = \mathbb{Z} + \mathbb{Z}\zeta + \cdots + \mathbb{Z}\zeta^{d-1} = \mathbb{Z}[\zeta].$$

Proof: We first prove the proposition in the case where n is a prime power ℓ^ν. Since $d(1, \zeta, \ldots, \zeta^{d-1}) = \pm\ell^s$, (2.9) gives us

$$\ell^s\mathcal{O} \subseteq \mathbb{Z}[\zeta] \subseteq \mathcal{O}.$$

Putting $\lambda = 1 - \zeta$, lemma (10.1) tells us that $\mathcal{O}/\lambda\mathcal{O} \cong \mathbb{Z}/\ell\mathbb{Z}$, so that $\mathcal{O} = \mathbb{Z} + \lambda\mathcal{O}$, and *a fortiori*

$$\lambda\mathcal{O} + \mathbb{Z}[\zeta] = \mathcal{O}.$$

Multiplying this by λ and substituting the result $\lambda\mathcal{O} = \lambda^2\mathcal{O} + \lambda\mathbb{Z}[\zeta]$, we obtain

$$\lambda^2\mathcal{O} + \mathbb{Z}[\zeta] = \mathcal{O}.$$

Iterating this procedure, we find

$$\lambda^t\mathcal{O} + \mathbb{Z}[\zeta] = \mathcal{O} \quad \text{for all} \quad t \geq 1.$$

For $t = s\,\varphi(\ell^\nu)$ this implies, in view of $\ell\,\mathcal{O} = \lambda^{\varphi(\ell^\nu)}\mathcal{O}$ (see (10.1)), that

$$\mathcal{O} = \lambda^t\mathcal{O} + \mathbb{Z}[\zeta] = \ell^s\mathcal{O} + \mathbb{Z}[\zeta] = \mathbb{Z}[\zeta].$$

In the general case, let $n = \ell_1^{\nu_1}\cdots\ell_r^{\nu_r}$. Then $\zeta_i = \zeta^{n/\ell_i^{\nu_i}}$ is a primitive $\ell_i^{\nu_i}$-th root of unity, and one has

$$\mathbb{Q}(\zeta) = \mathbb{Q}(\zeta_1)\cdots\mathbb{Q}(\zeta_r)$$

and $\mathbb{Q}(\zeta_1)\cdots\mathbb{Q}(\zeta_{i-1}) \cap \mathbb{Q}(\zeta_i) = \mathbb{Q}$. By what we have just seen, for each $i = 1, \dots, r$, the elements $1, \zeta_i, \dots, \zeta_i^{d_i-1}$, where $d_i = \varphi(\ell_i^{\nu_i})$, form an integral basis of $\mathbb{Q}(\zeta_i)|\mathbb{Q}$. Since the discriminants $d(1, \zeta_i, \dots, \zeta_i^{d_i-1}) = \pm\ell_i^{s_i}$ are pairwise relatively prime, we conclude successively from (2.11) that the elements $\zeta_1^{j_1}\cdots\zeta_r^{j_r}$, with $j_i = 0, \dots, d_i - 1$, form an integral basis of $\mathbb{Q}(\zeta)|\mathbb{Q}$. But each one of these elements is a power of ζ. Therefore every $\alpha \in \mathcal{O}$ may be written as a polynomial $\alpha = f(\zeta)$ with coefficients in $\mathbb{Z}$. Since ζ has degree $\varphi(n)$ over $\mathbb{Q}$, the degree of the polynomial $f(\zeta)$ may be reduced to $\varphi(n) - 1$. In this way one obtains a representation

$$\alpha = a_0 + a_1\zeta + \cdots + a_{\varphi(n)-1}\zeta^{\varphi(n)-1}.$$

Thus $1, \zeta, \dots, \zeta^{\varphi(n)-1}$ is indeed an integral basis. □

Knowing that $\mathbb{Z}[\zeta]$ is the ring of integers of the field $\mathbb{Q}(\zeta)$ we are now in a position to state explicitly the law of decomposition of prime numbers p into prime ideals of $\mathbb{Q}(\zeta)$. It is of the most beautiful simplicity.

(10.3) Proposition. *Let $n = \prod_p p^{\nu_p}$ be the prime factorization of n and, for every prime number p, let f_p be the smallest positive integer such that*

$$p^{f_p} \equiv 1 \bmod n/p^{\nu_p}.$$

Then one has in $\mathbb{Q}(\zeta)$ the factorization

$$p = (\mathfrak{p}_1\cdots\mathfrak{p}_r)^{\varphi(p^{\nu_p})},$$

where $\mathfrak{p}_1, \dots, \mathfrak{p}_r$ are distinct prime ideals, all of degree f_p.

Proof: Since $\mathcal{O} = \mathbb{Z}[\zeta]$, the conductor of $\mathbb{Z}[\zeta]$ equals 1, and we may apply proposition (8.3) to any prime number p. As a consequence, every p

decomposes into prime ideals in exactly the same way as the minimal polynomial $\phi_n(X)$ of ζ factors into irreducible polynomials mod p. All we have to show is therefore that

$$\phi_n(X) \equiv \big(p_1(X)\cdots p_r(X)\big)^{\varphi(p^{\nu_p})} \bmod p,$$

where $p_1(X), \dots, p_r(X)$ are distinct irreducible polynomials over $\mathbb{Z}/p\mathbb{Z}$ of degree f_p. In order to see this, put $n = p^{\nu_p}m$. As ξ_i, resp. η_j, varies over primitive roots of unity of order m, resp. p^{ν_p}, the products $\xi_i\eta_j$ vary precisely over the primitive n-th roots of unity, i.e., one has the decomposition over $\mathcal{O}$:

$$\phi_n(X) = \prod_{i,j}(X - \xi_i\eta_j).$$

Since $X^{p^{\nu_p}} - 1 \equiv (X-1)^{p^{\nu_p}} \bmod p$, one has $\eta_j \equiv 1 \bmod \mathfrak{p}$, for any prime ideal $\mathfrak{p} \mid p$. In other words,

$$\phi_n(X) \equiv \prod_i (X - \xi_i)^{\varphi(p^{\nu_p})} = \phi_m(X)^{\varphi(p^{\nu_p})} \bmod \mathfrak{p}.$$

This implies the congruence

$$\phi_n(X) \equiv \phi_m(X)^{\varphi(p^{\nu_p})} \bmod p.$$

Observing that f_p is the smallest positive integer such that $p^{f_p} \equiv 1 \bmod m$, it is obvious that this congruence reduces us to the case where $p \nmid n$, and hence $\varphi(p^{\nu_p}) = \varphi(1) = 1$.

As the characteristic p of $\mathcal{O}/\mathfrak{p}$ does not divide n, the polynomials $X^n - 1$ and nX^{n-1} have no common root in $\mathcal{O}/\mathfrak{p}$. So $X^n - 1 \bmod \mathfrak{p}$ has no multiple roots. We therefore see that passing to the quotient $\mathcal{O} \to \mathcal{O}/\mathfrak{p}$ maps the group μ_n of n-th roots of unity bijectively onto the group of n-th roots of unity of $\mathcal{O}/\mathfrak{p}$. In particular, the primitive n-th root of unity ζ modulo $\mathfrak{p}$ remains a primitive n-th root of unity. The smallest extension field of $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$ containing it is the field $\mathbb{F}_{p^{f_p}}$, because its multiplicative group $\mathbb{F}^*_{p^{f_p}}$ is cyclic of order $p^{f_p}-1$. $\mathbb{F}_{p^{f_p}}$ is therefore the field of decomposition of the reduced cyclotomic polynomial

$$\overline{\phi}_n(X) = \phi_n(X) \bmod p.$$

Being a divisor of $X^n - 1 \bmod p$, this polynomial has no multiple roots, and if

$$\overline{\phi}_n(X) = \overline{p}_1(X)\cdots\overline{p}_r(X)$$

is its factorization into irreducibles over $\mathbb{F}_p$, then every $\overline{p}_i(X)$ is the minimal polynomial of a primitive n-th root of unity $\overline{\xi} \in \mathbb{F}^*_{p^{f_p}}$. Its degree is therefore f_p. This proves the proposition. □

Let us emphasize two special cases of the above law of decomposition:

(10.4) Corollary. *A prime number p is ramified in $\mathbb{Q}(\zeta)$ if and only if*

$$n \equiv 0 \bmod p,$$

except in the case where $p = 2 = (4, n)$. A prime number $p \neq 2$ is totally split in $\mathbb{Q}(\zeta)$ if and only if

$$p \equiv 1 \bmod n.$$

The completeness of these results concerning the integral basis and the decomposition of primes in the field $\mathbb{Q}(\zeta)$ will not be matched by our study of the group of units and the ideal class group. The problems arising in this context are in fact among the most difficult problems posed by algebraic number theory. At the same time one encounters here plenty of astonishing laws which are the subject of a theory which has been developed only recently, **Iwasawa theory**.

The law of decomposition (10.3) in the cyclotomic field provides the proper explanation of Gauss's reciprocity law (8.6). This is based on the following

(10.5) Proposition. *Let ℓ and p be odd prime numbers, $\ell^* = (-1)^{\frac{\ell-1}{2}}\ell$, and ζ a primitive ℓ-th root of unity. Then one has:*

$$p \text{ is totally split in } \mathbb{Q}\big(\sqrt{\ell^*}\,\big) \iff p \text{ splits in } \mathbb{Q}(\zeta) \text{ into an even number of prime ideals.}$$

Proof: The little computation in § 8, p. 51 has shown us that $\ell^* = \tau^2$ with $\tau = \sum_{a \in (\mathbb{Z}/\ell\mathbb{Z})^*}\left(\frac{a}{\ell}\right)\zeta^a$, so that $\mathbb{Q}(\sqrt{\ell^*}\,) \subseteq \mathbb{Q}(\zeta)$. If p is totally split in $\mathbb{Q}(\sqrt{\ell^*}\,)$, say $p = \mathfrak{p}_1\mathfrak{p}_2$, then some automorphism σ of $\mathbb{Q}(\zeta)$ such that $\sigma\mathfrak{p}_1 = \mathfrak{p}_2$ transforms the set of all prime ideals lying above $\mathfrak{p}_1$ bijectively into the set of prime ideals above $\mathfrak{p}_2$. Therefore the number of prime ideals of $\mathbb{Q}(\zeta)$ above p is even. Now assume conversely that this is the case. Then the index of the decomposition group $G_\mathfrak{p}$, or in other words, the degree $[Z_\mathfrak{p} : \mathbb{Q}]$ of the decomposition field of a prime ideal $\mathfrak{p}$ of $\mathbb{Q}(\zeta)$ over p, is even. Since $G(\mathbb{Q}(\zeta)|\mathbb{Q})$ is cyclic, it follows that $\mathbb{Q}(\sqrt{\ell^*}\,) \subseteq Z_\mathfrak{p}$. The inertia degree of $\mathfrak{p} \cap Z_\mathfrak{p}$ over $\mathbb{Q}$ is 1 by (9.3), hence also the inertia degree of $\mathfrak{p} \cap \mathbb{Q}(\sqrt{\ell^*}\,)$. This implies that p is totally split in $\mathbb{Q}(\sqrt{\ell^*}\,)$. □

From this proposition we obtain the reciprocity law for two odd prime numbers ℓ and p,

$$\left(\frac{\ell}{p}\right)\left(\frac{p}{\ell}\right) = (-1)^{\frac{\ell-1}{2}\frac{p-1}{2}}$$

as follows. It suffices to show that

$$\left(\frac{\ell^*}{p}\right) = \left(\frac{p}{\ell}\right).$$

In fact, the completely elementary result $\left(\frac{-1}{p}\right) = (-1)^{\frac{p-1}{2}}$ (see § 8, p. 51) then gives

$$\left(\frac{p}{\ell}\right) = \left(\frac{\ell^*}{p}\right) = \left(\frac{-1}{p}\right)^{\frac{\ell-1}{2}} \left(\frac{\ell}{p}\right) = \left(\frac{\ell}{p}\right)(-1)^{\frac{p-1}{2}\frac{\ell-1}{2}}.$$

By (8.5) and (10.5), we know that $\left(\frac{\ell^*}{p}\right) = 1$ if and only if p decomposes in the field $\mathbb{Q}(\zeta)$ of ℓ-th roots of unity into an even number of prime ideals. By (10.3), this number is $r = \frac{\ell-1}{f}$, where f is the smallest positive integer such that $p^f \equiv 1 \bmod \ell$, i.e., r is even if and only if f is a divisor of $\frac{\ell-1}{2}$. But this is tantamount to the condition $p^{(\ell-1)/2} \equiv 1 \bmod \ell$. Since an element in the cyclic group $\mathbb{F}_\ell^*$ has an order dividing $\frac{\ell-1}{2}$ if and only if it belongs to $\mathbb{F}_\ell^{*2}$, the last congruence is equivalent to $\left(\frac{p}{\ell}\right) = 1$. So we do have $\left(\frac{\ell^*}{p}\right) = \left(\frac{p}{\ell}\right)$ as claimed.

Historically, Gauss's reciprocity law marked the beginning of algebraic number theory. It was discovered by *Euler*, but first proven by *Gauss*. The quest for similar laws concerning higher power residues, i.e., the congruences $x^n \equiv a \bmod p$, with $n > 2$, dominated number theory for a long time. Since this problem required working with the n-th cyclotomic field, *Kummer*'s attempts to solve it led to his seminal discovery of ideal theory. We have developed the basics of this theory in the preceding sections and tested it successfully in the example of cyclotomic fields. The further development of this theory has led to a totally comprehensive generalization of Gauss's reciprocity law, **Artin's reciprocity law**, one of the high points in the history of number theory, and of compelling charm. This law is the main theorem of **class field theory**, which we will develop in chapters IV–VI.

Exercise 1. (Dirichlet's Prime Number Theorem). For every natural number n there are infinitely many prime numbers $p \equiv 1 \bmod n$.

Hint: Assume there are only finitely many. Let P be their product and consider the n-th cyclotomic polynomial ϕ_n. Not all numbers $\phi_n(xnP)$, for $x \in \mathbb{Z}$, can equal 1. Let $p \mid \phi_n(xnP)$ for suitable x. Deduce a contradiction from this. (Dirichlet's prime number theorem is valid more generally for prime numbers $p \equiv a \bmod n$, provided $(a,n) = 1$ (see VII, (5.14) and VII, § 13)).

Exercise 2. For every finite abelian group A there exists a Galois extension $L\,|\,\mathbb{Q}$ with Galois group $G(L\,|\,\mathbb{Q}) \cong A$.

Hint: Use exercise 1.

Exercise 3. Every quadratic number field $\mathbb{Q}(\sqrt{d}\,)$ is contained in some cyclotomic field $\mathbb{Q}(\zeta_n)$, ζ_n a primitive n-th root of unity.

Exercise 4. Describe the quadratic subfields of $\mathbb{Q}(\zeta_n)\,|\,\mathbb{Q}$, in the case where n is odd.

Exercise 5. Show that $\mathbb{Q}(\sqrt{-1}\,)$, $\mathbb{Q}(\sqrt{2}\,)$, $\mathbb{Q}(\sqrt{-2}\,)$ are the quadratic subfields of $\mathbb{Q}(\zeta_n)\,|\,\mathbb{Q}$ for $n = 2^q$, $q \geq 3$.

§ 11. Localization

To "localize" means to form quotients, the most familiar case being the passage from an integral domain A to its field of fractions

$$K = \left\{ \frac{a}{b} \;\middle|\; a \in A,\ b \in A \smallsetminus \{0\} \right\}.$$

More generally, choosing instead of $A \smallsetminus \{0\}$ any nonempty $S \subseteq A \smallsetminus \{0\}$ which is closed under multiplication, one again obtains a ring structure on the set

$$AS^{-1} = \left\{ \frac{a}{s} \in K \;\middle|\; a \in A,\ s \in S \right\}.$$

The most important special case of such a multiplicative subset is the complement $S = A \smallsetminus \mathfrak{p}$ of a prime ideal $\mathfrak{p}$ of A. In this case one writes $A_\mathfrak{p}$ instead of AS^{-1}, and one calls the ring $A_\mathfrak{p}$ the **localization** of A at $\mathfrak{p}$. When dealing with problems that involve a single prime ideal $\mathfrak{p}$ of A at a time it is often expedient to replace A by the localization $A_\mathfrak{p}$. This procedure forgets everything that has nothing to do with $\mathfrak{p}$, and brings out more clearly all the properties concerning $\mathfrak{p}$. For instance, the mapping

$$\mathfrak{q} \longmapsto \mathfrak{q}A_\mathfrak{p}$$

gives a 1–1-correspondence between the prime ideals $\mathfrak{q} \subseteq \mathfrak{p}$ of A and the prime ideals of $A_\mathfrak{p}$. More generally for any multiplicative set S, one has the

(11.1) Proposition. *The mappings*

$$\mathfrak{q} \longmapsto \mathfrak{q}S^{-1} \quad \textit{and} \quad \mathfrak{Q} \longmapsto \mathfrak{Q} \cap A$$

are mutually inverse 1–1-correspondences between the prime ideals $\mathfrak{q} \subseteq A \smallsetminus S$ of A and the prime ideals $\mathfrak{Q}$ of AS^{-1}.

Proof: If $\mathfrak{q} \subseteq A \smallsetminus S$ is a prime ideal of A, then

$$\mathfrak{Q} = \mathfrak{q}S^{-1} = \left\{ \frac{q}{s} \mid q \in \mathfrak{q},\ s \in S \right\}$$

is a prime ideal of AS^{-1}. Indeed, in obvious notation, the relation $\frac{a}{s}\frac{a'}{s'} \in \mathfrak{Q}$, i.e., $\frac{aa'}{ss'} = \frac{q}{s''}$, implies that $s''aa' = qss' \in \mathfrak{q}$. Therefore $aa' \in \mathfrak{q}$, because $s'' \notin \mathfrak{q}$, and hence a or a' belong to $\mathfrak{q}$, which shows that $\frac{a}{s}$ or $\frac{a'}{s'}$ belong to $\mathfrak{Q}$. Furthermore one has

$$\mathfrak{q} = \mathfrak{Q} \cap A,$$

since $\frac{q}{s} = a \in \mathfrak{Q} \cap A$ implies $q = as \in \mathfrak{q}$, whence $a \in \mathfrak{q}$ because $s \notin \mathfrak{q}$.

Conversely, let $\mathfrak{Q}$ be an arbitrary prime ideal of AS^{-1}. Then $\mathfrak{q} = \mathfrak{Q} \cap A$ is obviously a prime ideal of A, and one has $\mathfrak{q} \subseteq A \smallsetminus S$. In fact, if $\mathfrak{q}$ were to contain an $s \in S$, then we would have $1 = s \cdot \frac{1}{s} \in \mathfrak{Q}$ because $\frac{1}{s} \in AS^{-1}$. Furthermore one has

$$\mathfrak{Q} = \mathfrak{q}S^{-1}.$$

For if $\frac{a}{s} \in \mathfrak{Q}$, then $a = \frac{a}{s} \cdot s \in \mathfrak{Q} \cap A = \mathfrak{q}$, hence $\frac{a}{s} = a\frac{1}{s} \in \mathfrak{q}S^{-1}$. The mappings $\mathfrak{q} \mapsto \mathfrak{q}S^{-1}$ and $\mathfrak{Q} \mapsto \mathfrak{Q} \cap A$ are therefore inverses of each other, which proves the proposition. $\square$

Usually S will be the complement of a union $\bigcup_{\mathfrak{p}\in X} \mathfrak{p}$ over a set X of prime ideals of A. In this case one writes

$$A(X) = \left\{ \frac{f}{g} \mid f, g \in A,\ g \not\equiv 0 \bmod \mathfrak{p} \text{ for } \mathfrak{p} \in X \right\}$$

instead of AS^{-1}. The prime ideals of $A(X)$ correspond by (11.1) 1–1 to the prime ideals of A which are contained in $\bigcup_{\mathfrak{p}\in X} \mathfrak{p}$, all the others are being eliminated when passing from A to $A(X)$. For instance, if X is finite or omits only finitely many prime ideals of A, then only the prime ideals from X survive in $A(X)$.

In the case that X consists of only one prime ideal $\mathfrak{p}$, the ring $A(X)$ is the localization

$$A_{\mathfrak{p}} = \left\{ \frac{f}{g} \mid f, g \in A,\ g \not\equiv 0 \bmod \mathfrak{p} \right\}$$

of A at $\mathfrak{p}$. Here we have the

(11.2) Corollary. *If $\mathfrak{p}$ is a prime ideal of A, then $A_{\mathfrak{p}}$ is a* **local ring**, *i.e., $A_{\mathfrak{p}}$ has a unique maximal ideal, namely $\mathfrak{m}_{\mathfrak{p}} = \mathfrak{p}A_{\mathfrak{p}}$. There is a canonical embedding*

$$A/\mathfrak{p} \hookrightarrow A_{\mathfrak{p}}/\mathfrak{m}_{\mathfrak{p}},$$

identifying $A_{\mathfrak{p}}/\mathfrak{m}_{\mathfrak{p}}$ with the field of fractions of $A/\mathfrak{p}$. In particular, if $\mathfrak{p}$ is a maximal ideal of A, then one has

$$A/\mathfrak{p}^n \cong A_{\mathfrak{p}}/\mathfrak{m}_{\mathfrak{p}}^n \quad \textit{for } n \geq 1.$$

Proof: Since the ideals of $A_\mathfrak{p}$ correspond 1–1 to the ideals of A contained in $\mathfrak{p}$, the ideal $\mathfrak{m}_\mathfrak{p} = \mathfrak{p}A_\mathfrak{p}$ is the unique maximal ideal. Let us consider the homomorphism

$$f : A/\mathfrak{p}^n \longrightarrow A_\mathfrak{p}/\mathfrak{m}_\mathfrak{p}^n, \quad a \bmod \mathfrak{p}^n \longmapsto a \bmod \mathfrak{m}_\mathfrak{p}^n.$$

For $n = 1$, f is injective because $\mathfrak{p} = \mathfrak{m}_\mathfrak{p} \cap A$. Hence $A_\mathfrak{p}/\mathfrak{m}_\mathfrak{p}A_\mathfrak{p}$ becomes the field of fractions of $A/\mathfrak{p}$. Let $\mathfrak{p}$ be maximal and $n \geq 1$. For every $s \in A \smallsetminus \mathfrak{p}$ one has $\mathfrak{p}^n + sA = A$, i.e., $\bar{s} = s \bmod \mathfrak{p}^n$ is a unit in $A/\mathfrak{p}^n$. For $n = 1$ this is clear from the maximality of $\mathfrak{p}$, and for $n \geq 1$ it follows by induction: $A = \mathfrak{p}^{n-1} + sA \Rightarrow \mathfrak{p} = \mathfrak{p}A = \mathfrak{p}(\mathfrak{p}^{n-1} + sA) \subsetneqq \mathfrak{p}^n + sA \Rightarrow \mathfrak{p}^n + sA = A$.

Injectivity of f: let $a \in A$ be such that $a \in \mathfrak{m}_\mathfrak{p}^n$, i.e., $a = b/s$ with $b \in \mathfrak{p}^n$, $s \notin \mathfrak{p}$. Then $as = b \in \mathfrak{p}^n$, so that $\bar{a}\,\bar{s} = 0$ in $A/\mathfrak{p}^n$, and hence $\bar{a} = 0$ in $A/\mathfrak{p}^n$.

Surjectivity of f: let $a/s \in A_\mathfrak{p}$, $a \in A$, $s \notin \mathfrak{p}$. Then by the above, there exists an $a' \in A$ such that $a \equiv a's \bmod \mathfrak{p}^n$. Therefore $a/s \equiv a' \bmod \mathfrak{p}^n A_\mathfrak{p}$, i.e., $a/s \bmod \mathfrak{m}_\mathfrak{p}^n$ lies in the image of f. □

In a local ring with maximal ideal $\mathfrak{m}$, every element $a \notin \mathfrak{m}$ is a unit. Indeed, since the principal ideal (a) is not contained in any other maximal ideal, it has to be the whole ring. So we have

$$A^* = A \smallsetminus \mathfrak{m}.$$

The simplest local rings, except for fields, are **discrete valuation rings**.

(11.3) Definition. *A discrete valuation ring is a principal ideal domain $\mathcal{O}$ with a unique maximal ideal $\mathfrak{p} \neq 0$.*

The maximal ideal is of the form $\mathfrak{p} = (\pi) = \pi\mathcal{O}$, for some prime element π. Since every element not contained in $\mathfrak{p}$ is a unit, it follows that, up to associated elements, π is the only prime element of $\mathcal{O}$. Every nonzero element of $\mathcal{O}$ may therefore be written as $\varepsilon\pi^n$, for some $\varepsilon \in \mathcal{O}^*$, and $n \geq 0$. More generally, every element $a \neq 0$ of the field of fractions K may be uniquely written as

$$a = \varepsilon\pi^n, \quad \varepsilon \in \mathcal{O}^*, \quad n \in \mathbb{Z}.$$

The exponent n is called the **valuation** of a. It is denoted $v(a)$, and it is obviously characterized by the equation

$$(a) = \mathfrak{p}^{v(a)}.$$

The valuation is a function

$$v : K^* \longrightarrow \mathbb{Z}.$$

Extending it to K by the convention $v(0) = \infty$, a simple calculation shows that it satisfies the conditions

$$v(ab) = v(a) + v(b), \quad v(a+b) \geq \min\{ v(a), v(b)\}.$$

This innocuous looking function gives rise to a theory which will occupy all of the next chapter.

The discrete valuation rings arise as localizations of Dedekind domains. This is a consequence of the

(11.4) Proposition. *If $\mathcal{O}$ is a Dedekind domain, and $S \subseteq \mathcal{O} \smallsetminus \{0\}$ is a multiplicative subset, then $\mathcal{O} S^{-1}$ is also a Dedekind domain.*

Proof: Let $\mathfrak{A}$ be an ideal of $\mathcal{O}S^{-1}$ and $\mathfrak{a} = \mathfrak{A} \cap \mathcal{O}$. Then $\mathfrak{A} = \mathfrak{a} S^{-1}$, because if $\frac{a}{s} \in \mathfrak{A}$, $a \in \mathcal{O}$ and $s \in S$, then one has $a = s \cdot \frac{a}{s} \in \mathfrak{A} \cap \mathcal{O} = \mathfrak{a}$, so that $\frac{a}{s} = a \cdot \frac{1}{s} \in \mathfrak{a}S^{-1}$. As $\mathfrak{a}$ is finitely generated, so is $\mathfrak{A}$, i.e., $\mathcal{O}S^{-1}$ is noetherian. It follows from (11.1) that every prime ideal of $\mathcal{O}S^{-1}$ is maximal, because this holds in $\mathcal{O}$. Finally, $\mathcal{O}S^{-1}$ is integrally closed, for if $x \in K$ satisfies the equation

$$x^n + \frac{a_1}{s_1}x^{n-1} + \cdots + \frac{a_n}{s_n} = 0$$

with coefficients $\frac{a_i}{s_i} \in \mathcal{O}S^{-1}$, then multiplying it with the n-th power of $s = s_1 \ldots s_n$ shows that sx is integral over $\mathcal{O}$, whence $sx \in \mathcal{O}$ and therefore $x \in \mathcal{O}S^{-1}$. This shows that $\mathcal{O}S^{-1}$ is a Dedekind domain. $\square$

(11.5) Proposition. *Let $\mathcal{O}$ be a noetherian integral domain. $\mathcal{O}$ is a Dedekind domain if and only if, for all prime ideals $\mathfrak{p} \neq 0$, the localizations $\mathcal{O}_\mathfrak{p}$ are discrete valuation rings.*

Proof: If $\mathcal{O}$ is a Dedekind domain, then so are the localizations $\mathcal{O}_\mathfrak{p}$. The maximal ideal $\mathfrak{m} = \mathfrak{p}\mathcal{O}_\mathfrak{p}$ is the only nonzero prime ideal of $\mathcal{O}_\mathfrak{p}$. Therefore, choosing any $\pi \in \mathfrak{m} \smallsetminus \mathfrak{m}^2$, one necessarily finds $(\pi) = \mathfrak{m}$, and furthermore $\mathfrak{m}^n = (\pi^n)$. Thus $\mathcal{O}_\mathfrak{p}$ is a principal ideal domain, and hence a discrete valuation ring.

Letting $\mathfrak{p}$ vary over all prime ideals $\neq 0$ of $\mathcal{O}$, we find in any case that

$$\mathcal{O} = \bigcap_{\mathfrak{p}} \mathcal{O}_\mathfrak{p}.$$

For if $\frac{a}{b} \in \bigcap_\mathfrak{p} \mathcal{O}_\mathfrak{p}$, with $a, b \in \mathcal{O}$, then

$$\mathfrak{a} = \{ x \in \mathcal{O} \mid xa \in b\mathcal{O} \}$$

is an ideal which cannot be contained in any prime ideal of $\mathcal{O}$. In fact, for any $\mathfrak{p}$, we may write $\frac{a}{b} = \frac{c}{s}$ with $c \in \mathcal{O}$, $s \notin \mathfrak{p}$, so that $sa = bc$, hence $s \in \mathfrak{a} \smallsetminus \mathfrak{p}$. As $\mathfrak{a}$ is not contained in any maximal ideal, it follows that $\mathfrak{a} = \mathcal{O}$, hence $a = 1 \cdot a \in b\,\mathcal{O}$, i.e., $\frac{a}{b} \in \mathcal{O}$.

Suppose now that the $\mathcal{O}_\mathfrak{p}$ are discrete valuation rings. Being principal ideal domains, they are integrally closed (see § 2), so $\mathcal{O} = \bigcap_\mathfrak{p} \mathcal{O}_\mathfrak{p}$ is also integrally closed. Finally, from (11.1) it follows that every prime ideal $\mathfrak{p} \neq 0$ of $\mathcal{O}$ is maximal because this is so in $\mathcal{O}_\mathfrak{p}$. Therefore $\mathcal{O}$ is a Dedekind domain. □

For a Dedekind domain $\mathcal{O}$, we have for each prime ideal $\mathfrak{p} \neq 0$ the discrete valuation ring $\mathcal{O}_\mathfrak{p}$ and the corresponding valuation

$$v_\mathfrak{p} : K^* \longrightarrow \mathbb{Z}$$

of the field of fractions. The significance of these valuations lies in their relation to the prime ideal factorization. If $x \in K^*$ and

$$(x) = \prod_\mathfrak{p} \mathfrak{p}^{\nu_\mathfrak{p}}$$

is the prime factorization of the principal ideal (x), then, for each $\mathfrak{p}$, one has

$$\nu_\mathfrak{p} = v_\mathfrak{p}(x).$$

In fact, for a fixed prime ideal $\mathfrak{q} \neq 0$ of $\mathcal{O}$, the first equation above implies (because $\mathfrak{p}\,\mathcal{O}_\mathfrak{q} = \mathcal{O}_\mathfrak{q}$ for $\mathfrak{p} \neq \mathfrak{q}$) that

$$x\mathcal{O}_\mathfrak{q} = \Big(\prod_\mathfrak{p} \mathfrak{p}^{\nu_\mathfrak{p}}\Big)\mathcal{O}_\mathfrak{q} = \mathfrak{q}^{\nu_\mathfrak{q}}\mathcal{O}_\mathfrak{q} = \mathfrak{m}_\mathfrak{q}^{\nu_\mathfrak{q}}.$$

Hence indeed $v_\mathfrak{q}(x) = \nu_\mathfrak{q}$. In view of this relation, the valuations $v_\mathfrak{p}$ are also called **exponential valuations**.

The reader should check that the localization of the ring $\mathbb{Z}$ at the prime ideal $(p) = p\mathbb{Z}$ is given by

$$\mathbb{Z}_{(p)} = \Big\{ \frac{a}{b} \;\Big|\; a, b \in \mathbb{Z},\ p \nmid b \Big\}.$$

The maximal ideal $p\mathbb{Z}_{(p)}$ consists of all fractions a/b satisfying $p \,|\, a$, $p \nmid b$, and the group of units consists of all fractions a/b satisfying $p \nmid ab$. The valuation associated to $\mathbb{Z}_{(p)}$,

$$v_p : \mathbb{Q} \longrightarrow \mathbb{Z} \cup \{\infty\},$$

is called the ***p*-adic valuation** of $\mathbb{Q}$. The valuation $v_p(x)$ of an element $x \in \mathbb{Q}^*$ is given by

$$v_p(x) = \nu,$$

where $x = p^\nu a/b$ with integers a, b relatively prime to p.

To end this section, we now want to compare a Dedekind domain $\mathcal{O}$ to the ring

$$\mathcal{O}(X) = \left\{ \frac{f}{g} \mid f, g \in \mathcal{O},\ g \not\equiv 0 \bmod \mathfrak{p} \ \text{ for } \ \mathfrak{p} \in X \right\},$$

where X is a set of prime ideals $\neq 0$ of $\mathcal{O}$ which contains almost all prime ideals of $\mathcal{O}$. By (11.1), the prime ideals $\neq 0$ of $\mathcal{O}(X)$ are given as $\mathfrak{p}_X = \mathfrak{p}\mathcal{O}(X)$, for $\mathfrak{p} \in X$, and it is easily checked that $\mathcal{O}$ and $\mathcal{O}(X)$ have the same localizations

$$\mathcal{O}_{\mathfrak{p}} = \mathcal{O}(X)_{\mathfrak{p}_X}.$$

We denote by $Cl(\mathcal{O})$, resp. $Cl(\mathcal{O}(X))$, the ideal class groups of $\mathcal{O}$, resp. $\mathcal{O}(X)$. They, as well as the groups of units $\mathcal{O}^*$ and $\mathcal{O}(X)^*$, are related by the following

(11.6) Proposition. *There is a canonical exact sequence*

$$1 \longrightarrow \mathcal{O}^* \longrightarrow \mathcal{O}(X)^* \longrightarrow \bigoplus_{\mathfrak{p} \notin X} K^*/\mathcal{O}_{\mathfrak{p}}^* \longrightarrow Cl(\mathcal{O}) \longrightarrow Cl(\mathcal{O}(X)) \longrightarrow 1,$$

and one has $K^*/\mathcal{O}_{\mathfrak{p}}^* \cong \mathbb{Z}$.

Proof: The first arrow is inclusion and the second one is induced by the inclusion $\mathcal{O}(X)^* \to K^*$, followed by the projections $K^* \to K^*/\mathcal{O}_{\mathfrak{p}}^*$. If $a \in \mathcal{O}(X)^*$ belongs to the kernel, then $a \in \mathcal{O}_{\mathfrak{p}}$ for $\mathfrak{p} \notin X$, and also for $\mathfrak{p} \in X$ because $\mathcal{O}_{\mathfrak{p}} = \mathcal{O}(X)_{\mathfrak{p}_X}$, hence $a \in \bigcap_{\mathfrak{p}} \mathcal{O}_{\mathfrak{p}}^* = \mathcal{O}^*$ (see the argument in the proof of (11.5)). This shows the exactness at $\mathcal{O}(X)^*$. The arrow

$$\bigoplus_{\mathfrak{p} \notin X} K^*/\mathcal{O}_{\mathfrak{p}}^* \longrightarrow Cl(\mathcal{O})$$

is induced by mapping

$$\bigoplus_{\mathfrak{p} \notin X} \alpha_{\mathfrak{p}} \bmod \mathcal{O}_{\mathfrak{p}}^* \longmapsto \prod_{\mathfrak{p} \notin X} \mathfrak{p}^{v_{\mathfrak{p}}(\alpha_{\mathfrak{p}})},$$

where $v_{\mathfrak{p}} : K^* \to \mathbb{Z}$ is the exponential valuation of K associated to $\mathcal{O}_{\mathfrak{p}}$. Let $\bigoplus_{\mathfrak{p} \notin X} \alpha_{\mathfrak{p}} \bmod \mathcal{O}_{\mathfrak{p}}^*$ be an element in the kernel, i.e.,

$$\prod_{\mathfrak{p} \notin X} \mathfrak{p}^{v_{\mathfrak{p}}(\alpha_{\mathfrak{p}})} = (\alpha) = \prod_{\mathfrak{p}} \mathfrak{p}^{v_{\mathfrak{p}}(\alpha)},$$

for some $\alpha \in K^*$. Because of unique prime factorization, this means that $v_{\mathfrak{p}}(\alpha) = 0$ for $\mathfrak{p} \in X$, and $v_{\mathfrak{p}}(\alpha_{\mathfrak{p}}) = v_{\mathfrak{p}}(\alpha)$ for $\mathfrak{p} \notin X$. It follows that $\alpha \in \bigcap_{\mathfrak{p} \in X} \mathcal{O}_{\mathfrak{p}}^* = \mathcal{O}(X)^*$ and $\alpha \equiv \alpha_{\mathfrak{p}} \bmod \mathcal{O}_{\mathfrak{p}}^*$. This shows exactness in the middle. The arrow

$$Cl(\mathcal{O}) \longrightarrow Cl(\mathcal{O}(X))$$

comes from mapping $\mathfrak{a} \mapsto \mathfrak{a}\mathcal{O}(X)$. The classes of prime ideals $\mathfrak{p} \in X$ are mapped onto the classes of prime ideals of $\mathcal{O}(X)$. Since $Cl(\mathcal{O}(X))$ is generated by these classes, the arrow is surjective. For $\mathfrak{p} \notin X$ we have $\mathfrak{p}\mathcal{O}(X) = (1)$, and this means that the kernel consists of the classes of the ideals $\prod_{\mathfrak{p}\notin X} \mathfrak{p}^{\nu_\mathfrak{p}}$. This, however, is visibly the image of the preceding arrow. Therefore the whole sequence is exact. Finally, the valuation $v_\mathfrak{p} : K^* \to \mathbb{Z}$ produces the isomorphism $K^*/\mathcal{O}_\mathfrak{p}^* \cong \mathbb{Z}$. □

For the ring of integers $\mathcal{O}_K$ of an algebraic number field K, the proposition yields the following results. Let S denote a finite set of prime ideals of $\mathcal{O}_K$ (not any more a multiplicative subset), and let X be the set of all prime ideals that do not belong to S. We put

$$\mathcal{O}_K^S = \mathcal{O}_K(X).$$

The units of this ring are called the **S-units**, and the group $Cl_K^S = Cl(\mathcal{O}_K^S)$ the **S-class group** of K.

(11.7) Corollary. *For the group $K^S = (\mathcal{O}_K^S)^*$ of S-units of K there is an isomorphism*

$$K^S \cong \mu(K) \times \mathbb{Z}^{\#S+r+s-1},$$

where r and s are defined as in § 5, p. 30.

Proof: The torsion subgroup of K^S is the group $\mu(K)$ of roots of unity in K. Since $Cl(\mathcal{O})$ is finite, we obtain the following identities from the exact sequence (11.6) and from (7.4):

$$\operatorname{rank}(K^S) = \operatorname{rank}(\mathcal{O}_K^*) + \operatorname{rank}\Big(\bigoplus_{\mathfrak{p}\in S} \mathbb{Z}\Big) = \#S + r + s - 1.$$

This proves the corollary. □

(11.8) Corollary. *The S-class group $Cl_K^S = Cl(\mathcal{O}_K^S)$ is finite.*

Exercise 1. Let A be an arbitrary ring, not necessarily an integral domain, let M be an A-module and S a multiplicatively closed subset of A such that $0 \notin S$. In $M \times S$ consider the equivalence relation

$$(m,s) \sim (m',s') \iff \exists s'' \in S \text{ such that } s''(s'm - sm') = 0.$$

Show that the set M_S of equivalence classes $\overline{(m,s)}$ forms an A-module, and that $M \to M_S$, $a \mapsto \overline{(a,1)}$, is a homomorphism. In particular, A_S is a ring. It is called the **localization** of A with respect to S.

Exercise 2. Show that, in the above situation, the prime ideals of A_S correspond 1–1 to the prime ideals of A which are disjoint from S. If $\mathfrak{p} \subseteq A$ and $\mathfrak{p}_S \subseteq A_S$ correspond in this way, then $A_S/\mathfrak{p}_S$ is the localization of $A/\mathfrak{p}$ with respect to the image of S.

Exercise 3. Let $f : M \to N$ be a homomorphism of A-modules. Then the following conditions are equivalent:

(i) f is injective (surjective).

(ii) $f_{\mathfrak{p}} : M_{\mathfrak{p}} \to N_{\mathfrak{p}}$ is injective (surjective) for every prime ideal $\mathfrak{p}$.

(iii) $f_{\mathfrak{m}} : M_{\mathfrak{m}} \to N_{\mathfrak{m}}$ is injective (surjective) for every maximal ideal $\mathfrak{m}$.

Exercise 4. Let S and T be two multiplicative subsets of A, and T^* the image of T in A_S. Then one has $A_{ST} \cong (A_S)_{T^*}$.

Exercise 5. Let $f : A \to B$ be a homomorphism of rings and S a multiplicatively closed subset such that $f(S) \subseteq B^*$. Then f induces a homomorphism $A_S \to B$.

Exercise 6. Let A be an integral domain. If the localization A_S is integral over A, then $A_S = A$.

Exercise 7 (Nakayama's Lemma). Let A be a local ring with maximal ideal $\mathfrak{m}$, let M be an A-module and $N \subseteq M$ a submodule such that M/N is finitely generated. Then one has the implication:

$$M = N + \mathfrak{m}M \implies M = N .$$

§ 12. Orders

The ring $\mathcal{O}_K$ of integers of an algebraic number field K is our chief interest because of its excellent property of being a Dedekind domain. Due to important theoretical as well as practical circumstances, however, one is pushed to devise a theory of greater generality which comprises also the theory of rings of algebraic integers which, like the ring

$$\mathcal{O} = \mathbb{Z} + \mathbb{Z}\sqrt{5} \subseteq \mathbb{Q}(\sqrt{5}),$$

are not necessarily integrally closed. These rings are the so-called **orders**.

(12.1) Definition. *Let $K\,|\,\mathbb{Q}$ be an algebraic number field of degree n. An order of K is a subring $\mathcal{O}$ of $\mathcal{O}_K$ which contains an integral basis of length n. The ring $\mathcal{O}_K$ is called the* **maximal order** *of K.*

In concrete terms, orders are obtained as rings of the form

$$\mathcal{O} = \mathbb{Z}[\alpha_1, \dots, \alpha_r],$$

where $\alpha_1, \dots, \alpha_r$ are integers such that $K = \mathbb{Q}(\alpha_1, \dots, \alpha_r)$. Being a submodule of the free $\mathbb{Z}$-module $\mathcal{O}_K$, $\mathcal{O}$ does of course admit a $\mathbb{Z}$-basis which, as $\mathbb{Q}\mathcal{O} = K$, has to be at the same time a basis of $K|\mathbb{Q}$, and therefore has length n. Orders arise often as rings of multipliers, and as such have their practical applications. For instance, if $\alpha_1, \dots, \alpha_n$ is any basis of $K|\mathbb{Q}$ and $M = \mathbb{Z}\alpha_1 + \dots + \mathbb{Z}\alpha_n$, then

$$\mathcal{O}_M = \{\alpha \in K \mid \alpha M \subseteq M\}$$

is an order. The theoretical significance of orders, however, lies in the fact that they admit "singularities", which are excluded as long as only Dedekind domains with their "regular" localizations $\mathcal{O}_\mathfrak{p}$ are considered. We will explain what this means in the next section.

In the preceding section we studied the localizations of a Dedekind domain $\mathcal{O}_K$. They are extension rings of $\mathcal{O}_K$ which are integrally closed, yet no longer integral over $\mathbb{Z}$. Now we study orders. They are subrings of $\mathcal{O}_K$ which are integral over $\mathbb{Z}$, yet no longer integrally closed. As a common generalization of both types of rings let us consider for now all **one-dimensional noetherian integral domains**. These are the noetherian integral domains in which every prime ideal $\mathfrak{p} \neq 0$ is a maximal ideal. The term "one-dimensional" refers to the general definition of the **Krull dimension** of a ring as being the maximal length d of a chain of prime ideals $\mathfrak{p}_0 \subsetneqq \mathfrak{p}_1 \subsetneqq \cdots \subsetneqq \mathfrak{p}_d$.

(12.2) Proposition. *An order $\mathcal{O}$ of K is a one-dimensional noetherian integral domain.*

Proof: Since $\mathcal{O}$ is a finitely generated $\mathbb{Z}$-module of rank $n = [K : \mathbb{Q}]$, every ideal $\mathfrak{a}$ is also a finitely generated $\mathbb{Z}$-module, and *a fortiori* a finitely generated $\mathcal{O}$-module. This shows that $\mathcal{O}$ is noetherian. If $\mathfrak{p} \neq 0$ is a prime ideal and $a \in \mathfrak{p} \cap \mathbb{Z}$, $a \neq 0$, then $a\mathcal{O} \subseteq \mathfrak{p} \subseteq \mathcal{O}$, i.e., $\mathfrak{p}$ and $\mathcal{O}$ have the same rank n. Therefore $\mathcal{O}/\mathfrak{p}$ is a finite integral domain, hence a field, and thus $\mathfrak{p}$ is a maximal ideal. □

In what follows, we always let $\mathcal{O}$ be a one-dimensional noetherian integral domain and K its field of fractions. We set out by proving the following stronger version of the Chinese remainder theorem.

(12.3) Proposition. *If $\mathfrak{a} \neq 0$ is an ideal of $\mathcal{O}$, then*

$$\mathcal{O}/\mathfrak{a} \cong \bigoplus_{\mathfrak{p}} \mathcal{O}_{\mathfrak{p}}/\mathfrak{a}\mathcal{O}_{\mathfrak{p}} = \bigoplus_{\mathfrak{p}\supseteq\mathfrak{a}} \mathcal{O}_{\mathfrak{p}}/\mathfrak{a}\mathcal{O}_{\mathfrak{p}}.$$

Proof: Let $\tilde{\mathfrak{a}}_{\mathfrak{p}} = \mathcal{O} \cap \mathfrak{a}\mathcal{O}_{\mathfrak{p}}$. For almost all $\mathfrak{p}$ one has $\mathfrak{p} \not\supseteq \mathfrak{a}$ and therefore $\mathfrak{a}\mathcal{O}_{\mathfrak{p}} = \mathcal{O}_{\mathfrak{p}}$, hence $\tilde{\mathfrak{a}}_{\mathfrak{p}} = \mathcal{O}$. Furthermore, one has $\mathfrak{a} = \bigcap_{\mathfrak{p}} \tilde{\mathfrak{a}}_{\mathfrak{p}} = \bigcap_{\mathfrak{p}\supseteq\mathfrak{a}} \tilde{\mathfrak{a}}_{\mathfrak{p}}$. Indeed, for any $a \in \bigcap_{\mathfrak{p}} \tilde{\mathfrak{a}}_{\mathfrak{p}}$, the ideal $\mathfrak{b} = \{x \in \mathcal{O} \mid xa \in \mathfrak{a}\}$ does not belong to any of the maximal ideals $\mathfrak{p}$ (in fact, one has $s_{\mathfrak{p}}a \in \mathfrak{a}$ for any $s_{\mathfrak{p}} \notin \mathfrak{p}$), consequently, $\mathfrak{b} = \mathcal{O}$, i.e., $a = 1 \cdot a \in \mathfrak{a}$, as claimed. (11.1) implies that, if $\mathfrak{p} \supseteq \mathfrak{a}$, then $\mathfrak{p}$ is the only prime ideal containing $\tilde{\mathfrak{a}}_{\mathfrak{p}}$. Therefore, given two distinct prime ideals $\mathfrak{p}$ and $\mathfrak{q}$ of $\mathcal{O}$, the ideal $\tilde{\mathfrak{a}}_{\mathfrak{p}}+\tilde{\mathfrak{a}}_{\mathfrak{q}}$ cannot be contained in any maximal ideal, whence $\tilde{\mathfrak{a}}_{\mathfrak{p}} + \tilde{\mathfrak{a}}_{\mathfrak{q}} = \mathcal{O}$. The Chinese remainder theorem (3.6) now gives the isomorphism

$$\mathcal{O}/\mathfrak{a} \cong \bigoplus_{\mathfrak{p}\supseteq\mathfrak{a}} \mathcal{O}/\tilde{\mathfrak{a}}_{\mathfrak{p}},$$

and we have $\mathcal{O}/\tilde{\mathfrak{a}}_{\mathfrak{p}} = \mathcal{O}_{\mathfrak{p}}/\mathfrak{a}\mathcal{O}_{\mathfrak{p}}$, because $\bar{\mathfrak{p}} = \mathfrak{p} \bmod \tilde{\mathfrak{a}}_{\mathfrak{p}}$ is the only maximal ideal of $\mathcal{O}/\tilde{\mathfrak{a}}_{\mathfrak{p}}$. □

For the ring $\mathcal{O}$, the fractional ideals of $\mathcal{O}$, in other words, the finitely generated nonzero $\mathcal{O}$-submodules of the field of fractions K, no longer form a group – unless $\mathcal{O}$ happens to be Dedekind. The way out is to restrict attention to the **invertible ideals**, i.e., to those fractional ideals $\mathfrak{a}$ of $\mathcal{O}$ for which there exists a fractional ideal $\mathfrak{b}$ such that

$$\mathfrak{a}\mathfrak{b} = \mathcal{O}.$$

These form an abelian group, for trivial reasons. The inverse of $\mathfrak{a}$ is still the fractional ideal

$$\mathfrak{a}^{-1} = \left\{x \in K \mid x\mathfrak{a} \subseteq \mathcal{O}\right\},$$

because it is the biggest ideal such that $\mathfrak{a}\mathfrak{a}^{-1} \subseteq \mathcal{O}$. The invertible ideals of $\mathcal{O}$ may be characterized as those fractional ideals which are "locally" principal:

(12.4) Proposition. *A fractional ideal $\mathfrak{a}$ of $\mathcal{O}$ is invertible if and only if, for every prime ideal $\mathfrak{p} \neq 0$,*

$$\mathfrak{a}_{\mathfrak{p}} = \mathfrak{a}\mathcal{O}_{\mathfrak{p}}$$

is a fractional principal ideal of $\mathcal{O}_{\mathfrak{p}}$.

Proof: Let $\mathfrak{a}$ be an invertible ideal and $\mathfrak{a}\mathfrak{b} = \mathcal{O}$. Then $1 = \sum_{i=1}^{r} a_i b_i$ with $a_i \in \mathfrak{a}$, $b_i \in \mathfrak{b}$, and not all $a_i b_i \in \mathcal{O}_\mathfrak{p}$ can lie in the maximal ideal $\mathfrak{p}\mathcal{O}_\mathfrak{p}$. Suppose $a_1 b_1$ is a unit in $\mathcal{O}_\mathfrak{p}$. Then $\mathfrak{a}_\mathfrak{p} = a_1 \mathcal{O}_\mathfrak{p}$ because, for $x \in \mathfrak{a}_\mathfrak{p}$, $x b_1 \in \mathfrak{a}_\mathfrak{p}\mathfrak{b} = \mathcal{O}_\mathfrak{p}$, hence $x = x b_1 (b_1 a_1)^{-1} a_1 \in a_1 \mathcal{O}_\mathfrak{p}$.

Conversely, assume $\mathfrak{a}_\mathfrak{p} = \mathfrak{a}\mathcal{O}_\mathfrak{p}$ is a principal ideal $a_\mathfrak{p}\mathcal{O}_\mathfrak{p}$, $a_\mathfrak{p} \in K^*$, for every $\mathfrak{p}$. Then we may and do assume that $a_\mathfrak{p} \in \mathfrak{a}$. We claim that the fractional ideal $\mathfrak{a}^{-1} = \{x \in K \mid x\mathfrak{a} \subseteq \mathcal{O}\}$ is an inverse for $\mathfrak{a}$. If this were not the case, then we would have a maximal ideal $\mathfrak{p}$ such that $\mathfrak{a}\mathfrak{a}^{-1} \subseteq \mathfrak{p} \subset \mathcal{O}$. Let $a_1, \dots, a_n$ be generators of $\mathfrak{a}$. As $a_i \in a_\mathfrak{p}\, \mathcal{O}_\mathfrak{p}$, we may write $a_i = a_\mathfrak{p} \frac{b_i}{s_i}$, with $b_i \in \mathcal{O}$, $s_i \in \mathcal{O} \smallsetminus \mathfrak{p}$. Then $s_i a_i \in a_\mathfrak{p}\mathcal{O}$. Putting $s = s_1 \cdots s_n$, we have $s a_i \in a_\mathfrak{p}\, \mathcal{O}$ for $i = 1, \dots, n$, hence $s a_\mathfrak{p}^{-1}\mathfrak{a} \subseteq \mathcal{O}$ and therefore $s a_\mathfrak{p}^{-1} \in \mathfrak{a}^{-1}$. Consequently, $s = s a_\mathfrak{p}^{-1} a_\mathfrak{p} \in \mathfrak{a}^{-1}\mathfrak{a} \subseteq \mathfrak{p}$, a contradiction. □

We denote the group of invertible ideals of $\mathcal{O}$ by $J(\mathcal{O})$. It contains the group $P(\mathcal{O})$ of fractional principal ideals $a\mathcal{O}$, $a \in K^*$.

(12.5) Definition. *The quotient group*

$$Pic(\mathcal{O}) = J(\mathcal{O})/P(\mathcal{O})$$

is called the **Picard group** *of the ring* $\mathcal{O}$.

In the case where $\mathcal{O}$ is a Dedekind domain, the Picard group is of course nothing but the ideal class group Cl_K. In general, we have the following description for $J(\mathcal{O})$ and $Pic(\mathcal{O})$.

(12.6) Proposition. *The correspondence* $\mathfrak{a} \mapsto (\mathfrak{a}_\mathfrak{p}) = (\mathfrak{a}\mathcal{O}_\mathfrak{p})$ *yields an isomorphism*

$$J(\mathcal{O}) \cong \bigoplus_{\mathfrak{p}} P(\mathcal{O}_\mathfrak{p}).$$

Identifying the subgroup $P(\mathcal{O})$ *with its image in the direct sum one gets*

$$Pic(\mathcal{O}) \cong \Big(\bigoplus_{\mathfrak{p}} P(\mathcal{O}_\mathfrak{p})\Big)/P(\mathcal{O}).$$

Proof: For every $\mathfrak{a} \in J(\mathcal{O})$, $\mathfrak{a}_\mathfrak{p} = \mathfrak{a}\mathcal{O}_\mathfrak{p}$ is a principal ideal by (12.4), and we have $\mathfrak{a}_\mathfrak{p} = \mathcal{O}_\mathfrak{p}$ for almost all $\mathfrak{p}$ because $\mathfrak{a}$ lies in only finitely many maximal ideals $\mathfrak{p}$. We therefore obtain a homomorphism

$$J(\mathcal{O}) \longrightarrow \bigoplus_{\mathfrak{p}} P(\mathcal{O}_\mathfrak{p}), \quad \mathfrak{a} \longmapsto (\mathfrak{a}_\mathfrak{p}).$$

It is injective, for if $\mathfrak{a}_\mathfrak{p} = \mathcal{O}_\mathfrak{p}$ for all $\mathfrak{p}$, then $\mathfrak{a} \subseteq \bigcap_\mathfrak{p} \mathcal{O}_\mathfrak{p} = \mathcal{O}$ (see the proof of (11.5)), and one has to have $\mathfrak{a} = \mathcal{O}$ because otherwise there would exist a maximal ideal $\mathfrak{p}$ such that $\mathfrak{a} \subseteq \mathfrak{p} \subset \mathcal{O}$, i.e., $\mathfrak{a}_\mathfrak{p} \subseteq \mathfrak{p}\mathcal{O}_\mathfrak{p} \neq \mathcal{O}_\mathfrak{p}$. In order to prove surjectivity, let $(a_\mathfrak{p}\mathcal{O}_\mathfrak{p}) \in \bigoplus_\mathfrak{p} P(\mathcal{O}_\mathfrak{p})$ be given. Then the $\mathcal{O}$-submodule

$$\mathfrak{a} = \bigcap_\mathfrak{p} a_\mathfrak{p}\mathcal{O}_\mathfrak{p}$$

of K is a fractional ideal. Indeed, since $a_\mathfrak{p}\mathcal{O}_\mathfrak{p} = \mathcal{O}_\mathfrak{p}$ for almost all $\mathfrak{p}$, there is some $c \in \mathcal{O}$ such that $ca_\mathfrak{p} \in \mathcal{O}_\mathfrak{p}$ for all $\mathfrak{p}$, i.e., $c\mathfrak{a} \subseteq \bigcap_\mathfrak{p} \mathcal{O}_\mathfrak{p} = \mathcal{O}$. We have to show that one has

$$\mathfrak{a}\mathcal{O}_\mathfrak{p} = a_\mathfrak{p}\mathcal{O}_\mathfrak{p}$$

for every $\mathfrak{p}$. The inclusion $\subseteq$ is trivial. In order to show that $a_\mathfrak{p}\mathcal{O}_\mathfrak{p} \subseteq \mathfrak{a}\mathcal{O}_\mathfrak{p}$, let us choose $c \in \mathcal{O}$, $c \neq 0$, such that $ca_\mathfrak{p}^{-1}a_\mathfrak{q} \in \mathcal{O}$ for the finitely many $\mathfrak{q}$ which satisfy $a_\mathfrak{p}^{-1}a_\mathfrak{q} \notin \mathcal{O}_\mathfrak{q}$. By the Chinese remainder theorem (12.3), we may find $a \in \mathcal{O}$ such that

$$a \equiv c \bmod \mathfrak{p} \quad \text{and} \quad a \in ca_\mathfrak{p}^{-1}a_\mathfrak{q}\mathcal{O}_\mathfrak{q} \quad \text{for} \quad \mathfrak{q} \neq \mathfrak{p}.$$

Then $\varepsilon = ac^{-1}$ is a unit in $\mathcal{O}_\mathfrak{p}$ and $a_\mathfrak{p}\varepsilon \in \bigcap_\mathfrak{q} a_\mathfrak{q}\mathcal{O}_\mathfrak{q} = \mathfrak{a}$, hence

$$a_\mathfrak{p}\mathcal{O}_\mathfrak{p} = (a_\mathfrak{p}\varepsilon)\mathcal{O}_\mathfrak{p} \subseteq \mathfrak{a}\mathcal{O}_\mathfrak{p}. \qquad \square$$

Passing from the ring $\mathcal{O}$ to its **normalization** $\tilde{\mathcal{O}}$, i.e., to the integral closure of $\mathcal{O}$ in K, one obtains a Dedekind domain. This is not all that easy to prove, however, because $\tilde{\mathcal{O}}$ is in general not a finitely generated $\mathcal{O}$-module. But at any rate we have the

(12.7) Lemma. *Let $\mathcal{O}$ be a one-dimensional noetherian integral domain and $\tilde{\mathcal{O}}$ its normalization. Then, for each ideal $\mathfrak{a} \neq 0$ of $\mathcal{O}$, the quotient $\tilde{\mathcal{O}}/\mathfrak{a}\tilde{\mathcal{O}}$ is a finitely generated $\mathcal{O}$-module.*

Proof: Let $a \in \mathfrak{a}$, $a \neq 0$. Then $\tilde{\mathcal{O}}/\mathfrak{a}\tilde{\mathcal{O}}$ is a quotient of $\tilde{\mathcal{O}}/a\tilde{\mathcal{O}}$. It thus suffices to show that $\tilde{\mathcal{O}}/a\tilde{\mathcal{O}}$ is a finitely generated $\mathcal{O}$-module. With this end, consider in $\mathcal{O}$ the descending chain of ideals containing $a\mathcal{O}$

$$\mathfrak{a}_m = (a^m\tilde{\mathcal{O}} \cap \mathcal{O}, a\mathcal{O}).$$

This chain becomes stationary. In fact, the prime ideals of the ring $\mathcal{O}/a\mathcal{O}$ are not only maximal but also minimal in the sense that $\mathcal{O}/a\mathcal{O}$ is a zero-dimensional noetherian ring. In such a ring every descending chain of ideals becomes stationary (see §3, exercise 7). If the chain $\bar{\mathfrak{a}}_m = \mathfrak{a}_m \bmod a\mathcal{O}$ is stationary at n, then so is the chain $\mathfrak{a}_m$. We show that, for this n, we have

$$\tilde{\mathcal{O}} \subseteq a^{-n}\mathcal{O} + a\tilde{\mathcal{O}}.$$

Let $\beta = \frac{b}{c} \in \tilde{\mathcal{O}}$, $b, c \in \mathcal{O}$. Apply the descending chain condition to the ring $\mathcal{O}/c\mathcal{O}$ and the chain of ideals $(\bar{a}^m)$, where $\bar{a} = a \bmod c\mathcal{O}$. Then $(\bar{a}^h) = (\bar{a}^{h+1})$, i.e., we find some $x \in \mathcal{O}$ such that $a^h \equiv xa^{h+1} \bmod c\mathcal{O}$, hence $(1 - xa)a^h \in c\mathcal{O}$, and therefore

$$\beta = \frac{b}{c}(1 - xa) + \beta xa = \frac{b}{a^h}\frac{(1 - xa)a^h}{c} + \beta xa \in a^{-h}\mathcal{O} + a\tilde{\mathcal{O}}.$$

Let h be the smallest positive integer such that $\beta \in a^{-h}\mathcal{O} + a\tilde{\mathcal{O}}$. It then suffices to show that $h \le n$. Assume $h > n$. Writing

$$(*) \qquad \beta = \frac{u}{a^h} + a\tilde{u} \quad \text{with } u \in \mathcal{O},\ \tilde{u} \in \tilde{\mathcal{O}},$$

we have $u = a^h(\beta - a\tilde{u}) \in a^h\tilde{\mathcal{O}} \cap \mathcal{O} \subseteq \mathfrak{a}_h = \mathfrak{a}_{h+1}$ because $h > n$, hence $u = a^{h+1}\tilde{u}' + au'$, $u' \in \mathcal{O}$, $\tilde{u}' \in \tilde{\mathcal{O}}$. Substituting this into $(*)$ gives

$$\beta = \frac{u'}{a^{h-1}} + a(\tilde{u} + \tilde{u}') \in a^{1-h}\mathcal{O} + a\tilde{\mathcal{O}}.$$

This contradicts the minimality of h. So we do have $\tilde{\mathcal{O}} \subseteq a^{-n}\mathcal{O} + a\tilde{\mathcal{O}}$.

$\tilde{\mathcal{O}}/a\tilde{\mathcal{O}}$ thus becomes a submodule of the $\mathcal{O}$-module $(a^{-n}\mathcal{O} + a\tilde{\mathcal{O}})/a\tilde{\mathcal{O}}$ generated by $a^{-n} \bmod a\tilde{\mathcal{O}}$. It is therefore itself a finitely generated $\mathcal{O}$-module, q.e.d. □

(12.8) Proposition (*Krull-Akizuki*). *Let $\mathcal{O}$ be a one-dimensional noetherian integral domain with field of fractions K. Let $L|K$ be a finite extension and $\mathcal{O}$ the integral closure of $\mathcal{O}$ in L. Then $\mathcal{O}$ is a Dedekind domain.*

Proof: The facts that $\mathcal{O}$ is integrally closed and that every nonzero prime ideal is maximal, are deduced as in (3.1). It remains to show that $\mathcal{O}$ is noetherian. Let $\omega_1, \dots, \omega_n$ be a basis of $L|K$ which is contained in $\mathcal{O}$. Then the ring $\mathcal{O}_0 = \mathcal{O}[\omega_1, \dots, \omega_n]$ is a finitely generated $\mathcal{O}$-module and in particular is noetherian since $\mathcal{O}$ is noetherian. We argue as before that $\mathcal{O}_0$ is one-dimensional and are thus reduced to the case $L = K$. So let $\mathfrak{A}$ be an ideal of $\mathcal{O}$ and $a \in \mathfrak{A} \cap \mathcal{O}$, $a \neq 0$; then by the above lemma $\mathcal{O}/a\mathcal{O}$ is a finitely generated $\mathcal{O}$-module. Since $\mathcal{O}$ is noetherian, so is the $\mathcal{O}$-submodule $\mathfrak{A}/a\mathcal{O}$, and also the $\mathcal{O}$-module $\mathfrak{A}$. □

Remark: The above proof is taken from *Kaplansky*'s book [82] (see also [101]). It shows at the same time that proposition (8.1), which we had proved only in the case of a separable extension $L|K$, is valid for general finite extensions of the field of fractions of a Dedekind domain.

Next we want to compare the one-dimensional noetherian integral domain $\mathcal{O}$ with its normalization $\tilde{\mathcal{O}}$. The fact that $\tilde{\mathcal{O}}$ is a Dedekind domain is evident and does not require the lengthy proof of (12.8) provided we make the following hypothesis:

(∗) $\mathcal{O}$ is an integral domain whose normalization $\tilde{\mathcal{O}}$ is a finitely generated $\mathcal{O}$-module.

This condition will be assumed for all that follows. It avoids pathological situations and is satisfied in all interesting cases, in particular for the orders in an algebraic number field.

The groups of units and the Picard groups of $\mathcal{O}$ and $\tilde{\mathcal{O}}$ are compared with each other by the following

(12.9) Proposition. *One has the canonical exact sequence*

$$1 \longrightarrow \mathcal{O}^* \longrightarrow \tilde{\mathcal{O}}^* \longrightarrow \bigoplus_{\mathfrak{p}} \tilde{\mathcal{O}}_{\mathfrak{p}}^*/\mathcal{O}_{\mathfrak{p}}^* \longrightarrow Pic(\mathcal{O}) \longrightarrow Pic(\tilde{\mathcal{O}}) \longrightarrow 1\,.$$

In the sum, $\mathfrak{p}$ varies over the prime ideals $\neq 0$ of $\mathcal{O}$ and $\tilde{\mathcal{O}}_{\mathfrak{p}}$ denotes the integral closure of $\mathcal{O}_{\mathfrak{p}}$ in K.

Proof: If $\tilde{\mathfrak{p}}$ varies over the prime ideals of $\tilde{\mathcal{O}}$, we know from (12.6) that

$$J(\tilde{\mathcal{O}}) \cong \bigoplus_{\tilde{\mathfrak{p}}} P(\tilde{\mathcal{O}}_{\tilde{\mathfrak{p}}})\,.$$

If $\mathfrak{p}$ is a prime ideal of $\mathcal{O}$, then $\mathfrak{p}\tilde{\mathcal{O}}$ splits in the Dedekind domain $\tilde{\mathcal{O}}$ into a product

$$\mathfrak{p}\tilde{\mathcal{O}} = \tilde{\mathfrak{p}}_1^{e_1} \cdots \tilde{\mathfrak{p}}_r^{e_r}\,,$$

i.e., there are only finitely many prime ideals of $\tilde{\mathcal{O}}$ above $\mathfrak{p}$. The same holds for the integral closure $\tilde{\mathcal{O}}_{\mathfrak{p}}$ of $\mathcal{O}_{\mathfrak{p}}$. Since every nonzero prime ideal of $\tilde{\mathcal{O}}_{\mathfrak{p}}$ has to lie above $\mathfrak{p}\mathcal{O}_{\mathfrak{p}}$, the localization $\tilde{\mathcal{O}}_{\mathfrak{p}}$ has only a finite number of prime ideals and is therefore a principal ideal domain (see §3, exercise 4). In view of (12.6), it follows that

$$P(\tilde{\mathcal{O}}_{\mathfrak{p}}) = J(\tilde{\mathcal{O}}_{\mathfrak{p}}) \cong \bigoplus_{\tilde{\mathfrak{p}} \supseteq \mathfrak{p}} P(\tilde{\mathcal{O}}_{\tilde{\mathfrak{p}}})$$

and therefore

$$J(\tilde{\mathcal{O}}) \cong \bigoplus_{\mathfrak{p}} \bigoplus_{\tilde{\mathfrak{p}} \supseteq \mathfrak{p}} P(\tilde{\mathcal{O}}_{\tilde{\mathfrak{p}}}) \cong \bigoplus_{\mathfrak{p}} P(\tilde{\mathcal{O}}_{\mathfrak{p}})\,.$$

Observing that $P(R) \cong K^*/R^*$ for any integral domain R with field of fractions K, we obtain the commutative exact diagram

$$\begin{array}{ccccccccc}
1 & \longrightarrow & K^*/\mathcal{O}^* & \longrightarrow & \bigoplus\limits_{\mathfrak{p}} K^*/\mathcal{O}_{\mathfrak{p}}^* & \longrightarrow & Pic(\mathcal{O}) & \longrightarrow & 1 \\
 & & \downarrow{\scriptstyle\alpha} & & \downarrow{\scriptstyle\beta} & & \downarrow{\scriptstyle\gamma} & & \\
1 & \longrightarrow & K^*/\tilde{\mathcal{O}}^* & \longrightarrow & \bigoplus\limits_{\mathfrak{p}} K^*/\tilde{\mathcal{O}}_{\mathfrak{p}}^* & \longrightarrow & Pic(\tilde{\mathcal{O}}) & \longrightarrow & 1.
\end{array}$$

For such a diagram one has in complete generality the well-known **snake lemma**: the diagram gives in a canonical way an exact sequence

$$\begin{aligned}
1 \longrightarrow \ker(\alpha) \longrightarrow \ker(\beta) \longrightarrow \ker(\gamma)& \\
\xrightarrow{\delta} \operatorname{coker}(\alpha) \longrightarrow \operatorname{coker}(\beta) \longrightarrow \operatorname{coker}(\gamma) \longrightarrow 1&
\end{aligned}$$

relating the kernels and cokernels of α, β, γ (see [23], chap. III, §3, lemma 3.3). In our particular case, α, β, and therefore also γ, are surjective, whereas

$$\ker(\alpha) = \tilde{\mathcal{O}}^*/\mathcal{O}^* \quad \text{and} \quad \ker(\beta) = \bigoplus_{\mathfrak{p}} \tilde{\mathcal{O}}_{\mathfrak{p}}^*/\mathcal{O}_{\mathfrak{p}}^*.$$

This then yields the exact sequence

$$1 \longrightarrow \mathcal{O}^* \longrightarrow \tilde{\mathcal{O}}^* \longrightarrow \bigoplus_{\mathfrak{p}} \tilde{\mathcal{O}}_{\mathfrak{p}}^*/\mathcal{O}_{\mathfrak{p}}^* \longrightarrow Pic(\mathcal{O}) \longrightarrow Pic(\tilde{\mathcal{O}}) \longrightarrow 1. \qquad \square$$

A prime ideal $\mathfrak{p} \neq 0$ of $\mathcal{O}$ is called **regular** if $\mathcal{O}_{\mathfrak{p}}$ is integrally closed, and thus a discrete valuation ring. For the regular prime ideals, the summands $\tilde{\mathcal{O}}_{\mathfrak{p}}^*/\mathcal{O}_{\mathfrak{p}}^*$ in (12.9) are trivial. There are only finitely many non-regular prime ideals of $\mathcal{O}$, namely the divisors of the **conductor** of $\mathcal{O}$. This is by definition the biggest ideal of $\tilde{\mathcal{O}}$ which is contained in $\mathcal{O}$, in other words,

$$\mathfrak{f} = \left\{a \in \tilde{\mathcal{O}} \;\middle|\; a\tilde{\mathcal{O}} \subseteq \mathcal{O}\right\}.$$

Since $\tilde{\mathcal{O}}$ is a finitely generated $\mathcal{O}$-module, we have $\mathfrak{f} \neq 0$.

(12.10) Proposition. *For any prime ideal $\mathfrak{p} \neq 0$ of $\mathcal{O}$ one has*

$$\mathfrak{p} \nmid \mathfrak{f} \iff \mathfrak{p} \textit{ is regular.}$$

If this is the case, then $\tilde{\mathfrak{p}} = \mathfrak{p}\tilde{\mathcal{O}}$ is a prime ideal of $\tilde{\mathcal{O}}$ and $\mathcal{O}_{\mathfrak{p}} = \tilde{\mathcal{O}}_{\tilde{\mathfrak{p}}}$.

Proof: Assume $\mathfrak{p} \nmid \mathfrak{f}$, i.e., $\mathfrak{p} \not\supseteq \mathfrak{f}$, and let $t \in \mathfrak{f} \setminus \mathfrak{p}$. Then $t\tilde{\mathcal{O}} \subseteq \mathcal{O}$, hence $\tilde{\mathcal{O}} \subseteq \frac{1}{t}\mathcal{O} \subseteq \mathcal{O}_{\mathfrak{p}}$. If $\mathfrak{m} = \mathfrak{p}\mathcal{O}_{\mathfrak{p}}$ is the maximal ideal of $\mathcal{O}_{\mathfrak{p}}$ then, putting $\tilde{\mathfrak{p}} = \mathfrak{m} \cap \tilde{\mathcal{O}}$, $\tilde{\mathfrak{p}}$ is a prime ideal of $\tilde{\mathcal{O}}$ such that $\mathfrak{p} \subseteq \tilde{\mathfrak{p}} \cap \mathcal{O}$, hence $\mathfrak{p} = \tilde{\mathfrak{p}} \cap \mathcal{O}$ because $\mathfrak{p}$ is maximal. Trivially, $\mathcal{O}_{\mathfrak{p}} \subseteq \tilde{\mathcal{O}}_{\tilde{\mathfrak{p}}}$, and if conversely $\frac{a}{s} \in \tilde{\mathcal{O}}_{\tilde{\mathfrak{p}}}$, for $a \in \tilde{\mathcal{O}}$, $s \in \tilde{\mathcal{O}} \setminus \tilde{\mathfrak{p}}$, then $ta \in \mathcal{O}$ and $ts \in \mathcal{O} \setminus \mathfrak{p}$, hence $\frac{a}{s} = \frac{ta}{ts} \in \mathcal{O}_{\mathfrak{p}}$. Therefore $\mathcal{O}_{\mathfrak{p}} = \tilde{\mathcal{O}}_{\tilde{\mathfrak{p}}}$. Thus, by (11.5), $\mathcal{O}_{\mathfrak{p}}$ is a valuation ring, i.e., $\mathfrak{p}$ is regular.

One has furthermore that $\tilde{\mathfrak{p}} = \mathfrak{p}\tilde{\mathcal{O}}$. In fact, $\tilde{\mathfrak{p}}$ is the only prime ideal of $\tilde{\mathcal{O}}$ above $\mathfrak{p}$. For if $\tilde{\mathfrak{q}}$ is another one, then $\tilde{\mathcal{O}}_{\tilde{\mathfrak{p}}} = \mathcal{O}_{\mathfrak{p}} \subseteq \tilde{\mathcal{O}}_{\tilde{\mathfrak{q}}}$, and therefore

$$\tilde{\mathfrak{p}} = \tilde{\mathcal{O}} \cap \tilde{\mathfrak{p}}\mathcal{O}_{\tilde{\mathfrak{p}}} \subseteq \tilde{\mathcal{O}} \cap \tilde{\mathfrak{q}}\mathcal{O}_{\tilde{\mathfrak{q}}} = \tilde{\mathfrak{q}},$$

hence $\tilde{\mathfrak{p}} = \tilde{\mathfrak{q}}$. Consequently, $\mathfrak{p}\tilde{\mathcal{O}} = \tilde{\mathfrak{p}}^e$, with $e \geq 1$, and furthermore $\mathfrak{m} = \mathfrak{p}\mathcal{O}_{\mathfrak{p}} = (\mathfrak{p}\tilde{\mathcal{O}})\mathcal{O}_{\mathfrak{p}} = \tilde{\mathfrak{p}}^e\mathcal{O}_{\mathfrak{p}} = \mathfrak{m}^e$, i.e., $e = 1$ and thus $\tilde{\mathfrak{p}} = \mathfrak{p}\tilde{\mathcal{O}}$.

Conversely, assume $\mathcal{O}_{\mathfrak{p}}$ is a discrete valuation ring. Being a principal ideal domain, it is integrally closed, and since $\tilde{\mathcal{O}}$ is integral over $\mathcal{O}$, hence *a fortiori* over $\mathcal{O}_{\mathfrak{p}}$, we have $\tilde{\mathcal{O}} \subseteq \mathcal{O}_{\mathfrak{p}}$. Let $x_1, \dots, x_n$ be a system of generators of the $\mathcal{O}$-module $\tilde{\mathcal{O}}$. We may write $x_i = \frac{a_i}{s_i}$, with $a_i \in \mathcal{O}$, $s_i \in \mathcal{O} \setminus \mathfrak{p}$. Setting $s = s_1 \cdots s_n \in \mathcal{O} \setminus \mathfrak{p}$, we find $sx_1, \dots, sx_n \in \mathcal{O}$ and therefore $s\tilde{\mathcal{O}} \subseteq \mathcal{O}$, i.e., $s \in \mathfrak{f} \setminus \mathfrak{p}$. It follows that $\mathfrak{p} \nmid \mathfrak{f}$. □

We now obtain the following simple description for the sum $\bigoplus_{\mathfrak{p}} \tilde{\mathcal{O}}_{\mathfrak{p}}^*/\mathcal{O}_{\mathfrak{p}}^*$ in (12.9).

(12.11) Proposition. $\bigoplus_{\mathfrak{p}} \tilde{\mathcal{O}}_{\mathfrak{p}}^*/\mathcal{O}_{\mathfrak{p}}^* \cong (\tilde{\mathcal{O}}/\mathfrak{f})^*/(\mathcal{O}/\mathfrak{f})^*$.

Proof: We apply the Chinese remainder theorem (12.3) repeatedly. We have

$$(1) \qquad \mathcal{O}/\mathfrak{f} \cong \bigoplus_{\mathfrak{p}} \mathcal{O}_{\mathfrak{p}}/\mathfrak{f}\mathcal{O}_{\mathfrak{p}}.$$

The integral closure $\tilde{\mathcal{O}}_{\mathfrak{p}}$ of $\mathcal{O}_{\mathfrak{p}}$ possesses only the finitely many prime ideals that lie above $\mathfrak{p}\mathcal{O}_{\mathfrak{p}}$. They give the localizations $\tilde{\mathcal{O}}_{\tilde{\mathfrak{p}}}$, where $\tilde{\mathfrak{p}}$ varies over the prime ideals above $\mathfrak{p}$ of the ring $\tilde{\mathcal{O}}$. At the same time, $\tilde{\mathcal{O}}_{\mathfrak{p}}$ is the localization of $\tilde{\mathcal{O}}$ with respect to the multiplicative subset $\tilde{\mathcal{O}} \setminus \tilde{\mathfrak{p}}$. Since $\mathfrak{f}$ is an ideal of $\tilde{\mathcal{O}}$, it follows that $\mathfrak{f}\tilde{\mathcal{O}}_{\mathfrak{p}} = \mathfrak{f}\mathcal{O}_{\mathfrak{p}}$. The Chinese remainder theorem yields

$$\tilde{\mathcal{O}}_{\mathfrak{p}}/\mathfrak{f}\tilde{\mathcal{O}}_{\mathfrak{p}} \cong \bigoplus_{\tilde{\mathfrak{p}} \supseteq \mathfrak{p}} \tilde{\mathcal{O}}_{\tilde{\mathfrak{p}}}/\mathfrak{f}\tilde{\mathcal{O}}_{\tilde{\mathfrak{p}}}$$

and

$$(2) \qquad \tilde{\mathcal{O}}/\mathfrak{f} \cong \bigoplus_{\mathfrak{p}} \bigoplus_{\tilde{\mathfrak{p}} \supseteq \mathfrak{p}} \tilde{\mathcal{O}}_{\tilde{\mathfrak{p}}}/\mathfrak{f}\tilde{\mathcal{O}}_{\tilde{\mathfrak{p}}} \cong \bigoplus_{\mathfrak{p}} \tilde{\mathcal{O}}_{\mathfrak{p}}/\mathfrak{f}\tilde{\mathcal{O}}_{\mathfrak{p}}.$$

Passing to unit groups, we get from (1) and (2) that

$$(3) \qquad (\tilde{\mathcal{O}}/\mathfrak{f})^*/(\mathcal{O}/\mathfrak{f})^* \cong \bigoplus_{\mathfrak{p}} (\tilde{\mathcal{O}}_{\mathfrak{p}}/\mathfrak{f}\tilde{\mathcal{O}}_{\mathfrak{p}})^*/(\mathcal{O}_{\mathfrak{p}}/\mathfrak{f}\mathcal{O}_{\mathfrak{p}})^* .$$

For $\mathfrak{f} \subseteq \mathfrak{p}$ we now consider the homomorphism

$$\varphi : \tilde{\mathcal{O}}_{\mathfrak{p}}^* \to (\tilde{\mathcal{O}}_{\mathfrak{p}}/\mathfrak{f}\tilde{\mathcal{O}}_{\mathfrak{p}})^*/(\mathcal{O}_{\mathfrak{p}}/\mathfrak{f}\mathcal{O}_{\mathfrak{p}})^* .$$

It is surjective. In fact, if $\varepsilon \bmod \mathfrak{f}\tilde{\mathcal{O}}_{\mathfrak{p}}$ is a unit in $\tilde{\mathcal{O}}_{\mathfrak{p}}/\mathfrak{f}\tilde{\mathcal{O}}_{\mathfrak{p}}$, then ε is a unit in $\tilde{\mathcal{O}}_{\mathfrak{p}}$. This is so because the units in any ring are precisely those elements that are not contained in any maximal ideal, and the preimages of the maximal ideals of $\tilde{\mathcal{O}}_{\mathfrak{p}}/\mathfrak{f}\tilde{\mathcal{O}}_{\mathfrak{p}}$ give precisely all the maximal ideals of $\tilde{\mathcal{O}}_{\mathfrak{p}}$, since $\mathfrak{f}\tilde{\mathcal{O}}_{\mathfrak{p}} \subseteq \mathfrak{p}\tilde{\mathcal{O}}_{\mathfrak{p}}$. The kernel of φ is a subgroup of $\tilde{\mathcal{O}}_{\mathfrak{p}}^*$ which is contained in $\mathcal{O}_{\mathfrak{p}}$, and which contains $\mathcal{O}_{\mathfrak{p}}^*$. It is therefore equal to $\mathcal{O}_{\mathfrak{p}}^*$. We now conclude that

$$\tilde{\mathcal{O}}_{\mathfrak{p}}^*/\mathcal{O}_{\mathfrak{p}}^* \cong (\tilde{\mathcal{O}}_{\mathfrak{p}}/\mathfrak{f}\tilde{\mathcal{O}}_{\mathfrak{p}})^*/(\mathcal{O}_{\mathfrak{p}}/\mathfrak{f}\mathcal{O}_{\mathfrak{p}})^* .$$

This remains true also for $\mathfrak{p} \not\supseteq \mathfrak{f}$ because then both sides are equal to 1 according to (12.10). The claim of the proposition now follows from (3). □

Our study of one-dimensional noetherian integral domains was motivated by the consideration of *orders*. For them, (12.9) and (12.11) imply the following generalization of Dirichlet's unit theorem and of the theorem on the finiteness of the class group.

(12.12) Theorem. *Let $\mathcal{O}$ be an order in an algebraic number field K, $\mathcal{O}_K$ the maximal order, and $\mathfrak{f}$ the conductor of $\mathcal{O}$.*

Then the groups $\mathcal{O}_K^*/\mathcal{O}^*$ and $Pic(\mathcal{O})$ are finite and one has

$$\# Pic(\mathcal{O}) = \frac{h_K}{(\mathcal{O}_K^* : \mathcal{O}^*)} \frac{\#(\mathcal{O}_K/\mathfrak{f})^*}{\#(\mathcal{O}/\mathfrak{f})^*} ,$$

where h_K is the class number of K. In particular, one has that

$$\operatorname{rank}(\mathcal{O}^*) = \operatorname{rank}(\mathcal{O}_K^*) = r + s - 1 .$$

Proof: By (12.9) and (12.11), and since $Pic(\mathcal{O}_K) = Cl_K$, we have the exact sequence

$$1 \longrightarrow \mathcal{O}_K^*/\mathcal{O}^* \longrightarrow (\mathcal{O}_K/\mathfrak{f})^*/(\mathcal{O}/\mathfrak{f})^* \longrightarrow Pic(\mathcal{O}) \longrightarrow Cl_K \longrightarrow 1 .$$

This gives the claim. □

The definition of the Picard group of a one-dimensional noetherian integral domain $\mathcal{O}$ avoids the problem of the uniqueness of prime ideal decomposition by restricting attention to the invertible ideals, and thus leaving aside the information carried by noninvertibles. But there is another important generalization of the ideal class group which does take into account *all* prime ideals of $\mathcal{O}$. It is based on an artificial re-introduction of the uniqueness of prime decomposition. This group is called the **divisor class group**, or **Chow group** of $\mathcal{O}$. Its definition starts from the free abelian group

$$Div(\mathcal{O}) = \bigoplus_{\mathfrak{p}} \mathbb{Z}\mathfrak{p}$$

on the set of all maximal ideals $\mathfrak{p}$ of $\mathcal{O}$ (i.e., the set of all prime ideals $\neq 0$). This group is called the **divisor group** of $\mathcal{O}$. Its elements are formal sums

$$D = \sum_{\mathfrak{p}} n_{\mathfrak{p}}\mathfrak{p}$$

with $n_{\mathfrak{p}} \in \mathbb{Z}$ and $n_{\mathfrak{p}} = 0$ for almost all $\mathfrak{p}$, called **divisors** (or 0-**cycles**). Corollary (3.9) simply says that, in the case of a Dedekind domain, the divisor group $Div(\mathcal{O})$ and the group of ideals are canonically isomorphic. The additive notation and the name of the group stem from function theory where divisors for analytic functions play the same rôle as ideals do for algebraic numbers (see chap. III, § 3).

In order to define the divisor class group we have to associate to every $f \in K^*$ a "principal divisor" $\operatorname{div}(f)$. We use the case of a Dedekind domain to guide us. There the principal ideal (f) was given by

$$(f) = \prod_{\mathfrak{p}} \mathfrak{p}^{v_{\mathfrak{p}}(f)},$$

where $v_{\mathfrak{p}} : K^* \to \mathbb{Z}$ is the $\mathfrak{p}$-adic exponential valuation associated to the valuation ring $\mathcal{O}_{\mathfrak{p}}$. In general, $\mathcal{O}_{\mathfrak{p}}$ is not anymore a discrete valuation ring. Nevertheless, $\mathcal{O}_{\mathfrak{p}}$ defines a homomorphism

$$\operatorname{ord}_{\mathfrak{p}} : K^* \to \mathbb{Z}$$

which generalizes the valuation function. If $f = a/b \in K^*$, with $a, b \in \mathcal{O}$, then we put

$$\operatorname{ord}_{\mathfrak{p}}(f) = \ell_{\mathcal{O}_{\mathfrak{p}}}(\mathcal{O}_{\mathfrak{p}}/a\mathcal{O}_{\mathfrak{p}}) - \ell_{\mathcal{O}_{\mathfrak{p}}}(\mathcal{O}_{\mathfrak{p}}/b\mathcal{O}_{\mathfrak{p}}),$$

where $\ell_{\mathcal{O}_{\mathfrak{p}}}(M)$ denotes the **length** of an $\mathcal{O}_{\mathfrak{p}}$-module M, i.e., the maximal length of a strictly decreasing chain

$$M = M_0 \supsetneqq M_1 \supsetneqq \cdots \supsetneqq M_\ell = 0$$

of $\mathcal{O}_{\mathfrak{p}}$-submodules. In the special case where $\mathcal{O}_{\mathfrak{p}}$ is a discrete valuation ring with maximal ideal $\mathfrak{m}$, the value $\nu = v_{\mathfrak{p}}(a)$ of $a \in \mathcal{O}_{\mathfrak{p}}$, for $a \neq 0$, is given by the equation

$$a\mathcal{O}_{\mathfrak{p}} = \mathfrak{m}^{\nu}.$$

It is equal to the length of the $\mathcal{O}_\mathfrak{p}$-module $\mathcal{O}_\mathfrak{p}/\mathfrak{m}^\nu$, because the longest chain of submodules is

$$\mathcal{O}_\mathfrak{p}/\mathfrak{m}^\nu \supset \mathfrak{m}/\mathfrak{m}^\nu \supset \cdots \supset \mathfrak{m}^\nu/\mathfrak{m}^\nu = (0).$$

Thus the function $\mathrm{ord}_\mathfrak{p}$ agrees with the exponential valuation $v_\mathfrak{p}$ in this case.

The property of the function $\mathrm{ord}_\mathfrak{p}$ to be a homomorphism follows from the fact (which is easily proved) that the length function $\ell_{\mathcal{O}_\mathfrak{p}}$ is multiplicative on short exact sequences of $\mathcal{O}_\mathfrak{p}$-modules.

Using the functions $\mathrm{ord}_\mathfrak{p} : K^* \to \mathbb{Z}$, we can now associate to every element $f \in K^*$ the divisor

$$\mathrm{div}(f) = \sum_\mathfrak{p} \mathrm{ord}_\mathfrak{p}(f)\mathfrak{p},$$

and thus obtain a canonical homomorphism

$$\mathrm{div} : K^* \longrightarrow Div(\mathcal{O}).$$

The elements $\mathrm{div}(f)$ are called **principal divisors**. They form a subgroup $\mathcal{P}(\mathcal{O})$ of $Div(\mathcal{O})$. Two divisors D and D' which differ only by a principal divisor are called **rationally equivalent**.

(12.13) Definition. *The quotient group*

$$CH^1(\mathcal{O}) = Div(\mathcal{O})/\mathcal{P}(\mathcal{O})$$

is called the **divisor class group** *or* **Chow group** *of* $\mathcal{O}$.

The Chow group is related to the Picard group by a canonical homomorphism

$$\mathrm{div} : Pic(\mathcal{O}) \longrightarrow CH^1(\mathcal{O})$$

which is defined as follows. If $\mathfrak{a}$ is an invertible ideal, then, by (12.4), $\mathfrak{a}\mathcal{O}_\mathfrak{p}$, for any prime ideal $\mathfrak{p} \neq 0$, is a principal ideal $a_\mathfrak{p}\mathcal{O}_\mathfrak{p}$, $a_\mathfrak{p} \in K^*$, and we put

$$\mathrm{div}(\mathfrak{a}) = \sum_\mathfrak{p} - \mathrm{ord}_\mathfrak{p}(a_\mathfrak{p})\mathfrak{p}.$$

This gives us a homomorphism

$$\mathrm{div} : J(\mathcal{O}) \longrightarrow Div(\mathcal{O})$$

of the ideal group $J(\mathcal{O})$ which takes principal ideals into principal divisors, and therefore induces a homomorphism

$$\mathrm{div} : Pic(\mathcal{O}) \longrightarrow CH^1(\mathcal{O}).$$

In the special case of a Dedekind domain we obtain:

(12.14) Proposition. *If $\mathcal{O}$ is a Dedekind domain, then*

$$\operatorname{div} : Pic(\mathcal{O}) \longrightarrow CH^1(\mathcal{O})$$

is an isomorphism.

Exercise 1. Show that

$$\mathbb{C}[X,Y]/(XY-X), \quad \mathbb{C}[X,Y]/(XY-1),$$

$$\mathbb{C}[X,Y]/(X^2-Y^3), \quad \mathbb{C}[X,Y]/(Y^2-X^2-X^3)$$

are one-dimensional noetherian rings. Which ones are integral domains? Determine their normalizations.

Hint: For instance in the last example, put $t = X/Y$ and show that the homomorphism $\mathbb{C}[X,Y] \to \mathbb{C}[t]$, $X \mapsto t^2-1$, $Y \mapsto t(t^2-1)$, has kernel $(Y^2-X^2-X^3)$.

Exercise 2. Let a and b be positive integers that are not perfect squares. Show that the fundamental unit of the order $\mathbb{Z}+\mathbb{Z}\sqrt{a}$ of the field $\mathbb{Q}(\sqrt{a})$ is also the fundamental unit of the order $\mathbb{Z}+\mathbb{Z}\sqrt{a}+\mathbb{Z}\sqrt{-b}+\mathbb{Z}\sqrt{a}\,\sqrt{-b}$ in the field $\mathbb{Q}(\sqrt{a},\sqrt{-b})$.

Exercise 3. Let K be a number field of degree $n = [K:\mathbb{Q}]$. A complete module in K is a subgroup of the form

$$M = \mathbb{Z}\alpha_1 + \cdots + \mathbb{Z}\alpha_n,$$

where $\alpha_1, \dots, \alpha_n$ are linearly independent elements of K. Show that the ring of multipliers

$$\mathcal{O} = \{\alpha \in K \mid \alpha M \subseteq M\}$$

is an order in K, but in general not the maximal order.

Exercise 4. Determine the ring of multipliers $\mathcal{O}$ of the complete module $M = \mathbb{Z}+\mathbb{Z}\sqrt{2}$ in $\mathbb{Q}(\sqrt{2})$. Show that $\varepsilon = 1+\sqrt{2}$ is a fundamental unit of $\mathcal{O}$. Determine all integer solutions of "Pell's equation"

$$x^2 - 2y^2 = 7.$$

Hint: $N(x+y\sqrt{2}) = x^2-2y^2$, $N(3+\sqrt{2}) = N(5+3\sqrt{2}) = 7$.

Exercise 5. In a one-dimensional noetherian integral domain the regular prime ideals $\neq 0$ are precisely the invertible prime ideals.

§ 13. One-dimensional Schemes

The first approach to the theory of algebraic number fields is dominated by the methods of arithmetic and algebra. But the theory may also be treated fundamentally from a geometric point of view, which will bring out novel aspects in a variety of ways. This geometric interpretation hinges on the possibility of viewing numbers as functions on a topological space.

In order to explain this, let us start from polynomials

$$f(x) = a_n x^n + \cdots + a_0$$

with complex coefficients $a_i \in \mathbb{C}$, which may be immediately interpreted as functions on the complex plane. This property may be formulated in a purely algebraic way as follows. Let $a \in \mathbb{C}$ be a point in the complex plane. The set of all functions $f(x)$ in the polynomial ring $\mathbb{C}[x]$ which vanish at the point a forms the maximal ideal $\mathfrak{p} = (x - a)$ of $\mathbb{C}[x]$. In this way the points of the complex plane correspond 1-1 to the maximal ideals of $\mathbb{C}[x]$. We denote the set of all these maximal ideals by

$$M = \mathrm{Max}\big(\mathbb{C}[x]\big).$$

We may view M as a new kind of space and may interpret the elements $f(x)$ of the ring $\mathbb{C}[x]$ as functions on M as follows. For every point $\mathfrak{p} = (x - a)$ of M we have the canonical isomorphism

$$\mathbb{C}[x]/\mathfrak{p} \xrightarrow{\sim} \mathbb{C},$$

which sends the residue class $f(x) \bmod \mathfrak{p}$ to $f(a)$. We may thus view this residue class

$$f(\mathfrak{p}) := f(x) \bmod \mathfrak{p} \in \kappa(\mathfrak{p})$$

in the residue class field $\kappa(\mathfrak{p}) = \mathbb{C}[x]/\mathfrak{p}$ as the "value" of f at the point $\mathfrak{p} \in M$. The topology on $\mathbb{C}$ cannot be transferred to M by algebraic means. All that can be salvaged algebraically are the point sets defined by equations of the form

$$f(x) = 0$$

(i.e., only the finite sets and M itself). These sets are defined to be the closed subsets. In the new formulation they are the sets

$$V(f) = \big\{\mathfrak{p} \in M \bigm| f(\mathfrak{p}) = 0\big\} = \big\{\mathfrak{p} \in M \bigm| \mathfrak{p} \supseteq (f(x))\big\}.$$

The algebraic interpretation of functions given above leads to the following geometric perception of completely general rings. For an arbitrary ring $\mathcal{O}$, one introduces the **spectrum**

$$X = \mathrm{Spec}(\mathcal{O})$$

as being the set of all prime ideals $\mathfrak{p}$ of $\mathcal{O}$. The **Zariski topology** on X is defined by stipulating that the sets

$$V(\mathfrak{a}) = \big\{\mathfrak{p} \bigm| \mathfrak{p} \supseteq \mathfrak{a}\big\}$$

be the closed sets, where $\mathfrak{a}$ varies over the ideals of $\mathcal{O}$. This does make X into a topological space (observe that $V(\mathfrak{a}) \cup V(\mathfrak{b}) = V(\mathfrak{a}\mathfrak{b})$) which, however, is usually not Hausdorff. The closed points correspond to the maximal ideals of $\mathcal{O}$.

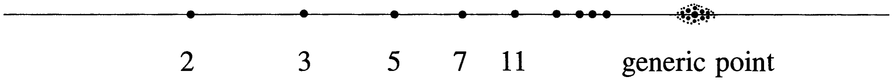
The elements $f \in \mathcal{O}$ now play the rôle of functions on the topological space X: the "value" of f at the point $\mathfrak{p}$ is defined to be

$$f(\mathfrak{p}) := f \bmod \mathfrak{p}$$

and is an element of the residue class field $\kappa(\mathfrak{p})$, i.e., in the field of fractions of $\mathcal{O}/\mathfrak{p}$. So the values of f do not in general lie in a single field.

Admitting also the non-maximal prime ideals as non-closed points, turns out to be extremely useful – and has an intuitive reason as well. For instance in the case of the ring $\mathcal{O} = \mathbb{C}[x]$, the point $\mathfrak{p} = (0)$ has residue class field $\kappa(\mathfrak{p}) = \mathbb{C}(x)$. The "value" of a polynomial $f \in \mathbb{C}[x]$ at this point is $f(x)$ itself, viewed as an element of $\mathbb{C}(x)$. This element should be thought of as the value of f at the **unknown** place x – which one may imagine to be everywhere or nowhere at all. This intuition complies with the fact that the closure of the point $\mathfrak{p} = (0)$ in the Zariski topology of X is the total space X. This is why $\mathfrak{p}$ is also called the **generic point** of X.

Example: The space $X = \operatorname{Spec}(\mathbb{Z})$ may be represented by a line.



For every prime number one has a closed point, and there is also the generic point (0), the closure of which is the total space X. The nonempty open sets in X are obtained by throwing out finitely many prime numbers $p_1, \dots, p_n$. The integers $a \in \mathbb{Z}$ are viewed as functions on X by defining the value of a at the point (p) to be the residue class

$$a(p) = a \bmod p \in \mathbb{Z}/p\mathbb{Z}.$$

The fields of values are then

$$\mathbb{Z}/2\mathbb{Z},\ \mathbb{Z}/3\mathbb{Z},\ \mathbb{Z}/5\mathbb{Z},\ \mathbb{Z}/7\mathbb{Z},\ \mathbb{Z}/11\mathbb{Z},\dots,\mathbb{Q}.$$

Thus every prime field occurs exactly once.

An important refinement of the geometric interpretation of elements of the ring $\mathcal{O}$ as functions on the space $X = \operatorname{Spec}(\mathcal{O})$ is obtained by forming the **structure sheaf** $\mathcal{O}_X$. This means the following. Let $U \neq \emptyset$ be an open subset of X. If $\mathcal{O}$ is a one-dimensional integral domain, then the ring of "regular functions" on U is given by

$$\mathcal{O}(U) = \Big\{ \frac{f}{g} \mid g(\mathfrak{p}) \neq 0 \quad \text{for all } \mathfrak{p} \in U \Big\},$$

in other words, it is the localization of $\mathcal{O}$ with respect to the multiplicative set $S = \mathcal{O} \setminus \bigcup_{\mathfrak{p}\in U} \mathfrak{p}$ (see § 11). In the general case, $\mathcal{O}(U)$ is defined to consist of all elements

$$s = (s_\mathfrak{p}) \in \prod_{\mathfrak{p}\in U} \mathcal{O}_\mathfrak{p}$$

which locally are quotients of two elements of $\mathcal{O}$. More precisely, this means that for every $\mathfrak{p} \in U$, there exists a neighbourhood $V \subseteq U$ of $\mathfrak{p}$, and elements $f, g \in \mathcal{O}$ such that, for each $\mathfrak{q} \in V$, one has $g(\mathfrak{q}) \neq 0$ and $s_\mathfrak{q} = f/g$ in $\mathcal{O}_\mathfrak{q}$. These quotients have to be understood in the more general sense of commutative algebra (see § 11, exercise 1). We leave it to the reader to check that one gets back the above definition in the case of a one-dimensional integral domain $\mathcal{O}$.

If $V \subseteq U$ are two open sets of X, then the projection

$$\prod_{\mathfrak{p}\in U} \mathcal{O}_\mathfrak{p} \longrightarrow \prod_{\mathfrak{p}\in V} \mathcal{O}_\mathfrak{p}$$

induces a homomorphism

$$\rho_{UV} : \mathcal{O}(U) \longrightarrow \mathcal{O}(V),$$

called the **restriction** from U to V. The system of rings $\mathcal{O}(U)$ and mappings ρ_{UV} is a **sheaf** on X. This notion means the following.

(13.1) Definition. *Let X be a topological space. A* **presheaf** *$\mathcal{F}$ of abelian groups (rings, etc.) consists of the following data.*

(1) *For every open set U, an abelian group (a ring, etc.) $\mathcal{F}(U)$ is given.*

(2) *For every inclusion $U \subseteq V$, a homomorphism $\rho_{UV} : \mathcal{F}(U) \to \mathcal{F}(V)$ is given, which is called restriction.*

These data are subject to the following conditions:

(a) $\mathcal{F}(\emptyset) = 0$,

(b) ρ_{UU} *is the identity* $\mathrm{id} : \mathcal{F}(U) \to \mathcal{F}(U)$,

(c) $\rho_{UW} = \rho_{VW} \circ \rho_{UV}$, *for open sets $W \subseteq V \subseteq U$.*

The elements $s \in \mathcal{F}(U)$ are called the **sections** of the presheaf $\mathcal{F}$ over U. If $V \subseteq U$, then one usually writes $\rho_{UV}(s) = s|_V$. The definition of a presheaf can be reformulated most concisely in the language of categories. The open sets of the topological space X form a category X_{top} in which only inclusions are admitted as morphisms. A presheaf of abelian groups (rings) is then simply a contravariant functor

$$\mathcal{F} : X_{\text{top}} \longrightarrow (ab),\ (rings)$$

into the category of abelian groups (resp. rings) such that $\mathcal{F}(\emptyset) = 0$.

(13.2) Definition. *A presheaf $\mathcal{F}$ on the topological space X is called a* **sheaf** *if, for all open coverings $\{U_i\}$ of the open sets U, one has:*

(i) *If $s, s' \in \mathcal{F}(U)$ are two sections such that $s|_{U_i} = s'|_{U_i}$ for all i, then $s = s'$.*

(ii) *If $s_i \in \mathcal{F}(U_i)$ is a family of sections such that*

$$s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$$

for all i, j, then there exists a section $s \in \mathcal{F}(U)$ such that $s|_{U_i} = s_i$ for all i.

The **stalk** of the sheaf $\mathcal{F}$ at the point $x \in X$ is defined to be the direct limit (see chap. IV, § 2)

$$\mathcal{F}_x = \varinjlim_{U \ni x} \mathcal{F}(U),$$

where U varies over all open neighbourhoods of x. In other words, two sections $s_U \in \mathcal{F}(U)$ and $s_V \in \mathcal{F}(V)$ are called equivalent in the disjoint union $\bigcup_{U \ni x} \mathcal{F}(U)$ if there exists a neighbourhood $W \subseteq U \cap V$ of x such that $s_U|_W = s_V|_W$. The equivalence classes are called **germs** of sections at x. They are the elements of $\mathcal{F}_x$.

We now return to the spectrum $X = \operatorname{Spec}(\mathcal{O})$ of a ring $\mathcal{O}$ and obtain the

(13.3) Proposition. *The rings $\mathcal{O}(U)$, together with the restriction mappings ρ_{UV}, form a* **sheaf** *on X. It is denoted by $\mathcal{O}_X$ and called the* **structure sheaf** *on X. The stalk of $\mathcal{O}_X$ at the point $\mathfrak{p} \in X$ is the localization $\mathcal{O}_{\mathfrak{p}}$, i.e., $\mathcal{O}_{X,\mathfrak{p}} \cong \mathcal{O}_{\mathfrak{p}}$.*

The proof of this proposition follows immediately from the definitions. The couple $(X, \mathcal{O}_X)$ is called an **affine scheme**. Usually, however, the structure sheaf $\mathcal{O}_X$ is dropped from the notation. Now let

$$\varphi : \mathcal{O} \longrightarrow \mathcal{O}'$$

be a homomorphism of rings and $X = \operatorname{Spec}(\mathcal{O})$, $X' = \operatorname{Spec}(\mathcal{O}')$. Then φ induces a continuous map

$$f : X' \longrightarrow X, \quad f(\mathfrak{p}') := \varphi^{-1}(\mathfrak{p}'),$$

and, for every open subset U of X, a homomorphism

$$f_U^* : \mathcal{O}(U) \longrightarrow \mathcal{O}(U'), \quad s \longmapsto s \circ f|_{U'},$$

where $U' = f^{-1}(U)$. The maps f_U^* have the following two properties.

a) If $V \subseteq U$ are open sets, then the diagram

$$\begin{tikzcd} \mathcal{O}(U) \arrow[r, "f_U^*"] \arrow[d, "\rho_{UV}"'] & \mathcal{O}(U') \arrow[d, "\rho_{U'V'}"] \\ \mathcal{O}(V) \arrow[r, "f_V^*"] & \mathcal{O}(V') \end{tikzcd}$$

is commutative.

b) For $\mathfrak{p}' \in U' \subseteq X'$ and $a \in \mathcal{O}(U)$ one has

$$a\big(f(\mathfrak{p}')\big) = 0 \implies f_U^*(a)(\mathfrak{p}') = 0 .$$

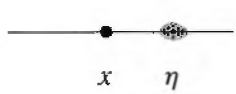
A continuous map $f : X' \to X$ together with a family of homomorphisms $f_U^* : \mathcal{O}(U) \to \mathcal{O}(U')$ which satisfy conditions a) and b) is called a **morphism** from the scheme X' to the scheme X. When referring to such a morphism, the maps f_U^* are usually not written explicitly. One can show that every morphism between two affine schemes $X' = \operatorname{Spec}(\mathcal{O}')$ and $X = \operatorname{Spec}(\mathcal{O})$ is induced in the way described above by a ring homomorphism $\varphi : \mathcal{O} \to \mathcal{O}'$.

The proofs of the above claims are easy, although some of them are a bit lengthy. The notion of scheme is the basis of a very extensive theory which occupies a central place in mathematics. As introductions into this important discipline let us recommend the books [51] and [104].

We will now confine ourselves to considering noetherian integral domains $\mathcal{O}$ of dimension ≤ 1, and propose to illustrate geometrically, via the scheme-theoretic interpretation, some of the facts treated in previous sections.

1. Fields. If K is a field, then the scheme $\operatorname{Spec}(K)$ consists of a single point (0) on top of which the field itself sits as the structure sheaf. One must not think that these one-point schemes are all the same because they differ essentially in their structure sheaves.

2. Valuation rings. If $\mathcal{O}$ is a discrete valuation ring with maximal ideal $\mathfrak{p}$, then the scheme $X = \operatorname{Spec}(\mathcal{O})$ consists of two points, the closed point $x = \mathfrak{p}$ with residue class field $\kappa(\mathfrak{p}) = \mathcal{O}/\mathfrak{p}$, and the generic point $\eta = (0)$ with residue class field $\kappa(\eta) = K$, the field of fractions of $\mathcal{O}$. One should think of X as a point x with an infinitesimal neighbourhood described by the generic point η:

X:


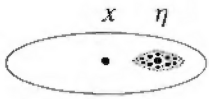
This intuition is justified by the following observation.

The discrete valuation rings arise as localizations

$$\mathcal{O}_\mathfrak{p} = \left\{ \frac{f}{g} \mid f, g \in \mathcal{O},\ g(\mathfrak{p}) \neq 0 \right\}$$

of Dedekind domains $\mathcal{O}$. There is no neighbourhood of $\mathfrak{p}$ in $X = \operatorname{Spec}(\mathcal{O})$ on which all functions $\frac{f}{g} \in \mathcal{O}_\mathfrak{p}$ are defined because, if $\mathcal{O}$ is not a local ring, we find by the Chinese remainder theorem for every point $\mathfrak{q} \neq \mathfrak{p}$, $\mathfrak{q} \neq 0$, an element $g \in \mathcal{O}$ satisfying $g \equiv 0 \bmod \mathfrak{q}$ and $g \equiv 1 \bmod \mathfrak{p}$. Then $\frac{1}{g} \in \mathcal{O}_\mathfrak{p}$ as a function is not defined at $\mathfrak{q}$. But every element $\frac{f}{g} \in \mathcal{O}_\mathfrak{p}$ is defined on a sufficiently small neighbourhood; hence one may say that all elements $\frac{f}{g}$ of the discrete valuation ring $\mathcal{O}_\mathfrak{p}$ are like functions defined on a "germ" of neighbourhoods of $\mathfrak{p}$. Thus $\operatorname{Spec}(\mathcal{O}_\mathfrak{p})$ may be thought of as such a "germ of neighbourhoods" of $\mathfrak{p}$.

We want to point out a small discrepancy of intuitions. Considering the spectrum of the one-dimensional ring $\mathbb{C}[x]$, the points of which constitute the complex plane, we will not want to visualize the infinitesimal neighbourhood $X_\mathfrak{p} = \operatorname{Spec}(\mathbb{C}[x]_\mathfrak{p})$ of a point $\mathfrak{p} = (x-a)$ as a small line segment, but rather as a little disc:

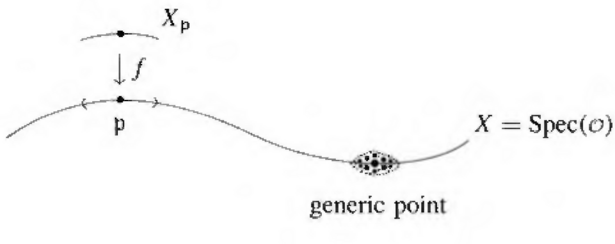


This two-dimensional nature is actually inherent in all discrete valuation rings with algebraically closed residue field. But the algebraic justification of this intuition is provided only by the introduction of a new topology, the **étale topology**, which is much finer than the Zariski topology (see [103], [132]).

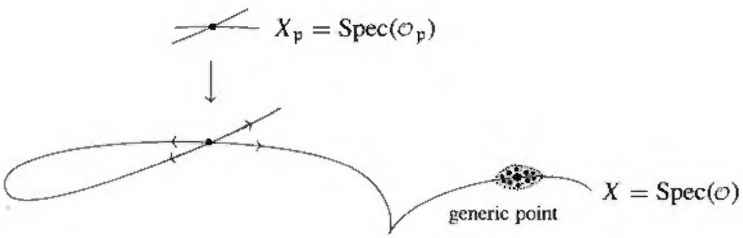
3. Dedekind rings. The spectrum $X = \operatorname{Spec}(\mathcal{O})$ of a Dedekind domain $\mathcal{O}$ is visualized as a smooth curve. At each point $\mathfrak{p}$ one may consider the **localization** $\mathcal{O}_\mathfrak{p}$. The inclusion $\mathcal{O} \hookrightarrow \mathcal{O}_\mathfrak{p}$ induces a morphism

$$f : X_\mathfrak{p} = \operatorname{Spec}(\mathcal{O}_\mathfrak{p}) \longrightarrow X,$$

which extracts the scheme $X_\mathfrak{p}$ from X as an "infinitesimal neighbourhood" of $\mathfrak{p}$:



4. Singularities. We now consider a one-dimensional noetherian integral domain $\mathcal{O}$ which is not a Dedekind domain, *e.g.*, an order in an algebraic number field which is different from the maximal order. Again we view the scheme $X = \operatorname{Spec}(\mathcal{O})$ as a curve. But now the curve will not be everywhere smooth, but will have singularities at certain points.

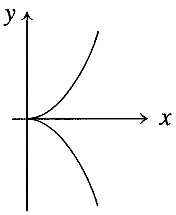


These are precisely the nongeneric points $\mathfrak{p}$ for which the localization $\mathcal{O}_\mathfrak{p}$ is no longer a discrete valuation ring, that is to say, the maximal ideal $\mathfrak{p}\mathcal{O}_\mathfrak{p}$ is not generated by a single element. For example, in the one-dimensional ring $\mathcal{O} = \mathbb{C}[x, y]/(y^2 - x^3)$, the closed points of the scheme X are given by the prime ideals

$$\mathfrak{p} = (x - a, y - b) \bmod (y^2 - x^3)$$

where (a, b) varies over the points of $\mathbb{C}^2$ which satisfy the equation

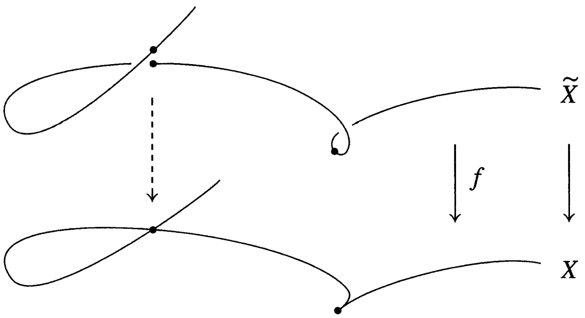
$$b^2 - a^3 = 0.$$



The only singular point is the origin. It corresponds to the maximal ideal $\mathfrak{p}_0 = (\bar{x}, \bar{y})$, where $\bar{x} = x \bmod (y^2 - x^3)$, $\bar{y} = y \bmod (y^2 - x^3) \in \mathcal{O}$. The maximal ideal $\mathfrak{p}_0\mathcal{O}_{\mathfrak{p}_0}$ of the local ring is generated by the elements $\bar{x}, \bar{y}$, and cannot be generated by a single element.

5. Normalization. Passing to the normalization $\tilde{\mathcal{O}}$ of a one-dimensional noetherian integral domain $\mathcal{O}$ means, in geometric terms, taking the *resolution* of the singularities that were just discussed. Indeed, if $X = \operatorname{Spec}(\mathcal{O})$ and

$\widetilde{X} = \operatorname{Spec}(\tilde{o})$, then the inclusion $o \hookrightarrow \tilde{o}$ induces a morphism $f : \widetilde{X} \to X$.



Since $\tilde{o}$ is a Dedekind domain, the scheme $\widetilde{X}$ is to be considered as smooth. If $\mathfrak{p}\tilde{o} = \tilde{\mathfrak{p}}_1^{e_r} \cdots \tilde{\mathfrak{p}}_r^{e_r}$ is the prime factorization of $\mathfrak{p}$ in $\tilde{o}$, then $\tilde{\mathfrak{p}}_1, \dots, \tilde{\mathfrak{p}}_r$ are the different points of $\widetilde{X}$ that are mapped to $\mathfrak{p}$ by f. One can show that $\mathfrak{p}$ is a regular point of X – in the sense that $o_{\mathfrak{p}}$ is a discrete valuation ring – if and only if $r = 1$, $e_1 = 1$ and $f_1 = (\tilde{o}/\tilde{\mathfrak{p}}_1 : o/\mathfrak{p}) = 1$.

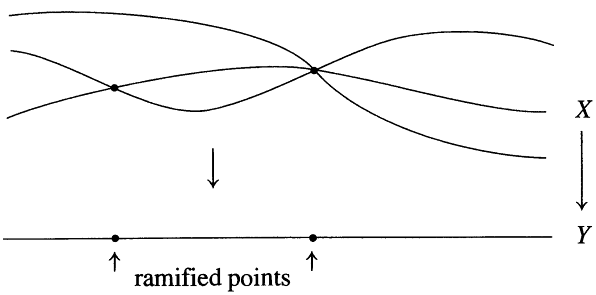
6. Extensions. Let o be a Dedekind domain with field of fractions K. Let $L|K$ be a finite separable extension, and $\mathcal{O}$ the integral closure of o in L. Let $Y = \operatorname{Spec}(o)$, $X = \operatorname{Spec}(\mathcal{O})$, and

$$f : X \longrightarrow Y$$

the morphism induced by the inclusion $o \hookrightarrow \mathcal{O}$. If $\mathfrak{p}$ is a maximal ideal of o and

$$\mathfrak{p}\mathcal{O} = \mathfrak{P}_1^{e_1} \cdots \mathfrak{P}_r^{e_r}$$

the prime decomposition of $\mathfrak{p}$ in $\mathcal{O}$, then $\mathfrak{P}_1, \dots, \mathfrak{P}_r$ are the different points of X which are mapped to $\mathfrak{p}$ by f. The morphism f is a "ramified covering." It is graphically represented by the following picture:



This picture, however, is a fair rendering of the algebraic situation only in the case where the residue class fields of o are algebraically closed (like for the ring $\mathbb{C}[x]$). Then, from the fundamental identity $\sum_i e_i f_i = n$, there are exactly $n = [L : K]$ points $\mathfrak{P}_1, \dots, \mathfrak{P}_n$ of X lying above each point $\mathfrak{p}$ of Y, except when $\mathfrak{p}$ is ramified in $\mathcal{O}$. At a point $\mathfrak{p}$ of ramification, several of the points $\mathfrak{P}_1, \dots, \mathfrak{P}_n$ coalesce. This also explains the terminology of ideals that "ramify."

If $L|K$ is Galois with Galois group $G = G(L|K)$, then every automorphism $\sigma \in G$ induces via $\sigma : \mathcal{O} \to \mathcal{O}$ an automorphism of schemes $\sigma : X \to X$. Since the ring $\mathcal{O}$ is fixed, the diagram

$$\begin{tikzcd} X \arrow[rr, "\sigma"] \arrow[dr, "f"'] & & X \arrow[dl, "f"] \\ & Y & \end{tikzcd}$$

is commutative. Such an automorphism is called a **covering transformation** of the ramified covering X/Y. The group of covering transformations is denoted by $\mathrm{Aut}_Y(X)$. We thus have a canonical isomorphism

$$G(L|K) \cong \mathrm{Aut}_Y(X).$$

In chap. II, § 7, we will see that the composite of two unramified extensions of K is again unramified. The composite $\tilde{K}$, taken inside some algebraic closure $\bar{K}$ of K, of *all* unramified extensions $L|K$ is called the **maximal unramified extension** of K. The integral closure $\tilde{\mathcal{O}}$ of $\mathcal{O}$ in $\tilde{K}$ is still a one-dimensional integral domain, but in general no longer noetherian, and, as a rule, there will be infinitely many prime ideals lying above a given prime ideal $\mathfrak{p} \neq 0$ of $\mathcal{O}$. The scheme $\tilde{Y} = \mathrm{Spec}(\tilde{\mathcal{O}})$ with the morphism

$$f : \tilde{Y} \longrightarrow Y$$

is called the **universal covering** of Y. It plays the same rôle for schemes that the universal covering space $\tilde{X} \to X$ of a topological space plays in topology. There the group of covering transformations $\mathrm{Aut}_X(\tilde{X})$ is canonically isomorphic to the fundamental group $\pi_1(X)$. Therefore we define in our present context the **fundamental group** of the scheme Y by

$$\pi_1(Y) = \mathrm{Aut}_Y(\tilde{Y}) = G(\tilde{K}|K).$$

This establishes a first link of Galois theory with classical topology. This link is pursued much further in **étale topology**.

The geometric point of view of algebraic number fields explained in this section is corroborated very convincingly by the theory of function fields of algebraic curves over a finite field $\mathbb{F}_p$. In fact, a very close analogy exists between both theories.

§ 14. Function Fields

We conclude this chapter with a brief sketch of the theory of **function fields**. They represent a striking analogy with algebraic number fields, and since they are immediately related to geometry, they actually serve as an important model for the theory of algebraic number fields.

The ring $\mathbb{Z}$ of integers with its field of fractions $\mathbb{Q}$ exhibits obvious analogies with the polynomial ring $\mathbb{F}_p[t]$ over the field $\mathbb{F}_p$ with p elements and its field of fractions $\mathbb{F}_p(t)$. Like $\mathbb{Z}$, $\mathbb{F}_p[t]$ is also a principal ideal domain. The prime numbers correspond to the monic irreducible polynomials $p(t) \in \mathbb{F}_p[t]$. Like the prime numbers they have finite fields $\mathbb{F}_{p^d}$, $d = \deg(p(t))$, as their residue class rings. The difference is, however, that now all these fields have the same characteristic. The geometric character of the ring $\mathbb{F}_p[t]$ becomes much more apparent in that, for an element $f = f(t) \in \mathbb{F}_p[t]$, the value of f at a point $\mathfrak{p} = (p(t))$ of the affine scheme $X = \operatorname{Spec}(\mathbb{F}_p[t])$ is actually given by the value $f(a) \in \mathbb{F}_p$, if $p(t) = t - a$, or more generally by $f(\alpha) \in \mathbb{F}_{p^d}$, if $\alpha \in \mathbb{F}_{p^d}$ is a zero of $p(t)$. This is due to the isomorphism

$$\mathbb{F}_p[t]/\mathfrak{p} \xrightarrow{\sim} \mathbb{F}_{p^d},$$

which takes the residue class $f(\mathfrak{p}) = f \bmod \mathfrak{p}$ to $f(\alpha)$. In the analogy between, on the one hand, the progression of the prime numbers $2, 3, 5, 7, \ldots$, and the growing of the cardinalities $p, p^2, p^3, p^4, \ldots$ of the residue fields $\mathbb{F}_{p^d}$ on the other, resides one of the most profound mysteries of arithmetic.

One obtains the same arithmetic theory for the finite extensions K of $\mathbb{F}_p(t)$ as for algebraic number fields. This is clear from what we have developed for arbitrary one-dimensional noetherian integral domains. But the crucial difference with the number field case is seen in that the function field K hides away a finite number of further prime ideals, besides the prime ideals of $\mathcal{O}$, which must be taken into account in a fully-fledged development of the theory.

This phenomenon appears already for the rational function field $\mathbb{F}_p(t)$, where it is due to the fact that the choice of the unknown t which determines the ring of integrality $\mathbb{F}_p[t]$ is totally arbitrary. A different choice, say $t' = 1/t$, determines a completely different ring $\mathbb{F}_p[1/t]$, and thus completely different prime ideals. It is therefore crucial to build a theory which is independent of such choices. This may be done either via the theory of valuations, or scheme theoretically, i.e., in a geometric way.

Let us first sketch the more naïve method, via the theory of valuations. Let K be a finite extension of $\mathbb{F}_p(t)$ and $\mathcal{O}$ the integral closure of $\mathbb{F}_p[t]$ in K.

By § 11, for every prime ideal $\mathfrak{p} \neq 0$ of $\mathcal{O}$ there is an associated normalized discrete valuation, i.e., a surjective function

$$v_{\mathfrak{p}} : K \longrightarrow \mathbb{Z} \cup \{\infty\}$$

satisfying the properties

(i) $v_{\mathfrak{p}}(0) = \infty$,

(ii) $v_{\mathfrak{p}}(ab) = v_{\mathfrak{p}}(a) + v_{\mathfrak{p}}(b)$,

(iii) $v_{\mathfrak{p}}(a+b) \geq \min\{v_{\mathfrak{p}}(a), v_{\mathfrak{p}}(b)\}$.

The relation between the valuations and the prime decomposition in the Dedekind domain $\mathcal{O}$ is given by

$$(a) = \prod_{\mathfrak{p}} \mathfrak{p}^{v_{\mathfrak{p}}(a)}.$$

The definition of a discrete valuation of K does not require the subring $\mathcal{O}$ to be given in advance, and in fact, aside from those arising from $\mathcal{O}$, there are finitely many other discrete valuations of K. In the case of the field $\mathbb{F}_p(t)$ there is one more valuation, besides the ones associated to the prime ideals $\mathfrak{p} = (p(t))$ of $\mathbb{F}_p[t]$, namely, the **degree valuation** v_∞. For $\frac{f}{g} \in \mathbb{F}_p(t)$, $f, g \in \mathbb{F}_p[t]$, it is defined by

$$v_\infty\left(\frac{f}{g}\right) = \deg(g) - \deg(f).$$

It is associated to the prime ideal $\mathfrak{p} = y\mathbb{F}_p[y]$ of the ring $\mathbb{F}_p[y]$, where $y = 1/t$. One can show that this exhausts all normalized valuations of the field $\mathbb{F}_p(t)$.

For an arbitrary finite extension K of $\mathbb{F}_p(t)$, instead of restricting attention to prime ideals, one now considers all normalized discrete valuations $v_{\mathfrak{p}}$ of K in the above sense, where the index $\mathfrak{p}$ has kept only a symbolic value. As an analogue of the ideal group we form the "divisor group", i.e., the free abelian group generated by these symbols,

$$Div(K) = \left\{ \sum_{\mathfrak{p}} n_{\mathfrak{p}}\mathfrak{p} \;\middle|\; n_{\mathfrak{p}} \in \mathbb{Z},\ n_{\mathfrak{p}} = 0 \text{ for almost all } \mathfrak{p} \right\}.$$

We consider the mapping

$$\operatorname{div} : K^* \longrightarrow Div(K), \quad \operatorname{div}(f) = \sum_{\mathfrak{p}} v_{\mathfrak{p}}(f)\mathfrak{p},$$

the image of which is written $\mathcal{P}(K)$, and we define the divisor class group of K by

$$Cl(K) = Div(K)/\mathcal{P}(K).$$

Unlike the ideal class group of an algebraic number field, this group is not finite. Rather, one has the canonical homomorphism

$$\deg : Cl(K) \longrightarrow \mathbb{Z},$$

which associates to the class of $\mathfrak{p}$ the degree $\deg(\mathfrak{p}) = [\kappa(\mathfrak{p}) : \mathbb{F}_p]$ of the residue class field of the valuation ring of $\mathfrak{p}$, and which associates to the class of an arbitrary divisor $\mathfrak{a} = \sum_{\mathfrak{p}} n_{\mathfrak{p}}\mathfrak{p}$ the sum

$$\deg(\mathfrak{a}) = \sum_{\mathfrak{p}} n_{\mathfrak{p}} \deg(\mathfrak{p}).$$

For a principal divisor $\operatorname{div}(f)$, $f \in K^*$, we find by an easy calculation that $\deg(\operatorname{div}(f)) = 0$, so that the mapping deg is indeed well-defined. As an analogue of the finiteness of the class number of an algebraic number field, one obtains here the fact that, if not $Cl(K)$ itself, the kernel $Cl^0(K)$ of deg is finite. The infinitude of the class group of function fields must not be considered as strange. On the contrary, it is rather the finiteness in the number field case that should be regarded as a deficiency which calls for correction. The adequate appreciation of this situation and its amendment will be explained in chap. III, § 1.

The ideal, completely satisfactory framework for the theory of function fields is provided by the notion of **scheme**. In the last section we introduced affine schemes as pairs $(X, \mathcal{O}_X)$ consisting of a topological space $X = \operatorname{Spec}(\mathcal{O})$ and a sheaf of rings $\mathcal{O}_X$ on X. More generally, a scheme is a topological space X with a sheaf of rings $\mathcal{O}_X$ such that, for every point of X, there exists a neighbourhood U which, together with the restriction $\mathcal{O}_U$ of the sheaf $\mathcal{O}_X$ to U, is isomorphic to an affine scheme in the sense of § 13. This generalization of affine schemes is the correct notion for a function field K. It shows all prime ideals at once, and misses none.

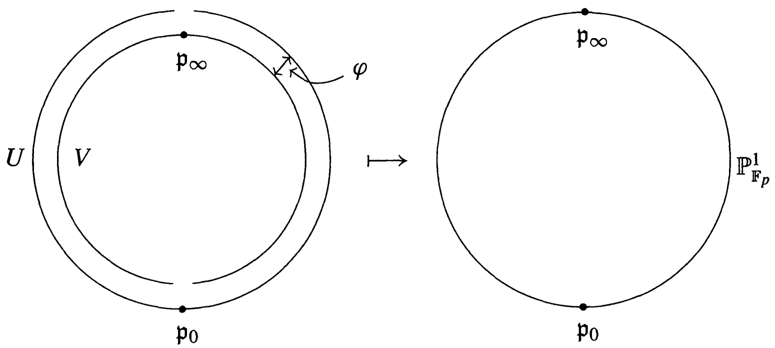
In the case $K = \mathbb{F}_p(t)$ for instance, the corresponding scheme $(X, \mathcal{O}_X)$ is obtained by gluing the two rings $A = \mathbb{F}_p[u]$ and $B = \mathbb{F}_p[v]$, or more precisely the two affine schemes $U = \operatorname{Spec}(A)$ and $V = \operatorname{Spec}(B)$. Removing from U the point $\mathfrak{p}_0 = (u)$, and the point $\mathfrak{p}_\infty = (v)$ from V, one has $U - \{\mathfrak{p}_0\} = \operatorname{Spec}(\mathbb{F}_p[u, u^{-1}])$, $V - \{\mathfrak{p}_\infty\} = \operatorname{Spec}(\mathbb{F}_p[v, v^{-1}])$, and the isomorphism $f : \mathbb{F}_p[u, u^{-1}] \to \mathbb{F}_p[v, v^{-1}]$, $u \mapsto v^{-1}$, yields a bijection

$$\varphi : V - \{\mathfrak{p}_\infty\} \longrightarrow U - \{\mathfrak{p}_0\}, \quad \mathfrak{p} \longmapsto f^{-1}(\mathfrak{p}).$$

We now identify in the union $U \cup V$ the points of $V - \{\mathfrak{p}_\infty\}$ with those of $U - \{\mathfrak{p}_0\}$ by means of φ, and obtain a topological space X. It is immediately obvious how to obtain a sheaf of rings $\mathcal{O}_X$ on X from the two sheaves $\mathcal{O}_U$ and $\mathcal{O}_V$. Removing from X the point $\mathfrak{p}_\infty$, resp. $\mathfrak{p}_0$, one gets canonical isomorphisms

$$\big(X - \{\mathfrak{p}_\infty\}, \mathcal{O}_{X - \{\mathfrak{p}_\infty\}}\big) \cong (U, \mathcal{O}_U), \quad \big(X - \{\mathfrak{p}_0\}, \mathcal{O}_{X - \{\mathfrak{p}_0\}}\big) \cong (V, \mathcal{O}_V).$$

The pair $(X, \mathcal{O}_X)$ is the scheme corresponding to the field $\mathbb{F}_p(t)$. It is called the **projective line** over $\mathbb{F}_p$ and denoted $\mathbb{P}^1_{\mathbb{F}_p}$.



More generally, one may similarly associate a scheme $(X, \mathcal{O}_X)$ to an arbitrary extension $K|\mathbb{F}_p(t)$. For the precise description of this procedure we refer the reader to [51].